

The Just-in-Time World

Six material supply chains, where their buffers sit, and how long a break takes to be felt

The Roundtable · Study 3 · research paper

Abstract

We follow six raw materials, each through the chain of one end product: copper to the building wire in a house, iron to a large power transformer, quartz sand to a power-management chip, salt to a bag of intravenous saline, crude oil to a tank of diesel, and lithium to a grid battery. For each we ask where the chain holds stock, how long a break at one link takes to be felt at the end, and whether the break passes or reorganizes the chain. Claims rest on a register of 235 rows; a sentence is printed only when the row under it has been validated, by a referee seat or, for a few facility facts, by the Author on the operator's own document.

The main finding is that the six chains do not answer the question in the same kind. Their answers are a curve, a transit time, a queue, and days of cover held at three different places; they are measured from different origins and cannot be ranked ¹. Where both ends are dated, stopping takes days and returning takes a season or years. One hurricane closed a quartz district and a saline plant about nine to twelve miles (fifteen to twenty kilometers) apart ²; the mine announced a restart in about fourteen days ³; the plant's first line took thirty-two, and its pre-storm output 141 days, with the shortage reported for about eleven months ⁴. US distillate cover was about 27 to 29 days in August 2026, against 32 to 34 in 1996 ⁵. Lithium brine spends 18 to 24 months in one operator's ponds ⁶, a delay that postpones an outage without absorbing it ⁷. On the supply paths on record, and even at the fastest order growth any maker has reported, a queue model's ceiling on large-transformer lead times is 24 to 25 months, below the 36 commonly quoted; the small share of orders from buyers who could be jumping the queue does not close the gap ⁸. When buyers can pay for priority, those who pay wait about the build time and everyone else holds the queue ⁹.

At four links, the instrument that lets residents see a harm was built after the harm was established ¹⁰. We report bands and floors, not forecasts, and we do not rank the chains by likelihood of failure.

1 • Introduction

1.1 The question

A new house outside Dallas has an electrical outlet in every room, and nothing about an outlet suggests it has a history. The wire behind it runs back through a distributor, a rod mill in Georgia, two container ports and an ocean, a smelter in China, a port with its own desalination plant, and an open pit in the Atacama Desert. This study is about arrangements of that kind, and about the fact that almost none of them keep any slack.

We follow six raw materials (copper, iron, sand, salt, oil and lithium), each through the supply chain of one end product: the building wire in a house, a large power transformer, a power-management chip, a bag of intravenous saline, a tank of diesel, and a grid battery. Of each chain we ask three things. Where does it hold stock, and how much? When one link breaks, how long is it before the break is felt at the end of the chain? And which breaks pass, and which reorganize the chain for good?

1.2 The frame, disclosed as framing

The six materials are those of Ed Conway's *Material World*, and the question is the one that book raises and does not settle: supply chains tuned for efficiency carry little inventory, so a break travels. That is an orienting frame and not a conclusion. At several points below the evidence cuts against its most dramatic form: a dominant node, the Spruce Pine quartz district, took a complete outage at its source, and both of its operators pointed to stocks held in the chain, one saying they would carry its customers through (Section 4.3); and refining utilization in 2025 stood below its 1998 peak (Section 4.5).

This is the third study in the Roundtable series. The first, *The Uneven Month*, priced a severed fuel link as a cost to households. This study measures the same kind of break in days, not dollars, and holds no price model (Section 4.5.3).

1.3 What this study is not

It is not a forecast, and it does not rank the six chains by how likely they are to fail. We set out to do that and could not: the cases here are in the study because they happened and were written down, which is no basis for a rate ¹¹. It forms no total across chains, of people or of days, for reasons Sections 2.5 and 5.1 give. Every waiting time it reports is a floor, and every magnitude is a band.

1.4 Architecture

Section 2 states the method. Section 3 reports a pattern in how the affected communities come to be measured at all, which governs how the later sections read. Section 4 takes the six chains in turn, each in the same four parts. Section 5 asks what a break feels like and why the six answers cannot be ranked. Section 6 accounts for one object in full. Sections 7 to 9 give limitations, what is and is not established, and how to reproduce the engines. Appendix A lists every row of the register; Appendix B the engines.

2 • Method

2.1 Scope

Each material is followed through one end product, and each product as one consignment from origin to disposal. The six consignments were chosen so that each ends at something a household meets: an outlet, the substation behind it, a charger, a hospital bag, a delivery truck, and the battery yard beside a substation. The grid battery was chosen over a phone battery because one phone failing is not a systemic event, and this study is about failures felt beyond the person they happen to.

A chain is described by six stages (extract, process, combine, install, distribute, dispose), and each stage is located as precisely as the record allows and no more precisely.

2.2 The register

Every claim in this paper rests on a row of the study's register. A row records one fact, figure, comparison or ruling, with its sources, the seat that owns it and the seat that validated it. A row's standing runs open, sourced, validated, integrated. A sentence is printed with a note only when the row under it is validated: a referee seat has read the source and the wording and accepted both. Five facility facts, handed back by a referee seat as outside its lane, were closed by the Author on the operator's or agency's own document. One conversion, the ship fuel for the two ocean crossings, was both made and passed by the Energy seat: its inputs are validated rows, and its arithmetic ships as a script that no second seat has re-run. Appendix A gives the validator recorded for every row.

Four seats refereed this study: Economics and Statistics, which audits scale and share claims; Energy and Biophysical Economics, which owns every conversion; Complex Systems and Polycrisis, which reviews dependencies between chains and the cases of reorganization; and Institutions and Commons, which reviews every statement about who works at a stage and who lives beside it.

A superscript number refers to the Notes, which name the register row or rows a sentence rests on, any condition the row carries, and the sources the row cites in the reference list. A note that names a ruling and no row marks a decision of a referee seat, which is not a claim's support. Appendix A lists every row with its standing and, for a validated row, what was validated, in words the validating seat has read.

2.3 Precision conventions

Places. In Figures 1, 3, 6, 7, 9 and 10 a solid dot marks a named site. A dashed ring marks a region or a kind of works and not a plant: the record names a state, a district or a class of producer, and the figure says only what the record says. A ring is a limit of the record and not a description of the place ¹².

Distances. Sea legs are the published port-to-port sailing distances. Days at sea are such a distance divided by the average speed at sea of the fleet that carries the cargo; the result is a band, and it leans long, because a scheduled ocean crossing tends to run faster than its fleet's annual mean. An overland leg between two sited ends is given as a great-circle floor: the route itself is longer. A floor is given only where the leg drawn is the object's whole path between its two ends. Floors are never summed, and a floor is never converted into days or fuel, because no speed or mode is sourced for the overland legs. Where one end is a ring, no floor is given: a straight line from the center of a region is not a floor on anything.

Waits. Every waiting time in this paper is a floor: the honest sentence is "at least this long" ¹³.

Units. The distance of a leg is in miles with kilometers in parentheses; mass is in tonnes.

2.4 Comparison objects

Energy is given in the unit its source uses, with the carrier named: electricity at the meter is not the same quantity as the primary energy behind it, and the two are never added. Where a comparison helps, a figure

is also given in a home's electricity use: a household-month is 899 kilowatt-hours, the average American residential customer's monthly electricity, and a home-year is twelve of them ¹⁴. The conversion standard is the Energy seat's, and a figure with no stage-compatible partner is given no comparison. Time and energy in this study each span many orders of magnitude, from a flooded plant's first line back to an aquifer's recovery, and from the silicon in one chip to a year of one industry's electricity.

2.5 Who fields

Each named site and each class of works carries a statement of who works there and, where the record holds it, who lives beside it; where the study holds nothing, the statement says so. A named site's statement is printed in its site record, and a class's in a line beginning *Who*. The units differ from row to row (a site, an operation, a company, a legal bargaining unit, a forecast, an exposed population), and no total can honestly be formed from them ¹². No headcount is summed anywhere in this paper.

2.6 Engines

Three computations stand behind the modeled figures. The transformer queue model (E3) is a queue of orders for large transformers against domestic build capacity and imports, with two classes of buyer, one of which pays to be served first. Its ceiling and its divergence boundary are read from a deterministic central path, and the ceiling is also reproduced as the median of seeded runs. The reserve-pool companion (E3-C) computes how many of its own failures a utility's spare transformers cover and how much a voluntary sharing pool adds, as the share of failures a stocked spare can serve rises. It is a seeded simulation checked against closed forms, and its pooling figures are upper bounds on the pool's reach (Figure 5). The σ fit is a deterministic fit that gives a wide envelope for the share of orders from buyers who could be jumping the queue. E3 and E3-C were each rebuilt by the Author in an independent construction; the σ fit was recomputed in its arithmetic and not in its choices. Appendix B gives the files, seeds and fingerprints, and states which objects were reproduced, by whom, and what each reproduction does and does not test.

3 · How we know: instruments that arrive late

Four times in this study, at four different links, the same institutional thing happens. An aquifer under the Atacama is drawn down for two decades before a court orders the baseline study that would have measured it. A refining complex dating from the early twentieth century begins reporting what crosses its fence line in 2019. The chemical plants that make the plastic in an intravenous bag were required to begin monitoring for vinyl chloride at the property line in July 2026, and on the day the requirement took effect a proclamation exempted a named list of them, including some of the largest, for two more years ¹⁵. Disposal wells that had been raising the ground pressure under West Texas for years are required, from 2023, to carry a gauge deep in the hole and report what it reads. The intervals run from about five years to more than a century, and in every case the instrument that would let the people living there see what is happening to them is built after the harm is established, not before ¹⁰.

On its own that is a complaint and not a finding. The finding is in what kind of instrument gets built. All four are public. The benzene readings from every American refinery are on an EPA dashboard; the Texas

injection pressures go to a university-run tool open to anyone; the Chilean settlement is supervised in open court with annual compliance hearings; and the chemical-plant rule is the refinery rule's descendant by EPA's own account. None of these had to be. A regulator can measure privately and act on what it finds, and the harm would be addressed just the same. Building the instrument in public does something else: it hands the people who live there a standing capability they did not have before, and, where the record is filed rather than merely served, one that does not depend on anyone's continued attention. Each quarter of refinery fence-line data carries a date by which it must be public, and a quarter once published cannot be unpublished. An enforcement action ends; a data stream keeps producing. What that changes is who is able to make the argument, not who wins it. Port Arthur's fence line has been publishing since 2019, and Port Arthur's air is still Port Arthur's air.

The boundary of the pattern is visible inside this study too. Around the Guixi smelter in Jiangxi, the arsenic and cadmium in the soil and the vegetables were established by university researchers and a non-governmental organization, not by an instrument anyone in those villages can consult. So the claim is narrower than it first looks: where a regulator eventually acts, the thing it builds is public, and where none does, the knowledge exists without belonging to anyone who could use it.

The count stays at four. Section 4.5.2 describes two state regulators responding to one hazard in one basin in the same year; that is one observation seen twice, not a fifth case ¹².

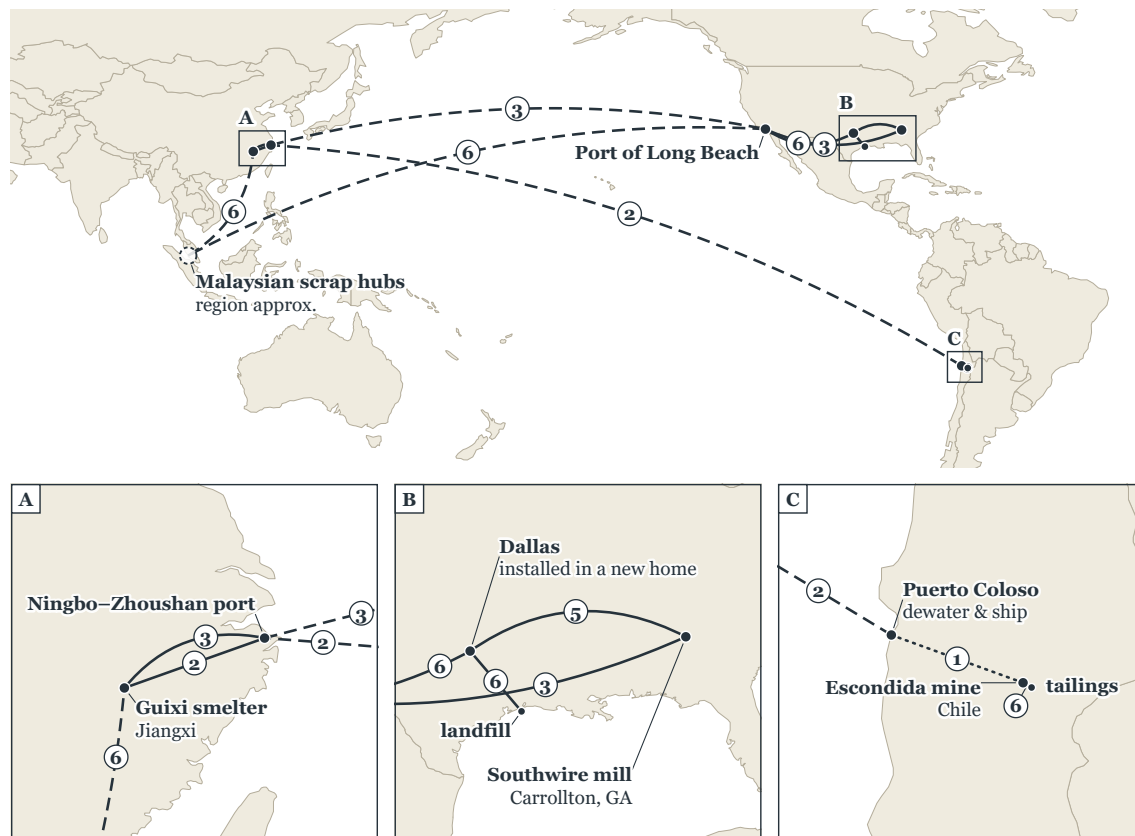
4 • The six chains

Each chain is reported in the same order: the consignment, in a table; the chain by stage; where its buffer sits; and what the study does not hold.

4.1 Copper: the building wire in a house

Table 1. The consignment: one spool of building wire.

Consignment	one spool NM-B building wire, 88 kg copper ¹⁶
Origin	Minera Escondida, Atacama Desert, Chile
Distance traveled	the ocean legs, port to port; the longest about 6,700 miles (10,760 km), Shanghai to Los Angeles. And overland, in straight lines that the routes exceed: in China, at least about 280 miles (450 km) from the port to the smelter and the same back; in the United States, at least about 1,850 miles (3,000 km) from Long Beach to the mill at Carrollton, and at least about 650 miles (1,050 km) from the mill to the Dallas area. The pipe from the mine to its own port is about 105 miles (170 km). The ocean distance is measured between the distance table's own ports; Figure 1 pins neighboring ports, Ningbo and Long Beach, and the study does not establish which of each pair this cargo used ¹⁷
Days in transit	48–52 days at sea, both crossings, at each fleet's average speed; time in port, at the smelter, overland and at the mill is not sourced ¹⁸
Energy to make	≈ one household-month of electricity (0.35–1.15 by boundary) ¹⁹
Ocean fuel, both crossings	four or five gallons (3.6–5.3; sea legs only; overland legs excluded) ²⁰
Buffer at destination	No figure: what a distributor keeps at the end of this chain is not published in the federal monthly survey. The parent category that it does publish measures electrical and electronics wholesalers in dollars of stock over dollars of sales, and on a chain priced in copper that is not a count of wire ²¹
Final disposition	three endings, and one of them is not an ending: tailings at the mine, accumulating since the day extraction started; insulation to landfill; and demolition scrap re-entering the smelting pool, a trade that rerouted twice in five years as import specifications tightened on both sides of the Pacific ²²



Drawn by `build_chain_maps.py` from the study's own map tables.

Country outlines: Natural Earth, public domain.

Each file's SHA-256 fingerprint is recorded inside this image.

Figure 1. The copper chain, from the mine to a house, with its three endings: tailings at the mine, a landfill, and the scrap trade back across the Pacific. A filled dot is a located plant, port or site; a dashed ring is a region or a kind of work, and not a plant. The circled number on a leg is its stage, from 1, extraction, to 6, disposal. A leg in long dashes goes by sea, a solid leg overland, a dotted leg by pipe; a leg in dashes and dots, in Figure 6 only, is one whose trade goes mostly by air. Each lettered box is enlarged in the panel of the same letter. A leg joins its two ends and does not trace a route. Figures 3, 6, 7, 9 and 10 use the same marks.

Extraction. The wire begins as ore under the Atacama Desert at Minera Escondida, one pit complex producing about one tonne in every eighteen of the world's mined copper (fiscal year 2025)²³. Its concentrate leaves not by truck but by a 105-mile (170 km) slurry pipeline to the mine's own private port²⁴. Across Chile, concentrating copper took about 13 gigajoules of electricity for every tonne of copper in the concentrate in 2024, up by nearly half since 2010; the state copper commission gives the reason as deeper pits and harder rock, and in the open pits the fuel burned per tonne of copper has doubled over the same years as ore grades fell²⁵.



Plate 1. Escondida from orbit: two pits, leach pads and tailings works spread across tens of kilometers of the driest desert on Earth.
NASA Earth Observatory · public domain

Site record 1. Minera Escondida, Atacama Desert, Chile.

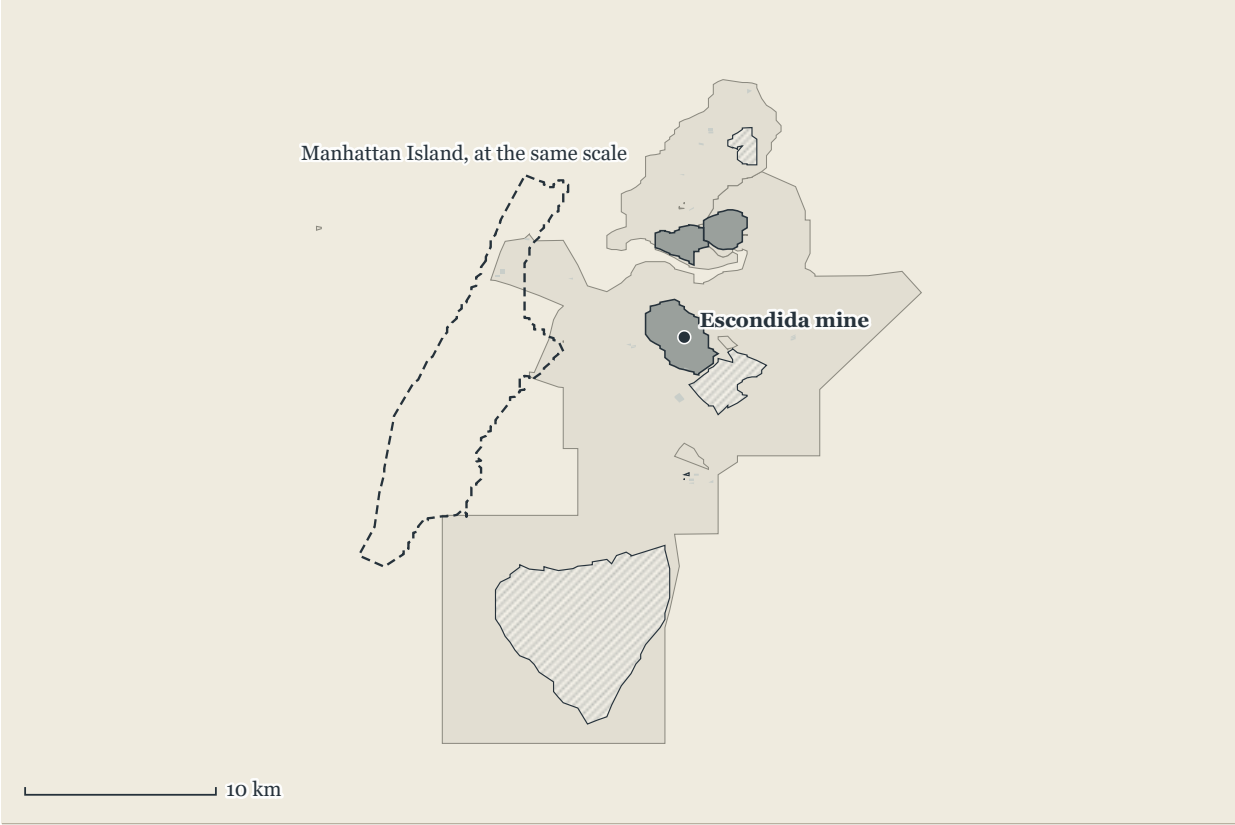
Who	BHP operates the mine with 57.5%, Rio Tinto holds 30% and the JECO companies 12.5% between them, 10 and 2.5. The company's own annual report gives 4,000 of its own employees and 6,532 operational contractors at the end of 2025, and does not say whom 'operational' leaves out ²⁶ . The Comunidad Atacameña de Peine is both litigant, having filed before the state did, and now co-governor through the standing governance table; the state is at once allocator and co-plaintiff, through the national water authority (DGA), the state's legal counsel (CDE), the environmental regulator (SMA) and the Environmental Tribunal.
Energy	Escondida and the Spence mine together contracted 6 TWh a year of renewable electricity from Enel Generación Chile (from August 2021) and Colbún (from January 2022); the volumes are contractual and annual, shared between two mines, and say nothing about hour-by-hour supply ²⁷ .
Ecosystem	The drawdown case on the MNT aquifer ended in a court-approved conciliation in December 2024, on a divided bench: about US\$47M and 14 measures, with implementation under tribunal supervision, and the court put the aquifer's own recovery at about the year 2200 ²⁸ . Since 2020 the mine has run on 100% desalinated seawater. It is pumped 105 miles (170 km) inland and 3,100 m up ²⁹ .
Comparison	One pit is about 5.7% of the world's mined copper, one tonne in every eighteen, in its 2025 fiscal year ²³ .

Site record 2. Puerto Coloso, Chile: the mine's port and its water supply.

Who	The port is Minera Escondida's own, and the desalination plant beside it is run by EWS. The Caleta Coloso fishing community's organizations have administered the cove under Chile's Caletas law since 2025, and the port's owner funds the cove's new facilities through its foundation ³⁰ .
Energy	Concentrate goes down the 105-mile (170 km) slurry pipe and water comes up the same corridor. Desalting costs 3.4 kWh a cubic meter and the pumping about 14, so the climb costs four times the desalting ²⁹ .
Ecosystem	One site carries two lifelines, the mine's outlet and its water supply: lose Coloso and the mine loses both, on whatever water buffer it holds ³¹ .
Comparison	The water piped up to the pit carries about 1.4 TWh a year, about as much electricity as 130,000 US homes use in a year ²⁹ .

The district spans tens of kilometers of pits, leach ponds and tailings works (Figure 2). The water that keeps it running is seawater now, desalinated at the coast and pumped 3,100 meters uphill ²⁹. The basin's Indigenous communities took the aquifer drawdown to court, the Peine community filing before the state did; the case ended in a court-approved conciliation (fourteen measures, a standing governance table,

implementation still under tribunal supervision), while the court put the aquifer's own recovery at around the year 2200 ²⁸.



Drawn by build_chain_maps.py from the study's own map tables.
Country outlines: Natural Earth, public domain. Footprints © OpenStreetMap contributors (ODbL).
Each file's SHA-256 fingerprint is recorded inside this image.

Figure 2. The Escondida district at true scale. Dark fill is the pits, hatching is tailings, the pale area is plant ground, and the small gray shapes are ponds. The dashed outline is Manhattan Island at the same scale. The footprints are OpenStreetMap's.

Processing. The concentrate ships across the Pacific to Guixi, in Jiangxi province, the only single plant on Earth that refines more than a million tonnes a year ³². The copper becomes cathode there. Farming villages lie in the smelter's deposition plume: about fifteen villages, some ten thousand people, with documented metals in their soil and vegetables ³³. What powers the smelter is not on public record. A 1,880-megawatt, four-unit coal station stands beside it (Global Energy Monitor's figures); whether any of that output crosses the fence is stated nowhere a reader can check ³⁴.

Site record 3. The Guixi smelter, Jiangxi, China.

Who	Jiangxi Copper Corporation is owned and operated by the provincial state, which is its environmental regulator at the same time; the group's holding stood at 44.06% at year end ³⁵ . The Guixi Power Generation Company stands next door, and about 15 farming villages, some 10,000 people, lie in the deposition plume ³³ .
Energy	The smelting stage takes about 9 GJ a tonne, on a 2017 Chinese average within a band of 9 to 14, while the electrolytic finish takes only 250 to 400 kWh a tonne: the heat, not the electricity, is the story of a smelter ³⁶ .
Ecosystem	Peer-reviewed work finds arsenic, cadmium, copper, lead and zinc in the soils and vegetables of named villages ³³ .
Comparison	It is the only single plant on Earth that refines more than a million tonnes a year: 1,012.5 kt in 2024 ³² .

Combination. The copper then crosses the same ocean again: cathode east to Long Beach, then overland to Carrollton, Georgia, where Southwire can cast it to rod at close to a tonne a minute³⁷ and its Building Wire Plant in the same town makes the wire³⁸. The insulation extruded around it is PVC, which is where the salt chain enters the copper chain³⁹.

Site record 4. Southwire, Carrollton, Georgia: rod mill and building-wire plant.

Who	Southwire Company LLC is family-owned and, by its own account, one of the leading manufacturers of wire and cable ⁴⁰ . Labor at the Carrollton works appears as a role and no more: rod-mill and building-wire-plant crews ⁴¹ . No public body is named at this works. This study holds nothing on who lives beside the works.
Energy	The rod mill is rated at 54 tonnes an hour, on the company's own figure ³⁷ , and the maker specifies its shaft furnaces for natural gas, LPG, or both ⁴² .
Ecosystem	This study holds no ecosystem record for this works. The house wire's insulation is PVC, and what making PVC costs, and whom, is followed in the salt section of this paper as a material and not as this wire's own supply ³⁹ .
Comparison	Cathode can become rod at close to a tonne a minute ³⁷ .

The crossings. The cathode's crossing takes 14 to 15 days at the container fleet's own average speed⁴³. It is the shorter of the two: the ore's run to the smelter takes more than twice as long⁴⁴, and the two together are roughly 48 to 52 days at sea. Transit is the longest time this study can put on the chain, and it is the part that cannot be made up after a break: a plant can be restarted, and an ocean cannot be shortened. The fuel is the only thing spent on the way, four or five gallons of bunker for this spool's share of the sea legs²⁰.

Energy. Making one home's worth of wire took roughly as much energy as that home uses in electricity in a month, and over its life the wire will deliver several hundred times the energy it cost to make¹⁹. That asymmetry, cheap to make next to what flows through it and slow to replace, is the shape the later sections look for, in time where they cannot find it in energy.

Installation and distribution. A distributor's warehouse and a contractor's van carry the spool to the house, where an electrician staples it into the stud bay and it is not seen again.

Disposal. The chain has three endings, and one of them is not an ending: tailings at the mine, which began accumulating the day extraction started; insulation to landfill; and demolition scrap re-entering the smelting pool, a trade that rerouted twice in five years as import specifications tightened on both sides of the Pacific²². The Malaysian end of that trade is drawn in Figure 1 as a ring: no facility is named in the record, so none is pinned.

There is a short loop as well as the long one. A tonne of clean scrap melted at the mill is ready to become rod again for roughly three to six percent of the energy it takes to bring a tonne from ore. That leaves out collecting, sorting and shipping the scrap, which nobody here has measured⁴⁵. The furnace's maker puts the ceiling at a fifth of the charge⁴², and it is not the road demolition scrap takes: mixed and insulated scrap goes back through a smelter, and this study has not measured that road's energy at all.

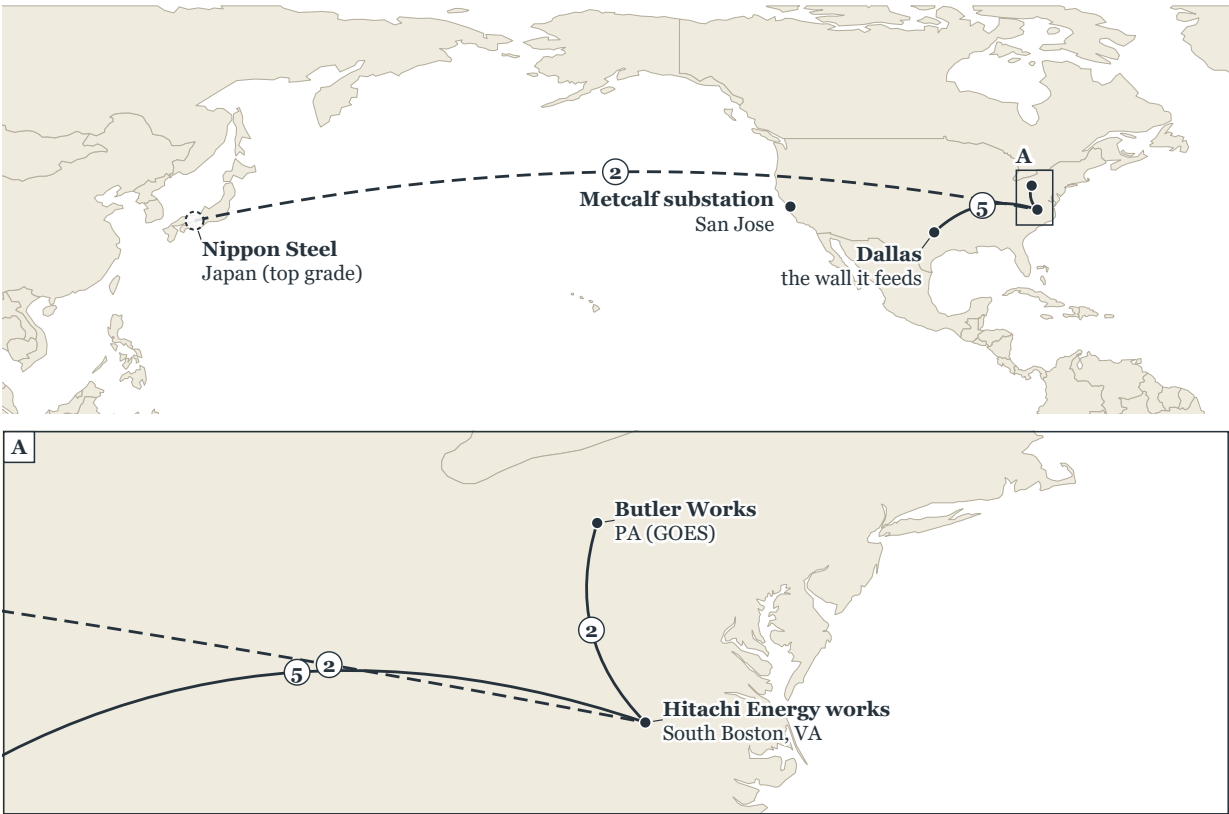
The buffer. What a distributor keeps at the end of this chain is not published in the federal monthly survey (Table 1). At the other end the buffer is water, and Section 5.3 takes it up: the mine holds a reservoir whose volume we have not found published, and an interruption would slow the concentrator before it stopped it.

Not held. Time in port, at the smelter, overland and at the mill; the smelter's power supply; the energy of the smelter road for mixed scrap; the distributor's stock.

4.2 Iron: the large power transformer

Table 2. The consignment: one large power transformer.

Consignment	one large power transformer, bespoke
Steel	one domestic mill; ~80% imported ⁴⁶
Copper	about 40 t at 500 MVA, one maker's declaration ⁴⁷
Build	~9 months intrinsic ⁴⁸
Wait	36 months commonly quoted, 60 recorded; floors, not levels ⁴⁹
Price	quoted ~+80% (NIAC), transaction +63–68% (federal PPI) ⁵⁰
Buffer at destination	utilities reported spares equal to 116 percent of the most-crucial units in 2016, which is the high-voltage machines standing in substations and not the 4,900 to 6,799 of the fleet; the two most formal sharing programs had never been used as of 2024 ⁵¹
Final disposition	refit; scrap steel recycles well ⁵² ; the copper windings go for scrap, and what recycling them takes is not measured in this study



Drawn by `build_chain_maps.py` from the study's own map tables.
Country outlines: Natural Earth, public domain.
Each file's SHA-256 fingerprint is recorded inside this image.

Figure 3. The iron chain: electrical steel to the transformer works (stage 2), and the transformer to the wall it feeds (stage 5). GOES is grain-oriented electrical steel. The ring in Japan marks a maker and not a located mill. The Metcalf substation, at San Jose, is the site of Section 4.2 (Fragility, and the reserve) and Site record 6; no leg of this consignment runs to it. Marks as in Figure 1.

The machine. The biggest of the machines this section follows weigh as much as a loaded 747 ⁵³, and every one is wound by hand. A worker guides copper wire, tens of tonnes of it in the big units (about forty in one

maker's declared 500 MVA design)⁴⁷, around a core of steel so specialized that sheets are rolled to nine-thousandths of an inch⁵⁴. The American grid holds thousands of them⁵⁵.



Plate 2. A large power transformer under tow in Cow Canyon, Utah, reportedly the heaviest load ever moved on Utah highways: seven trucks, five of them pushing, at five miles an hour.
Josh Ewing · CC BY 3.0

The steel. Transformer cores need grain-oriented electrical steel, and the United States has exactly one company that makes any: AK Steel, now Cleveland-Cliffs, at Butler, Pennsylvania⁵⁶. It produces an intermediate grade that American transformer builders mostly do not use; the top grade is made by one company, Nippon Steel, in Japan⁵⁶. About 80% of the grain-oriented steel in American transformers is imported⁵⁴. The one domestic mill is also being crowded out of its own product: the same machines can make electric-vehicle motor steel, which pays better, so the lithium chain competes with the iron chain for the same mill⁵⁴.

Site record 5. Butler Works, Pennsylvania.

Who	Cleveland-Cliffs, the former AK Steel works, is the only US maker of grain-oriented electrical steel, and in DOE's own words domestic GOES is “a major weak link... shrinking supply” ⁵⁶ . Labor at the Butler works appears as a role and no more: mill crews ⁵⁷ . This study holds nothing on who lives beside the works.
Energy	This study holds no energy figure for this works.
Ecosystem	This study holds no ecosystem record for this works.
Comparison	GOES sheets are rolled to nine-thousandths of an inch ⁵⁴ . GOES and the specialty copper inside a transformer are each about a quarter of its production cost ⁵⁴ , and the same machines can make EV-motor steel, which pays better: the lithium chain competing with the iron chain for the same mill ⁵⁴ .

The works. A large power transformer is not a product; it is a project. Every unit is engineered to its substation, in the government's words “usually custom-built to fit the specifications of their use case”⁵⁸, and the heart of the build is still manual: winding crews, paper insulation placed by hand, master welders⁵⁹. The intrinsic build time, with an open slot and no queue, was about nine months in the last slack market

⁶⁰. The works in Figure 3 is a real one: Hitachi Energy's plant at South Boston, Virginia, building transformers since 1968, with three dated expansion steps between October 2022 and a groundbreaking in June 2026. The record gives dates and dollars and not one capacity figure, so no capacity is drawn from it

⁶¹. This study holds nothing on the works' workforce or neighbors.

The move. Moving one is its own engineering discipline: 100 to 400 short tons, 200,000 to 800,000 pounds in the department's own words ⁵³, on one of a handful of special railcars in North America ⁵³, with the last five to ten miles the hardest of the journey ⁵³. The transport is itself a chokepoint: it binds exactly when many units must move at once.

In service. The average American large power transformer is around forty years old, and seventy percent are twenty-five or older ⁶². Forty years is the design life, not a cliff. On a hazard fitted to the primary sources, roughly four in five units pass forty and the median unit lives nearly sixty years; both are estimates, with bands ⁶³. The fleet is not about to fail; it is about to need replacing at a rate the queue cannot casually absorb.

The wait

Before the pandemic a large transformer could be ordered and received inside a year ⁶⁴. Today thirty-six months is the commonly quoted wait and sixty the recorded maximum ⁴⁹. In March 2025 GE Vernova's chief executive told an interviewer that the company's backlog of orders for gas turbines, power transformers and switchgear, taken together, stretched into 2028, and that he expected to be largely sold out through the end of that year by the summer ⁶⁵. Prices are up by about four-fifths on quoted prices, and by 63 to 68 percent on the federal transaction index, depending on the year ⁵⁰. Nothing was destroyed. Demand arrived faster than a bespoke, hand-wound, single-steel-supplier industry could grow ⁶⁶.

The queue model puts the mechanical ceiling of the primary-sourced system at roughly 24–25 months, even on the fastest order growth any maker has reported: the supply paths on record cannot produce the 36 the industry reports ⁶⁷. The gap is an adjudicated finding. The share of orders from buyers who could be jumping the queue is real but small, somewhere between one order in five and one in twenty, worth at most half a month at the median quote and not the missing twelve ⁶⁸. The residual belongs to three things this study cannot yet tell apart: how quotes themselves are formed, the domestic side of supply, whose path no primary source covers, and the backlog the boom began with ⁶⁹. The pressure on the queue is on record: power plants and batteries holding signed or draft interconnection agreements add up to about seven and a half times what the country connected in 2024 ⁷⁰.

The model's sharpest finding is distributional, and it is a mechanism and not a number. When buyers can pay to jump the queue, the priority class waits about the build time (nine months, the build-time estimate the model is given, not something it computes) at every priority share we ran, while the standard class holds the entire queue ⁹. Record deliveries and record waits at the same time are not a paradox: a queue grows whenever orders outrun deliveries, however many are delivered. What priority changes is who stands in it. *The Uneven Month* found that shocks bill the bottom of the income distribution; this section finds the same shape in industrial plumbing.

At what demand growth does a replacement queue stop being a wait and become a permanent state? The boundary is computed, and the country's published forecasts sit on the safe side of it ⁷¹: at the national

forecast rate the queue cannot become permanent under any capacity response. The divergence wedge opens between roughly 7¾ and 10½ percent a year, depending on what counts as cleared (the record gives a band, not a line), which is hotter than any published forecast (Figure 4).

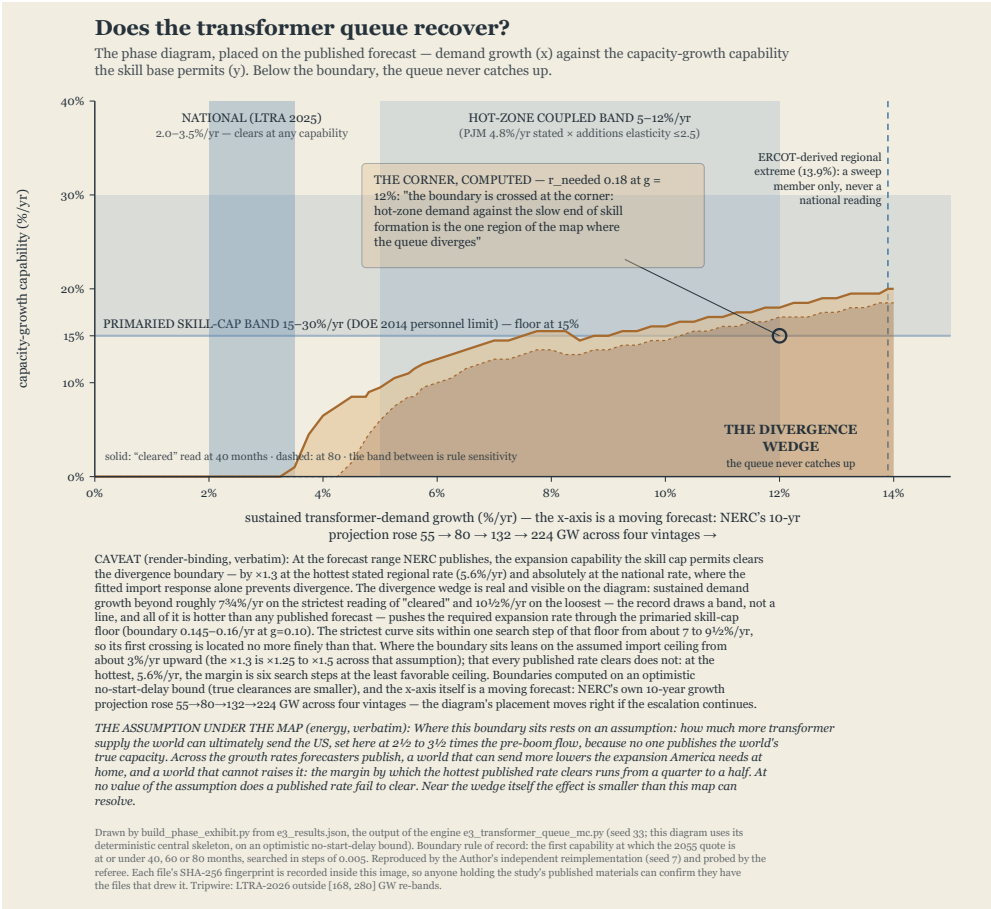


Figure 4. Demand growth against the capacity response. The wedge is where a replacement queue stops clearing; the published national forecast is placed against it. The figure's two scope notes are printed on it.

Fragility, and the reserve

In April 2013, rifle fire damaged a number of transformers at the Metcalf substation in California; the one federal account declines to say how many, and so does this study ⁷². Two things are true of that night. The grid re-dispatched around the damage and no blackout followed; and repairs to the equipment took nearly a month, with the power still on. Electrons reroute; iron does not. Only the public record is used here, and nothing in the queue model speaks to how outages start ⁷³.

Site record 6. Metcalf substation, California.

Who	PG&E (owner-operator); this study names no one else here, on purpose; public record only; no method detail is given ⁷⁴
Energy	The energy story at this node is the transmission layer that re-dispatched around the damage ⁷⁵ .
Ecosystem	This study holds no ecosystem record for this substation.
Comparison	both truths at once: no blackout (the system re-dispatched around the damage) AND nearly a month of repairs with the power still on: electrons reroute, iron does not ⁷⁵

In 2015 Congress asked whether the United States should keep a strategic reserve of transformers. The resulting study concluded what the industry already knew, that bespoke machines make a warehouse of spares nearly useless, and the reserve was shelved in favor of voluntary sharing programs ⁷⁶. The reserve-pool model turns that judgment into a curve: a reserve's value scales with standardization, the fraction of failures a stocked spare can serve, and that fraction is a choice, made a few hundred times a year by utilities each buying for its own system, none of whom is choosing it ⁷⁶. The buffer is not missing because nobody thought of it. The Department of Energy's own account of why is regulatory: utilities may raise rates to recover from a disaster, and not to prevent one ⁷⁶ (Figure 5).

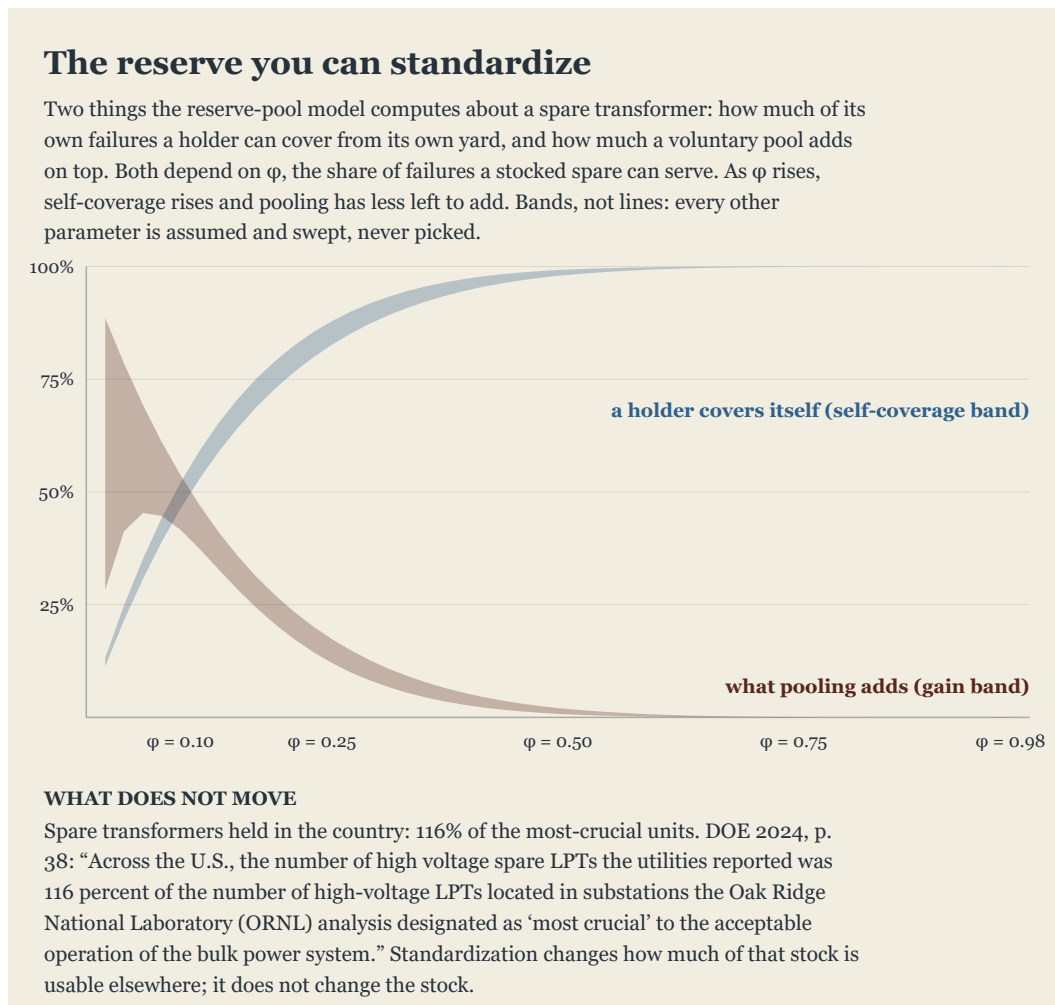


Figure 5, part 1 of 3; continues.

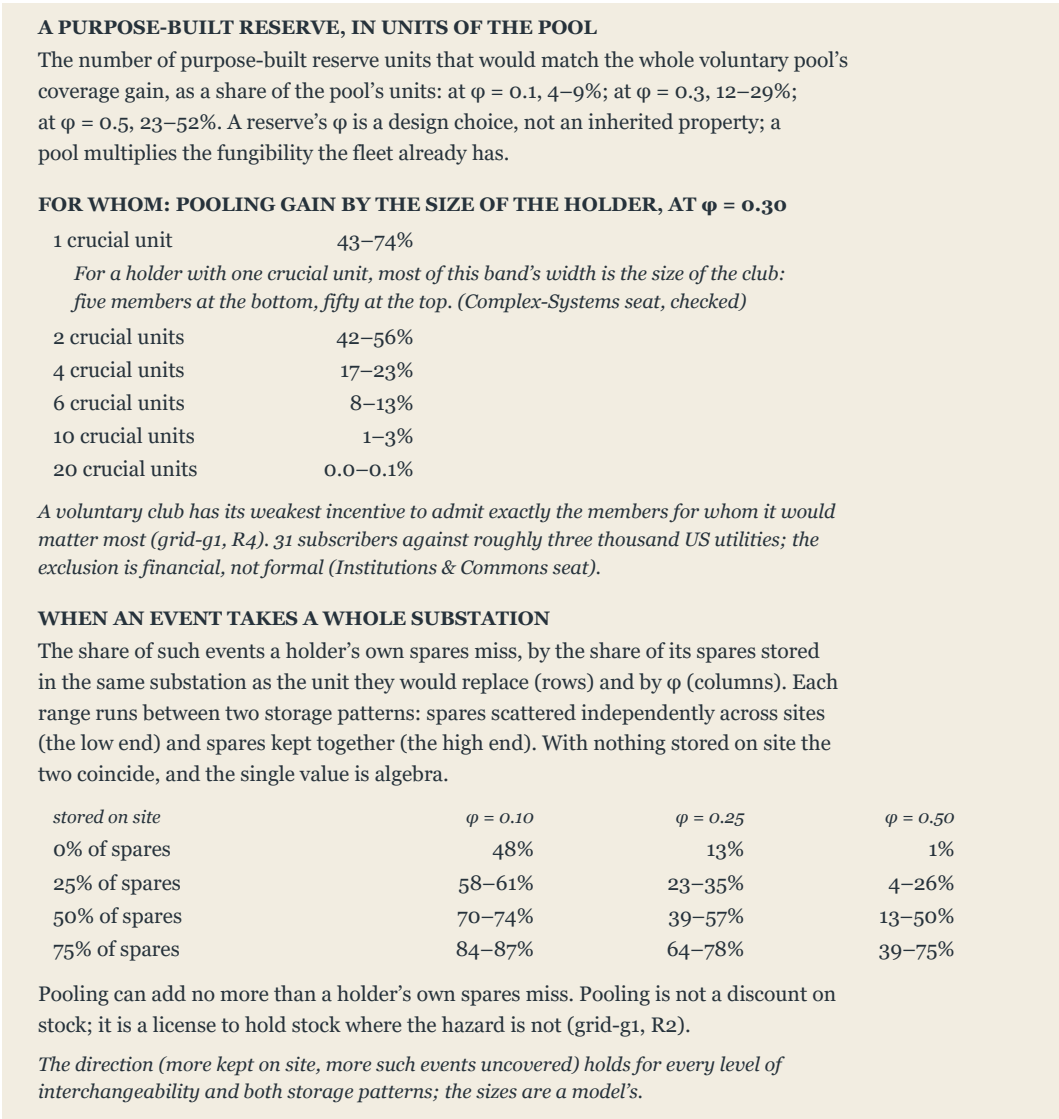


Figure 5, part 2 of 3; continues.

THE RECORD, IN DOE'S 2024 WORDS. THE STUDY TAKES NO POSITION ON WHETHER A FEDERAL RESERVE SHOULD EXIST.

"A 2017 Department of Energy (DOE) Report to Congress assessed options for a strategic transformer reserve and recommended a non-government solution driven by industry actions and compliance with North American Electric Reliability Corporation's Reliability Standard CIP-014-2 requirements." (p. 4) "While this report does not advocate for a change in this recommendation, it does offer data indicating significant progress within the industry in addressing this issue." (p. 4, July 2024) "In its 2017 Report to Congress, the DOE noted that a federally-owned LPT reserve program would require five or more geographically diverse sites nation-wide, and well over 100 transformers." (p. 43)

Stated direction of bias, from the engine: INDEPENDENT compatibility OVERSTATES pooling. Real compatibility is clustered by design family. Every pooling figure here is an UPPER BOUND on the voluntary substitute's reach.

That bound is on the pool's reach: whether some other member's spare fits. What pooling adds is reach multiplied by the hole a holder's own spares leave, and the same clustering moves the hole, in a direction this engine does not model. Read every gain here for its direction and its order, never for its size. (Complex-Systems seat)

Drawn by build_reserve_exhibit.py from reserve_pool_results.json, member_size_sweep.json and site_event_closed_form_check.json (seed 41, 40,000 paths; 2016 DOE-anchored edition; the 2023 derived edition is not drawn). Each file's SHA-256 fingerprint is recorded inside this image, so anyone holding the study's published materials can confirm they have the files that drew it, and run the first on the rest to draw it again. Register row grid-g1 (validated jointly by the Complex-Systems and Institutions & Commons seats). The tour carries the interactive edition.

Figure 5. Self-coverage and the gain from pooling, against the share of failures a stocked spare can serve. As standardization rises a holder covers more of its own failures and a pool has less left to add; the anchored spare stock does not move. The record is quoted and the study takes no position.

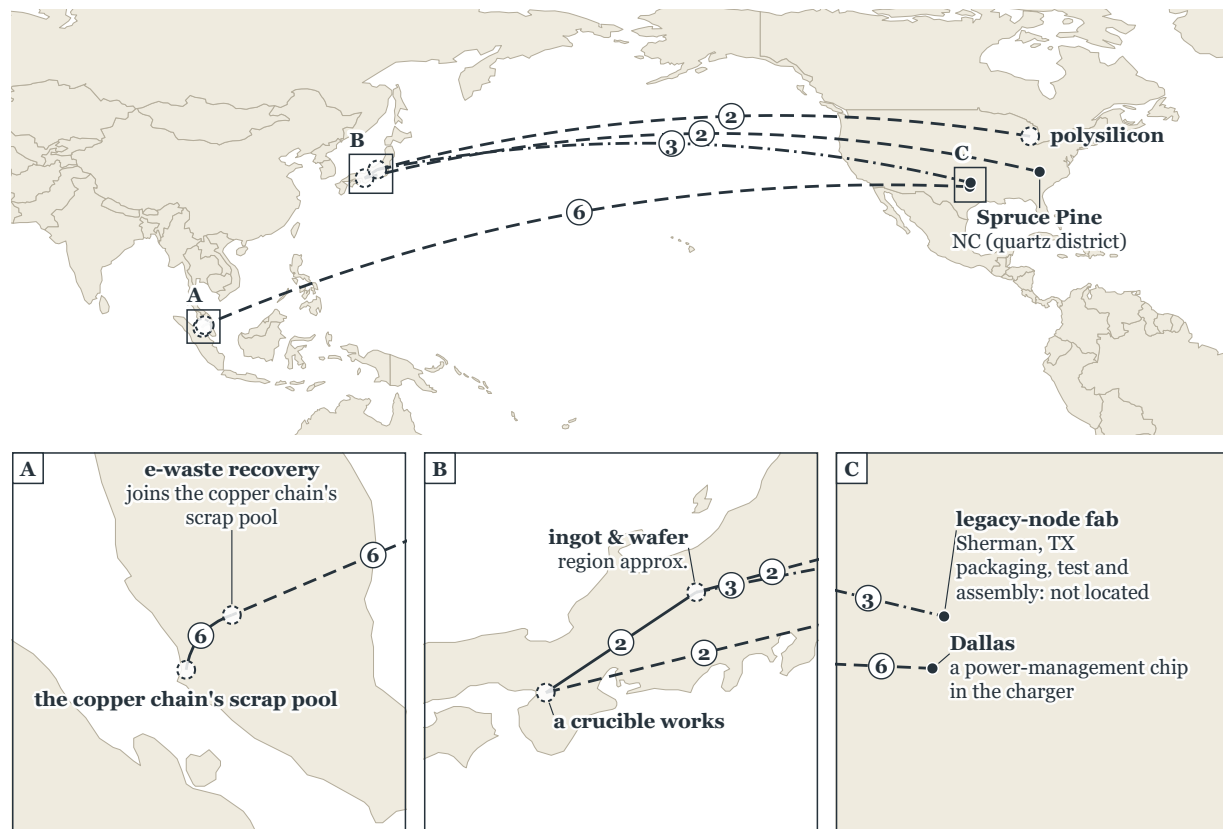
The buffer. This chain's answer to the buffer question is a queue and not a stock (Section 5.3), with a sharing club where a reserve might have been.

Not held. A capacity figure for any works; a published duration for a transformer failure; the inventory behind the one dedicated sharing program, or the inputs of the model that sizes it.

4.3 Sand: the power-management chip

Table 3. The consignment: one power-management chip.

Consignment	one power-management chip, in a phone charger
Origin	the Spruce Pine quartz district, North Carolina
Distance traveled	the Pacific crossing, anchored and not pinned: about 5,600 to 6,000 miles (8,962 to 9,603 km), between ports of the kind each end would use. Los Angeles is the assumed US end and the fab is inland at Sherman ⁷⁷
Days in transit	Not shown: most of this cargo does not sail: of doped wafers imported from Japan in 2024, four fifths by weight traveled by air. The sea crossing is computed, at about twelve to thirteen days, and deliberately not rendered as the transit time ⁷⁸
Energy to make	the silicon in one small chip, about a watt-hour, from solar-grade polysilicon at about 49 kWh a kilogram; the fab’s own electricity, which the Energy seat expects to be the larger term, is a figure this study named and did not obtain ⁷⁹
Fuel to cross	No figure: four fifths of this cargo by weight flew, and a sea figure for a cargo that flew is wrong in direction and not merely in size, because a ship sits near the least fuel-intensive mode and air near the most. An air-freight intensity is the one figure that would make this a finding ⁸⁰
Buffer at destination	buyers’ median inventory of the chips they highlighted fell from 40 days in 2019 to under 5 in 2021, a median across respondents and not a measure of any one chip ⁸¹
Final disposition	the copper on the board can return to the copper chain’s scrap pool; the silicon does not, and the federal summaries report silicon recycling in one word, insignificant ⁸²



Drawn by `build_chain_maps.py` from the study's own map tables.
 Country outlines: Natural Earth, public domain.
 Each file's SHA-256 fingerprint is recorded inside this image.

Figure 6. The sand chain, with the leg (stage 6, inset A) by which the copper on a circuit board returns to the copper chain's scrap pool. The wafer leg (stage 3) is drawn in dashes and dots: of doped wafers imported from Japan in 2024, four fifths by weight came by air (Table 3), which is a record of the trade and not of this consignment. The record says nothing of how the other Pacific legs travel, and they are drawn as sea legs. Packaging, test and assembly are not located and are not pinned. Marks as in Figure 1.

The product. Inside a phone charger is a power-management chip that turns wall current into what a phone can take. It is cheap, it is made on factory lines that are decades old, and in 2021 the world ran out of it ⁸¹. Its chain runs back to a furnace, a crucible, and a valley in North Carolina that a hurricane closed in the same week it closed the valley next door, about nine to twelve miles (fifteen to twenty kilometers) away ².

The valley. The crucible that holds molten silicon while a crystal is pulled from it is lined with quartz of a purity almost nothing else reaches, and most of it comes from one valley: Spruce Pine, North Carolina, with two operators, Sibelco and The Quartz Corp ⁸³. No agency publishes the valley's share of world supply. The United States Geological Survey carries no line for it at all; the 70-to-90 percent figures in circulation trace to industry-analyst estimates and to analyses repeating them ⁸⁴. What can be said, attributed, is that it is "the only plant in the world right now that serves the semiconductor industry in its entirety," in the words of a market-research chief who has tracked the chain for two decades. The careful counter-reading is that synthetic quartz exists at five to ten times the cost, China is scaling, and the operators hold stocks: a dominant source, not an irreplaceable linchpin. We report both. The quartz does not become the chip; it is made into the crucible that holds the melt ⁸⁵.



Plate 3. Spruce Pine, North Carolina: the town in the valley. The quartz operations sit in the district around it; the record of the halt and the restart is theirs.

Kubigula · public domain

Site record 7. The Spruce Pine quartz district, North Carolina.

Who	The district has two operators: Sibelco, which is Belgian, and The Quartz Corp, a 50/50 Imerys and Norsk Mineral joint venture whose purification plant is in Drag, Norway, so the buffer is an ownership artifact. A union representation petition was filed at Sibelco’s Crystal Plant in 2025: the union was not certified; the tally is unverified ⁸³ . There is no federal designation: silicon was added to the critical-minerals list in November 2025 and quartz was not.
Energy	This study holds no energy figure for this district.
Ecosystem	This study holds no ecosystem record for this district.
Comparison	How much of the world’s supply? No agency publishes the share : the USGS carries no line for it, and the 70 to 90% figures trace to analyst estimates and to analyses repeating them. Attributed: “the only plant in the world right now that serves the semiconductor industry in its entirety” (TechCet’s chief executive, via NBC). The counter-reading, in its own voice: synthetic quartz at 5 to 10 times the cost, China scaling, and the operators’ own held stocks, which make it <i>a dominant source, not an irreplaceable linchpin</i> ⁸⁶ .

The storm at the mine. On September 26, 2024 both operators halted, with power lines the critical path. About two weeks later a restart was announced: production and shipments resuming, "progressively ramping up to full capacity." How long the ramp took is not stated anywhere. The dated restart is Sibelco's: two operators halted, and one restart is dated ³. Six days after the storm one operator said: "strong levels of feed stock in Norway... coupled with safety stocks of finished products and those that exist at different levels throughout the supply chain, we do not anticipate any critical situation." The other said that its final product stock had not been impacted ⁸⁷.

This node took a hundred-percent outage at its source, and both operators pointed to stocks held in the chain, one saying they would carry its customers through, because someone had kept slack. The Quartz Corp says the decision was deliberate, taken after COVID; the feedstock sat in Norway because the firm is binational. How many weeks of cover that was, nobody has published. Buffers bought the weeks; whether

substitution could have bought the months was never tested ⁸⁸. Concentration survived here because at least one operator chose inventory over efficiency, and says so.

The crucible and the melt. The quartz goes to a crucible maker, drawn as a class because no plant is named ⁸⁹, where it becomes a short-lived vessel: the crucible degrades in use and cracks after it cools ⁸⁵. The silicon itself arrives by another path (quartzite, an arc furnace, the Siemens process) at about forty-nine kilowatt-hours per kilogram on the solar-grade basis the International Energy Agency's photovoltaic program publishes; the electronic-grade figure this chain runs on is higher, and nobody publishes it. Either way the silicon in one chip cost about a watt-hour, a few seconds of a home's electricity. The fab's electricity is expected to be the larger term, and this study did not obtain it ⁷⁹.

Who: polysilicon producers are chemical manufacturers and heavy users of electricity, as this study's own figure for them shows. This study names none of them and has not sourced where their plants are. The work is process operation, maintenance and electrical trades. No headcount is given, and this study holds nothing on the communities beside these plants. A class, standing in until a plant is named. ⁹⁰

Who: the makers of fused-quartz crucibles are specialist firms. One is named in this study, at company level only; no plant is identified as making the crucible this chain follows. The work is furnace operation, forming and quality control. No headcount is given, and this study holds nothing on the communities beside these works. A class, standing in until a plant is named. ⁹¹

The wafer and the fab. A crystal is pulled and sliced into wafers by a small number of firms, with Japan's SUMCO, by its own account, among the front-runners; we found no public agency that publishes the shares ⁹². The wafers go not to leading-edge fabs but to older, cheaper lines, the legacy-node factories where power-management chips are made and where through the 2010s almost nobody added capacity.

Who: the firms that pull silicon crystals and slice them into wafers. One, Japan's SUMCO, describes itself as among the front-runners; no public agency publishes the shares, and this study has not sourced where the works are. The work is crystal pulling, slicing and measurement. No headcount is given, and this study holds nothing on the communities beside these works. A class, standing in until a plant is named. ⁹³

The buffer. When the pandemic's demand wave hit those lines, the Commerce Department's supply-chain inquiry drew more than 150 responses and found that the median inventory of the chips buyers highlighted had fallen from forty days in 2019 to less than five days in 2021, naming, in its own words, "analog chips (used in power management...)" ⁸¹.

Reorganization. The legacy-node class then reorganized: Texas Instruments broke ground at Sherman, Texas in 2022 on up to four 300-millimeter fabs, and in December 2024 signed for up to \$1.6 billion of federal CHIPS funding across three ⁹⁴. This study holds nothing on the site's workforce, its local incentives, or who lives beside it. Announcement to first wafer runs about three years, the same shape as the transformer works' expansion delay. That the class stagnated before 2020 is this study's claim and not yet a primary-sourced fact: the trade body's capacity numbers are forecasts, and are labeled as such ⁹⁵.

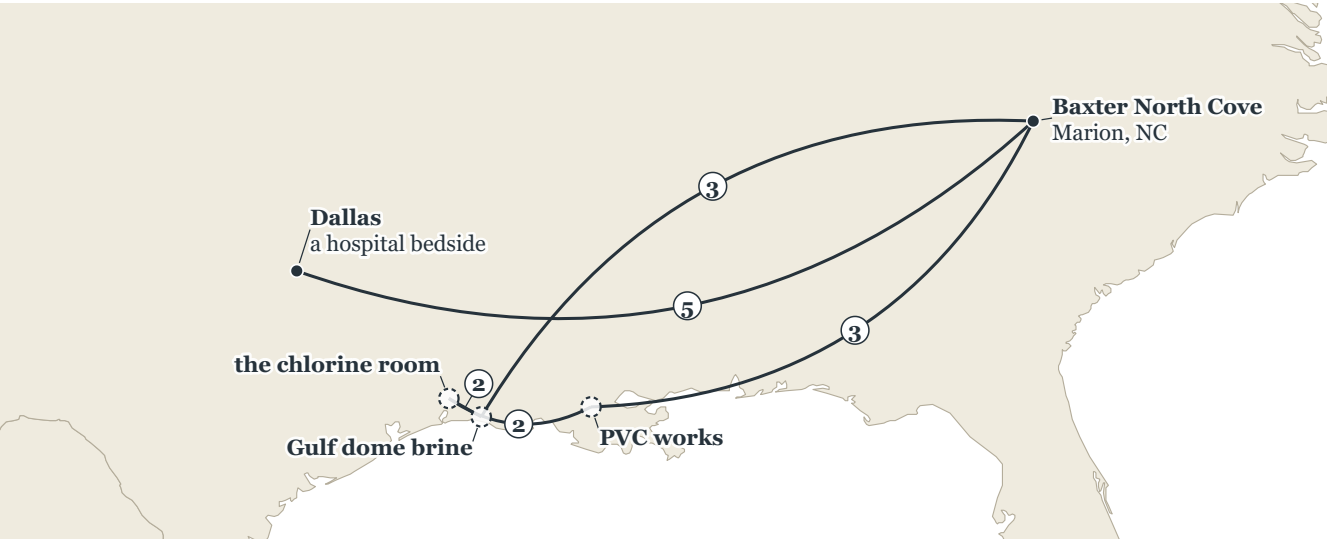
Disposal. When the charger fails, the board goes to an e-waste stream where the copper and gold are recovered and the silicon is lost, work done by a recovery workforce, formal and informal, in the places that do the world's recovering ⁹⁶. The copper on the board re-enters the copper chain's scrap pool, an arc drawn in Figure 6 ⁹⁷, and the chip itself ends.

Not held. The valley's share of world supply; the operators' weeks of cover; the ramp to full capacity; the fab's electricity; the wafer market's shares.

4.4 Salt: the bag of intravenous saline

Table 4. The consignment: one liter bag of intravenous saline.

Consignment	one liter bag of intravenous saline; one plant made about 1.5 million a day, in the hospital association's words and on its figure alone ⁹⁸
Origin	brine from a Gulf Coast salt dome; the bag filled at Marion, North Carolina
Distance traveled	the finished bag's last leg is at least about 850 miles (1,350 km) in a straight line, from the plant at Marion to the Dallas area; the route itself is longer. Before that the consignment is not one object: brine becomes chlorine, chlorine becomes film, film becomes a bag at Marion, and no leg upstream of the plant is named ⁹⁹
Days in transit	No figure: the transit time is not measured for the last leg, and the legs before the plant are not named ¹⁰⁰
Energy to make	No figure: the one energy this study measured is the chlorine room's, per tonne of chlorine and caustic made together; a bag's share would need the chlorine in its film and a split between the two products, and the water, the film and the sterilizing are not measured ¹⁰¹
Salt at the start	of about 40 million tonnes a year, brine is the cheapest form at about ten dollars a tonne against fifty-six for rock salt ¹⁰²
Buffer at destination	not a days-of-inventory figure: what the record holds is the days from the plant's closure to the first documented allocation of supplies ¹⁰³
Final disposition	single use; what becomes of the bag is not measured in this study



Drawn by build_chain_maps.py from the study's own map tables.
Country outlines: Natural Earth, public domain.
Each file's SHA-256 fingerprint is recorded inside this image.

Figure 7. The salt chain: brine from a Gulf Coast dome, the chlorine room, the vinyl works, and the bag plant at Marion, North Carolina. The three rings are kinds of works; the legs that leave them join classes and are not named routes (Table 4). Marks as in Figure 1.

The product. A liter of salt water in a plastic bag. One plant made about 1.5 million of them a day: "approximately 60% — or 1.5 million bags," in the hospital association's words, the "approximately"

governing both ¹⁰⁴. Two kinds of salt meet in it: the sodium chloride dissolved in the water, and the chlorine that became the bag.



Plate 4. A liter of lactated Ringer's, hung and checked. The bag is this section's product; the fill is one of the solutions the shortage listing covers.

Sgt. Timothy Stephens · public domain

Brine and chlorine. The United States produces about forty million tonnes of salt a year, and the cheapest form is brine pumped from domes along the Gulf Coast: about ten dollars a tonne, against fifty-six for rock salt ¹⁰². Roughly fourteen million tonnes a year flow to the chlor-alkali belt. The survey names no operations, so the origin is drawn as a ring, and the statement of who is there stays at that precision too: the dome operators, and the parishes and counties that sit over the domes ¹⁰⁵.

Electrolysis splits brine into chlorine, caustic soda and hydrogen: one tonne of chlorine, 1.1 tonnes of caustic, 30 kilograms of hydrogen, always in that ratio ¹⁰⁶. It costs about 2,600 to 2,900 kilowatt-hours per tonne of chlorine, roughly three months of a household's electricity for every tonne; the European industry alone draws 20 to 26 terawatt-hours a year, the electricity of 1.9 to 2.4 million American homes ¹⁰⁷. Chlorine and caustic are made together and sold apart, and chlorine cannot wait (it liquefies badly and stores worse) while caustic keeps in tanks. So production follows chlorine, and caustic floods or starves ¹⁰⁶.

The vinyl works. Where the chlorine goes next is a stage drawn as a ring and not examined in detail: the plants that turn it into vinyl, and the people beside them. This study names one operating vinyls plant, at Plaquemine, and one proposed for St. James Parish whose permits were vacated in 2022, reinstated in January 2024 and left standing when further review was refused that September; it is not built, and the case is described by where it stands, never as settled. The people along these reaches appear here through the public record only: that litigation, and the property-line monitoring these plants were to begin in July 2026, deferred for two years by a proclamation filed the morning it took effect ¹⁵. One community in this belt has largely ceased to exist as a place: Mossville, in Calcasieu Parish, founded in the 1790s by African Americans, by its residents' own account. From 2013 most of it was bought out under a voluntary purchase program for the areas beside Sasol's Westlake expansion; Sasol says those who declined still live there. That

buyout was Sasol's, whose plants there are not this chain's; it is reported as the limit case of the question of what passes and what is permanent, and not as this chain's doing ¹⁰⁸.

The bag works, and the storm. Marion, North Carolina: ten manufacturing lines, more than 2,500 employees, and the 1.5 million bags a day.

Site record 8. The North Cove plant, Marion, North Carolina.

Who	Baxter International runs the plant, with more than 2,500 employees on two separate accounts, in McDowell County. The FDA is the shortage authority, and in its own words “FDA does not have authority over allocation or distribution of these products”. HHS names group-purchasing contracts among the causes of thin hospital buffers, and the statutory safe harbor (42 U.S.C. 1320a-7b(b)(3)(C)) is what makes that contracting model lawful ¹⁰⁹ .
Energy	This study holds no energy figure for this plant.
Ecosystem	This study holds no ecosystem record for this plant.
Comparison	About 60%, or about 1.5 million bags, of the IV solutions used in the US every day, in the hospital association’s words, with the “approximately” governing both ¹⁰⁴ . Three dated milestones followed Helene: the first line at day 32, all ten lines at day 121, pre-hurricane output at day 141, and the shortage itself was not over for about eleven months ⁴ .

Hurricane Helene's rain came down the valley on the morning of September 27 ¹¹⁰; by September 29 the plant was "affected by flooding and currently closed for production." The recovery has three dated marks: the first line, the highest-throughput one and a quarter of the plant's output, back on October 31, thirty-two days in; all ten lines on January 28, one hundred and twenty-one days in; and pre-hurricane output on February 17, day one hundred and forty-one ⁴. The system was already short before the storm. Of the nine intravenous products on the FDA's shortage list after it, six had been listed before, the common dextrose solutions since February 2022 and normal saline since June 2018; three were new ¹¹¹.

Two valleys. The quartz district and the saline plant were closed by the same storm, about nine to twelve miles (fifteen to twenty kilometers) apart and three days apart, and came back on clocks that do not line up (Figure 8). Matched first production to first production, the plant took a bit over twice as long as the mine. Set the restart against all lines running, which is not a matched pair, and the gap is nearer nine times. The comparison most often reached for, two weeks against "fully recovered," cannot be made at all, because the mine never published a full-capacity date ¹¹². What separates a two-week node from a four-month node is not the storm. It is the depth of the damage and the regime that governs restart: a mine restarts when the power line is repaired; a sterile-pharmaceutical plant restarts when every line is requalified. It is also where the buffers sat. The sand chain had stock in Norway. The salt chain began rationing within days of the closure.

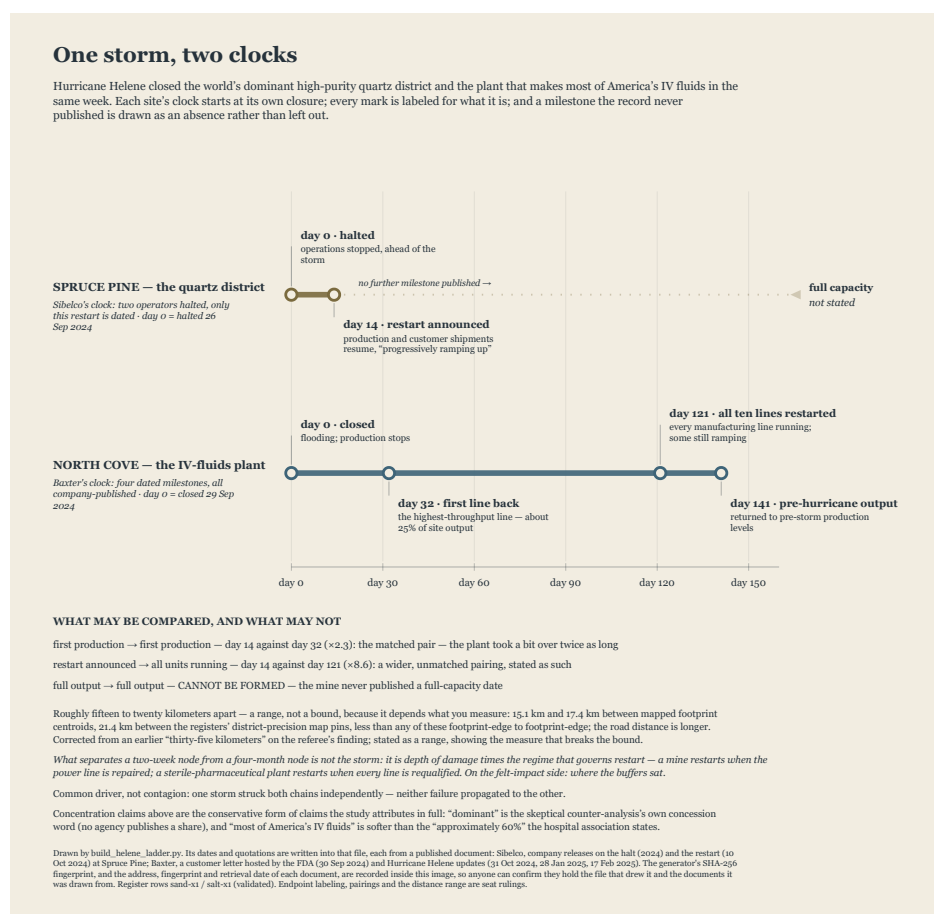


Figure 8. Restoration milestones for the Spruce Pine quartz district and the North Cove saline plant, each measured from its own closure. The mine's unpublished full-capacity date is drawn as an open bar.

The shortage. Nearly eighty-five percent of four hundred hospital pharmacists surveyed by their professional society in mid-October reported a moderate or critical impact, critical meaning treatments delayed or canceled; they converted infusions to injections, bought premixed drugs, used smaller bags, and reserved supplies for certain patients. In the same survey, seventeen percent of all participants reported canceling or delaying voluntary surgeries ¹¹³. The FDA authorized imports from nine of Baxter's own plants abroad, three weeks in, and by the end of the year, by the company's account, nearly two hundred airplanes had delivered nearly eighteen thousand tons of product, the ton being the company's word ¹¹⁴. The agency also said what it could not do: "FDA does not have authority over allocation or distribution of these products" ¹⁰⁹.

The buffer. The plant was back in about four months. The shortage was not over until August 2025: "approximately eleven months," in the hospital association's statement of record to the Senate. The seven months in between are the pipeline refilling and the allocations unwinding. The system recovers when the buffers refill, not when the plant restarts ⁴.

No reorganization. In 2017 Hurricane Maria hit the same company's plants in Puerto Rico and the country ran short of saline; the FDA Commissioner said the shortages "resulted from, or were worsened by" the storm. The shortage listing for normal saline that the FDA posted on June 21, 2018 was still open when Helene hit: one unbroken listing, deepened by two hurricanes seven years apart ¹¹⁵. The response was the

same both times (emergency imports, new entrants, extended expiry) and faster in 2024. The response layer reorganized and the capacity layer did not. Set beside Section 4.3: chips got plants, and bags got airlifts. Whether a shock de-concentrates a chain depends on the market structure around the node, not on the shock.

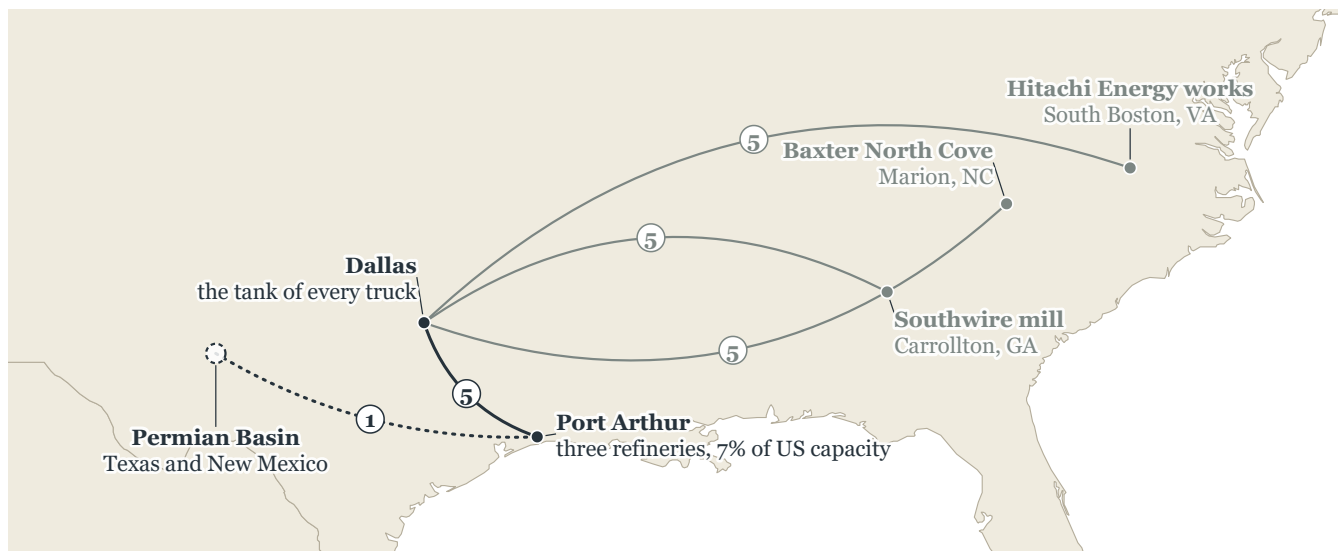
Between Maria and Helene, American institutions learned to respond: the same FDA toolkit, used faster, with an airlift and a Defense Production Act order for one remediation material. They did not learn to be less fragile. On the primary record: the law requires firms to keep a "redundancy risk management plan," kept by the firm and inspected only "as appropriate"; a proposal to pay hospitals for buffer stock was solicited and "not adopted"; the strategic reserve "focuses on... countermeasures that do not have a commercial market"; a six-month reserve bill sat unenacted in October 2025; and the 2024 federal white paper on preventing shortages never mentions Maria, Baxter, saline, or intravenous fluids ¹¹⁴. Section 4.2 found a reserve question with an owners' club to answer it. This is its medical counterpart, with the club missing.

Not held. Any leg upstream of the plant; the energy in one bag; a days-of-inventory figure at hospitals; what becomes of the bag after use.

4.5 Oil: the tank of diesel

Table 5. The consignment: one tank of diesel.

Consignment	one tank of diesel, the fuel in the truck that delivered the other four
Origin	the Permian Basin, west Texas and New Mexico: two states that were 59 percent of US crude production in 2025, against 23 in 2002 ¹¹⁶
Distance traveled	the diesel's last leg is at least about 250 miles (400 km) in a straight line, from Port Arthur, where Figure 9 places the refining, to the Dallas area; the route itself is longer. The crude's leg before it cannot be bounded: this study places its origin as a region, the Permian Basin, and a straight line from the region's center is not a floor ¹¹⁷
Days in transit	No figure: the transit time is not measured for either leg. The fuel in the pipes is counted inside the buffer below, as the part of the stock nobody can draw ¹¹⁸
Energy to make	No figure: a refinery makes diesel, gasoline and jet fuel from the same barrel, and one tank's share of its energy depends on how that energy is divided among them, a choice this study has not made ¹¹⁹
Refined at	the Gulf Coast holds 54.4 percent of operable US capacity, and Port Arthur's three refineries 7.0; why refining sits there, whether geology, ports or permits, a capacity table cannot say ¹²⁰
Buffer at destination	about 27 to 29 days of distillate cover in mid-August 2026, against about 32 to 34 in the same week of 1996; a ceiling on comfort, not a countdown ⁵
Final disposition	burned. What a severed link costs is this Roundtable's first study's finding: a median of about \$23 billion a month in American household costs, falling at 8.9 percent of after-tax income at the bottom against 1.6 at the top ¹²¹



Drawn by `build_chain_maps.py` from the study's own map tables.

Country outlines: Natural Earth, public domain.

Each file's SHA-256 fingerprint is recorded inside this image.

Figure 9. The oil chain: crude by pipe from the Permian Basin to the refineries at Port Arthur, and diesel to the tank of a truck. In gray, the final legs of three other chains, which run on this chain's product. Marks as in Figure 1.

Diesel is unlike the other five products. It is not a fifth chain beside the others; it is the last leg of each of them. Figure 9 draws the final leg of three of the other chains (the wire to the house, the transformer to the substation, the saline to the ward), and each ends on a truck running on this section's product. The chip's last leg is not drawn: the study does not locate where it starts. The other chains have chokepoints of their own. This one does not have to break to be felt everywhere, because it is already inside every other chain's last mile.

4.5.1 The chain

Extraction. The crude starts in west Texas and southeastern New Mexico. Two states produced 58.9% of American crude oil in 2025¹¹⁶. That share had fallen for two decades, to 22.6% in 2002, and then rose 2.61×. Concentration here is not the resting state of the system: it was reduced, and then it was rebuilt, inside one working lifetime¹¹⁶. The origin is drawn as a ring because the evidence is at state level: the record names two states and not a wellhead. That is a limit of the record and not a description of the place; the people who live above this rock are absent from a figure drawn at the wrong scale to hold them¹².

Ownership and the regulator. In New Mexico, some of this rock is state trust land, and the royalties have a named destination: the Land Grant Permanent Fund, set up when the state entered the union in 1912. It pays out more than a billion dollars a year (\$1.34 billion in 2024) to public schools, universities and other beneficiaries, and the money reaches it monthly¹²². Texas oil and gas is overseen by the Railroad Commission of Texas, created in 1891 to stop railroads overcharging farmers; it has not regulated a railroad since 2005, when the last rail duties went to the state highway department¹²². Two things are not claimed. The split between state trust, federal and private land is not sourced, so the fund is context and not attribution. And this study holds no count of the people who work here, so no headcount is given.

Refining. Crude becomes diesel on the Gulf Coast: the eight largest American refineries include seven in one region, and the first outside it is eighth, in Indiana ¹²⁰. Port Arthur, Texas alone holds three refineries, together 7% of national capacity, and the largest single refinery in the country is one of them ¹²⁰. Why refining sits there is a question a capacity table cannot answer. Three explanations are available and cannot be separated: the geology put the oil nearby, the ports made it cheap to move, and the permitting made it hard to build anywhere else. One reading, untested here, is that they ran in sequence, and that what looks like a choice is a position inherited and then frozen.



Plate 5. Valero's refinery at Port Arthur, Texas, at dusk: one of the three refineries in one town that this section's diesel comes through.

Carol M. Highsmith, Library of Congress · public domain

Who lives with it. The Energy Information Administration's capacity report names each refinery's parent: Motiva is wholly owned by Saudi Aramco and is the largest of the three; the second files under Premcor and belongs to Valero; the third is TotalEnergies ¹²³. Seven percent of American refining capacity, in one town, is held by a foreign state oil company, an American independent and a French major.

Port Arthur had 56,039 people at the 2020 census, in a county of 256,526. Median household income is \$45,752 against \$76,292 for Texas (three-fifths of the state figure), and the poverty rate is 29.1% against the state's 13.8% ¹²⁴. The 56,039 describes the town that hosts the complex. It is not a count of anyone's exposure to anything ¹²⁴.

The most documented environmental record here is not about the refineries. In 2021 the Port Arthur Community Action Network, with Lone Star Legal Aid and the Environmental Integrity Project, filed a civil rights complaint, not against a plant but against the state regulator, over its oversight of a petroleum-coke facility that calcines the refineries' own byproduct. The complaint is that the emissions were inadequately monitored or quantified, and so not properly weighed when permits were decided ¹²⁵. EPA investigated for two years, the state withdrew from negotiations, and the file was closed without a finding either way: re-

filable, with the agency saying it would keep gathering data. The neighborhood the complaint names is West Port Arthur, and this study puts no number beside that name. West Port Arthur has no boundary anyone tabulates: the census counts cities and tracts, and the harm has a neighborhood. Any figure printed here would be a line we had drawn ourselves, carrying the Census's authority ¹².

Slack in refining. In refining, the buffer is idle capacity. The expected account is that slack has been squeezed out steadily, and it is wrong ¹²⁶. Utilization climbed from 77.6% in 1985 to a peak of 95.6% in 1998, and in 2025 was 91.9%, below that peak. The tightening is an era, not a trend.

Utilization is throughput divided by capacity, so a change splits exactly into the two ¹²⁷. Which window is chosen decides the answer, so both are given. After 2020, demand brought utilization back: throughput rose 13.4% while capacity fell 2.6%, so 83% of the movement is the numerator ¹²⁷. Against 2019 the system is tighter, and not because it is busier: throughput has fallen 1.7% and capacity 3.4%. Capacity peaked at 18,808 thousand barrels a day in 2019, bottomed at 17,963 in 2022, and has recovered about a quarter of that, not steadily, and stands 635 below its 2019 peak ¹²⁷. Slack in this system is a residual, never a reserve, and when it persists it is removed.

The buffer. Downstream of the refinery the buffer is tanks. In mid-August 2026 the United States held about 27 to 29 days of distillate cover (stocks of distillate against distillate product supplied, averaged over four, thirteen and fifty-two weeks) against about 32 to 34 in the same week of 1996, measured the same way ⁵: four to five days less, and the lowest of any comparable August in a record that begins in 1991. The difference is taken basis by basis, not by subtracting the two ranges. The figure is a ceiling and not a countdown. Tank heels and pipeline linefill sit inside the number and are available to nobody; exports are netted out; and it is an August rate standing against a winter risk. All three point the same way: the real buffer is shorter than the figure.

4.5.2 The induced-seismicity record

Oil comes up with water, the water is injected back underground, and the injection has been causing earthquakes. The Railroad Commission's answer was to draw Seismic Response Areas on the basin and act inside them. In the Northern Culberson-Reeves area it suspended every deep disposal permit, 23 wells, effective January 12, 2024. Around Stanton it cut daily injection volumes and told individual wells to stop ¹²⁸.

Two things in that record matter more than the enforcement. The first is a delay. In the Culberson-Reeves area the Commission estimates twelve to eighteen months between injection rising and seismicity rising: its own estimate, not a measured interval. That figure is for injection going up. Asked how long a decrease takes to show, the Commission calls it uncertain and publishes no number. The physics says which way the uncertainty runs: shutting a well stops adding pressure but does not take back the pressure already in the rock, which keeps spreading for months before it decays. The wait for a fall is longer than the wait for a rise, so every figure below is measured early, with some of the decline still arriving ¹²⁹.

The second is that the Commission states where it sets its own buffer. To hold off a magnitude-5 event it acts when earthquakes are running between magnitude 3 and 4. And the instrument changed as it went. In January 2022 the Commission asked existing wells for six months of daily volume and surface pressure.

Thirteen months later it began requiring new wells to carry a bottomhole pressure gauge and report both pressures monthly ¹²⁸.

The response plan set a goal: that magnitude-3.5 earthquakes be decreasing by the end of 2023. Counts did fall inside the response areas, by about half around Culberson-Reeves and about three-fifths around Midland. But in a nearby part of the same basin with no response area at all, they fell by about seventy percent. The treated ground improved; the untreated ground improved more. And Culberson-Reeves had already been falling for a year before the suspension was signed ¹³⁰.

Whether the largest earthquakes responded is a question the record cannot answer. Magnitude-5 events run about three-quarters of one a year here. To detect a halving at that rate would need roughly sixty years before and sixty years after. There are four, and under three. The largest earthquake before the suspension was magnitude 5.4; the largest after it was magnitude 5.4, on May 4, 2025. Both are public record, and neither is evidence of anything ¹³⁰. A goal written on the number of magnitude-3.5 earthquakes can be checked in a few years; a goal written on the largest event cannot be checked for about a century. The Commission set its goal on the indicator it could verify and acted on the one it could not. When a hazard is heavy-tailed and the policy runs on a policy timescale, the thing that can be checked and the thing that needs to be known are different objects.

Which comparison tests the order. Earthquakes in a ring outside the response area rose while those inside it fell, which suggested the suspended water had moved. That was the wrong comparison. The Commission did not close disposal in an area; it closed deep disposal. The order is named for a place, but it selected on depth, and a comparison by location tests the label and not the mechanism. The right comparison is inside the treated area: the shallower rock under the same ground, with the same operators, the same instruments and the same regional activity. Split that way, the deep earthquakes nearly stop, down about 92%, while the shallow ones fall by about 48%, roughly what the whole basin was doing ¹³⁰. So the record can partly separate the order from the trend, by depth and not by location.

This is held loosely, for stated reasons: the depth of an earthquake is not the depth of a disposal well, the catalog resolves depth to about a kilometer, the deep counts are small, and the whole is measured early. The ring outside the area did rise, mostly in New Mexico, but had the deep water crossed the state line the new earthquakes there should be deep, and about one in eight is.

The displacement account was then tested directly. Had the suspended water crossed the line, somewhere had to open to take it. New deep disposal wells on the New Mexico side peaked at thirty-one a year in 2018 and 2019 and were down to five a year by 2021, a year before Texas wrote its first response plan. Across the four years before the interventions there were eighty new deep wells; across the four years after, twenty-one. Not one intervention is followed by an increase ¹³⁰. New Mexico's new earthquakes are shallow because New Mexico's new injection is shallow, and it is New Mexico's own.

One channel remains untested. The test covered new capacity: whether anywhere was built to take the water. It did not cover the volumes going into wells that already existed, and New Mexico's published disposal-well records carry no volume figures at all, for any of the 1,066 wells in them. That channel is untested, which is not the same as tested and clear ¹³⁰.

The other regulator. New Mexico was already regulating induced earthquakes of its own, on a different theory of cause ¹²⁸. Texas asks which rock is injected into: the suspension closed deep disposal, by formation. New Mexico asks how close a well is to what has already happened: two magnitude-2.5 earthquakes inside thirty days and ten miles, and monitoring starts; one magnitude-3.0, and injection rates come down by distance from it; a magnitude-3.5 within ten miles, and wells within three miles shut in. Texas published its first response-area map on October 21, 2021. New Mexico's protocol is dated November 23, its public map layers December 20, and Texas's statement of guidance January 31, 2022: four regulatory acts, two agencies, two states, inside fourteen weeks. Neither state was following the other. This is one professional response seen from two sides of a line, which is a different thing from a hazard migrating.

4.5.3 Two channels

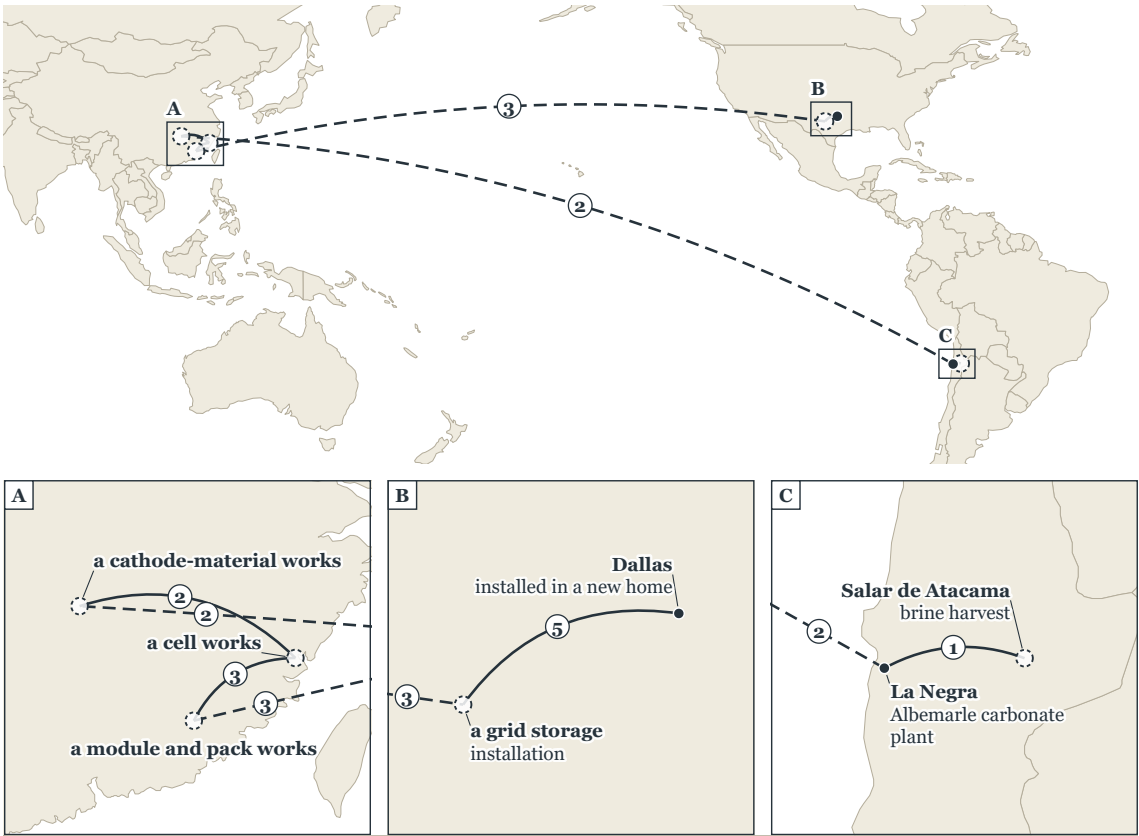
A severed diesel link is felt two ways, and the Roundtable has now measured one each. The price channel was *The Uneven Month*: a median \$23.3 billion a month in added household cost, falling hardest where there is least, 8.9% of after-tax income at the bottom against 1.6% at the top, regressive in every one of a hundred thousand bootstrap replicates ¹²¹. The physical channel is this section: not what it costs, but how many days of it there are. Neither stands in for the other. The first study holds no freight layer: it priced a shock and did not delay a delivery. This one holds no price model.

Not held. A floor on the crude's own leg; a transit time for either leg; one tank's share of a refinery's energy; the split of land ownership over the basin; a count of who works there.

4.6 Lithium: the grid battery

Table 6. The consignment: one grid battery.

Consignment	one grid battery: cells stacked into modules, modules into packs, packs into containers wired into a substation
Origin	brine under the Salar de Atacama, Chile
Distance traveled	outbound about 10,500 to 11,200 miles (16,946 to 17,964 km), northern Chile to China; inbound about 6,700 miles (10,760 km), the same crossing the copper section measures ¹³¹
Days in transit	outbound, about three to three and a half weeks at sea: the carbonate travels bagged and containerized, on the salt flat's other operator's own filing, which also names break bulk as an alternative. Inbound, 14 to 15 days: battery packs cross by sea in containers; on the federal trade record for 2024, 99 percent of this cargo by weight went by vessel, and 99 percent of that in a box ¹³²
Energy to make	No figure: a battery's making energy is counted per kilowatt-hour of capacity, and this one's size is not stated. The one step the study did measure, evaporation, the operator says the desert sun does, and it is limited by time, not fuel ¹³³
Assembled at	for the one product this study sourced, the maker's own annual filing names Lathrop, California and Shanghai; one product, not a statement about the industry ¹³⁴
Buffer at destination	at the pond, about as many months of its own output as the brine spends in it; at the destination, none published, in tonnes or in days ¹³⁵
Final disposition	Beyond this study: the chain is followed to the battery yard, where the battery stands in service, and no further. Whether any of its lithium comes back into the chain, and by what route, is not in this study's sources: no return leg is drawn in Figure 10, and a missing leg is not a finding that none exists



Drawn by `build_chain_maps.py` from the study's own map tables.
Country outlines: Natural Earth, public domain.
Each file's SHA-256 fingerprint is recorded inside this image.

Figure 10. The lithium chain. Rings mark stages the record does not locate as plants: of the six stages before the house, one is a named site. Marks as in Figure 1.

The brine. Under the crust of the Salar de Atacama sits water with so much salt dissolved in it that it is nearer syrup than water. The other five chains begin with a rock that is dug. This one begins with a liquid that is pumped, and a liquid moves, refills, and is shared with whatever else draws on the same ground. Almost every stage of this chain is drawn in Figure 10 as a ring: the record names a stage and not a plant. One of the six is a named site.



Plate 6. Lithium evaporation ponds on the Salar de Atacama, Chile, seen by Landsat 8 on November 4, 2018. The image does not distinguish one operator's ponds from another's.

NASA Earth Observatory · public domain

One aquifer, three companies. Copper is mined a short distance from the salt flat, and it too needs water. In December 2024 a Chilean environmental tribunal approved a settlement that named them together. A copper mine, a second copper operation and a lithium producer were sued jointly by the Comunidad Indígena Atacameña de Peine over damage to one named body of groundwater. The agreement binds all three to fourteen measures worth about US\$47 million, the first of which is that aquifer's recovery, and the sixth of which is a standing table where the community and the companies sit together ¹³⁶. The tribunal approved it on a divided bench: the dissenting minister's ground was that no measure in the agreement repairs the aquifer drawdown itself, with recovery estimated at around the year 2200 and only under an immediate cessation nobody signed. Implementation runs under the tribunal's continued supervision, and unevenly: the standing table functions, while the baseline study that most of the measures wait on is still at terms of reference ²⁸.

Two chains in this study are therefore linked, and the kind of link matters. They are linked by an instrument: one community, one proceeding, one shared bill. They are not shown here to be linked physically. Whether pumping brine and pumping fresh water act on the same water is disputed among the people who study it, and this study does not settle it ¹³⁶. The test was written down before the record was read: if the lithium producer had turned out not to be a party, there would be no link ¹³⁷.

The ponds. The standard method of getting lithium out of brine is to let the sun do it. The solution is moved through a series of shallow ponds and left while the desert takes the water off, until what remains is concentrated enough to move to a plant that turns it into carbonate. This is the first delay in the study that money cannot shorten. The transformer queue is long because there is not enough plant and not enough skill, and more of both would bring the wait down. Evaporation is not like that: more ponds yield more tonnes, and do not yield them sooner ¹².

Albemarle, describing its own ponds on this salt flat in its own annual report, gives about eighteen to twenty-four months, and adds that "if weather conditions are not favorable, the evaporation process may be prolonged" ⁶. Twenty-four months is therefore not a ceiling: the company declines to put one there, and so

do we. What it names as the thing that stretches the time is the weather, not plant, people or money. The figure usually given is a year to eighteen months; this study declined to print it while it was unsourced, and the operator's filed figure begins where that one ends. It is one company's ponds and not the salt flat's, which is a fact about the record and not a doubt about the number. The other operator on the salt flat describes the same pond sequence in more detail and never says how long it takes.

Who: Albemarle runs the carbonate plant, and it is also one of the companies in the Peine settlement at the salt flat. The company links the two places; the community does not. Peine's case is about groundwater at the salt flat, not about this plant. Around the plant is an industrial district of the Antofagasta region with people living in it, and this study can say no more about them than that: it holds no Chilean regulatory record for the plant itself. ¹³⁸

Mined is not refined. Which country is said to control lithium depends on a distinction that is rarely drawn. At the mine stage no one dominates: Australia leads with rather more than a third, Chile about a fifth, China about a sixth, and the top three together come to roughly three-quarters, shares of a world total in which the United States figure is withheld. The United States Geological Survey reports American mine production only as "W", so every share here divides by a total containing a withheld, non-zero producer ¹³⁹. At the refining stage the picture changes completely. On the same year's reading of the International Energy Agency's published chart, China holds around seventy percent of the world's refined lithium, and the top three between them hold nearly all of it ¹⁴⁰. The most-cited source in the field publishes mine production and reserves and does not report refining at all ¹³⁹.

Who: nobody, and the reason is worth a sentence. This stage is a world total, not a place, and adding up is the operation that removes people. Five producing countries stand behind the figure, with five workforces, five sets of neighbors and five regulators; the number sums their tonnes and nothing else. Where the chain reaches a place again, the people return. ¹⁴¹

Against copper. Concentration means something only beside a comparison, and one honest comparison is available: the same chart (the International Energy Agency's, of who refined what in 2024, on page 93 of its 2025 critical-minerals outlook), the same year, the same stage, the same way of counting. Refined lithium is about seventy percent in one country, with the top three at nearly all of it. Refined copper is about forty-four percent in that same country, with the top three about sixty ¹⁴⁰. The same chart is why this paper does not call lithium the most concentrated of anything. Battery-grade graphite sits near ninety-five percent in one country and magnet rare earths near ninety-one, both above lithium. Against copper, the one other of this study's six that the chart covers, lithium's refining is markedly the more concentrated; among critical minerals generally it is not close to the top ¹⁴⁰. These shares were read off a printed bar chart at high magnification and are good to a point or two. They are a reading, not a measurement.

The battery. Cells are stacked into modules, modules into packs, packs into containers about the size of a shipping crate, and a row of those containers is wired into a substation. The chain ends there, in a fenced yard of batteries wired into a substation. That places it beside the transformer of Section 4.2: the two are answers to the same question, what stands between the generator and the house. These are two findings, each established on its own evidence, and the likeness between them is offered as support for neither.

Who: the makers of cathode powder are specialty chemical producers. This study names none of them and has not sourced where their plants are. The work is preparing the precursor, firing it in kilns, and quality control. No headcount is given, and this study holds nothing on the communities beside these plants. A class, standing in until a plant is named. ¹⁴²

Who: the makers of lithium-ion cells. This study names none of them and has not sourced where their plants are. The work is coating electrodes, assembling cells, and the slow first charging and testing that follows. No headcount is given, and this study holds nothing on the communities beside these plants. A class, standing in until a plant is named. ¹⁴³

Who: the firms that build cells into modules, packs and containers. Where that is done is not established in general. In the one case this study has sourced, the maker's own annual filing names dedicated factories for the product in Lathrop, California and in Shanghai: neither where the cells are made nor where the battery is installed. Other makers may do it differently. The work is module assembly, enclosures, cooling and commissioning. No headcount is given. This study has not looked for a community record at either factory, so it states none. ¹⁴⁴

The buffer. Every tonne spends eighteen to twenty-four months in Albemarle's ponds, so at any moment those ponds hold eighteen to twenty-four months of their own output. The residence time is the inventory ¹⁴⁵. Because the company puts no ceiling on the one, there is none on the other.

Whether that is a buffer depends on what it is set against. If the brine stops arriving, the ponds keep delivering at full rate for the full residence time and then stop. Against an interruption behind them that is eighteen to twenty-four months in hand; against anything in front of them it is nothing at all, because a tonne cannot be pulled forward without sunlight it has not had ¹⁴⁵. And it buys time, not supply. Run a warehouse and a pond system against the same six-month interruption: the warehouse loses nothing, and the ponds lose all six of those months, but lose them eighteen to twenty-four months after the brine stopped. The outage still arrives, undiminished, and late ⁷. Postponement is worth a great deal. It is not absorption.

A warehouse differs from the ponds, but not by running down gently: anything drawn at full rate ends in a cliff, and rationing is a decision somebody makes and not a property of the stock. The difference is that in a warehouse the level is a choice, and in the ponds it is an identity, residence time multiplied by throughput. This is a derivation and not a measurement: a sourced residence time run through a standard identity. It stays a ratio permanently, because the ponds destroy most of their own mass on purpose, so a tonnage would first have to choose between raw brine, contained lithium and finished product, three answers orders of magnitude apart. It also assumes the ponds are running level; a growing operation holds less, a shrinking one more.

The question of how many days lie between a break and a change at the end of the chain has no single answer here, because there is no single break. Cover is not a property of a chain. It is a property of a cut ⁷. Asked cut by cut, the chain divides in two. How long until a break is felt can be computed from physics wherever somebody publishes a transit time. How much of the break is absorbed is unpublished at every link. This chain's delays are public, because they are physics and operators describe their own processes; its inventories are private, because they are commercial ⁷. The one cut the study can answer in full, it answers from the residence time and not from anyone's disclosure ¹⁴⁶.

Who: four kinds of party, because at this stage the unit changes. The owner, a utility or an independent operator. The public utility commission that approved the installation and decided how its cost is recovered, which is where the cost decision is actually made, and a body whose hearings the public can attend. The ratepayers of the territory it serves, who pay for it. And whoever lives beside the site, named separately, because living next to something is not the same as paying for it. No figure is put on anyone's bill: nothing in this study supports one. Every other 'who' in this study is somebody else: a mine's workforce, a smelter's neighbors, a town along a fence line. At the end of this chain the reader stops being a spectator and becomes a party, because the installation is paid for through the rate base they are in. ¹⁴⁷

Not held. A physical link between brine and fresh-water pumping; a ceiling on the residence time; a second operator's figure; any inventory in tonnes or days downstream of the ponds; whether the cathode's chemistry links this chain to the iron chain. That last test was never run: it rested on a premise the study could not source, and a link that cannot be sourced is not built ¹⁴⁸. That is not a finding that the two chains are separate.

5 · What a break feels like

5.1 Six chains, six kinds of answer

Asked how long a break takes to reach the end of the chain, the six do not answer the same way, and they do not answer the same kind of question. Physics decides what kind of answer a chain can give about its own disruption; the record decides whether we have it. More study fills the second and never converts one kind into another ¹.

Copper is a reservoir feeding a process that can be slowed, so its answer is a curve and not a number; we hold the shape and not the volume. Lithium is a pipeline transit nobody can hurry, computed from physics because no interruption of it is on record. Oil is days of cover in a drawable reservoir, and the only chain where an agency publishes both halves of that fraction. Iron is a queue, because the thing is built to order and nothing substitutes: a lead time, and no failure duration at all. Salt is days of cover at the point of use, observed, and the only chain with three endpoints dated. Sand is days of cover spread along the chain, observed, with a halt somebody chose.

These are not six sizes of the same thing, and they cannot be put in order. An ordering, like a ratio, presumes a common origin, and these six do not share one. A lead time runs from an order placed; a restart from a closure; a residence time from material entering a pond; days of cover from the moment resupply stops. Ordering them would order the definitions and not the chains. A common unit is the trap here and not the help: putting all six in days is exactly what makes a sort look available ¹.

Nor can the chains be ranked by likelihood of failure. The hazard data that exists covers hurricanes and earthquakes far better than it covers a fire at the only crucible works, so an ordering built on it would rank these chains by how well studied their hazards are ¹¹.

5.2 One storm, two valleys

In late September 2024 Hurricane Helene reached the southern Appalachians, and two of this study's chains lost a node in the same week: a quartz district in one valley and a saline plant in another, about nine to twelve miles (fifteen to twenty kilometers) apart ¹¹². Spruce Pine halted on September 26 and announced a restart about fourteen days later ³. The flooding arrived on the morning of the 27th ¹¹⁰, and North Cove closed on the 29th, back to its pre-storm output a hundred and forty-one days after that ⁴.

The plant coming back is not the shortage ending. The first line restarted at thirty-two days; all ten at a hundred and twenty-one; output was back where it began at a hundred and forty-one; and the shortage itself was still being reported to the Senate about eleven months after the flood ⁴.

This is one driver and two chains, not two storms' worth of evidence. The quartz mine and the saline plant share nothing but weather: different owners, different regulators, different customers, and neither reached the other. Neither Spruce Pine operator published a date for full capacity, so the two full recoveries cannot be compared at all: only the restarts, and only the first lines.

5.3 The cases by kind

A transit nobody can hurry. Lithium has no interruption on record, so its answer is computed and not observed. Brine spends about eighteen to twenty-four months in the evaporation ponds ⁶, so at any moment the ponds hold that much of their own output ¹⁴⁵. Against a six-month interruption a warehouse loses nothing, while the ponds lose all six months of it, not for another eighteen to twenty-four months, and then in full ⁷.

A mine that slows. The Escondida concentrator runs on desalinated water pumped up three thousand meters from the coast ²⁹. If that supply were interrupted the mine would not stop; it would slow. Cover is then a relationship between how much water is left and how hard the plant is being run, and the slowing is what would reach the market ¹. How much water is stored on site is a figure we do not have, so how long the slowing could last, or when it would become a stop, is a question this study cannot answer ³¹.

A stock somebody counts. A federal agency publishes the distillate stock and the rate at which it is consumed, every week, so the buffer can be divided out. Cover in 2026 runs at about twenty-seven to twenty-nine days, measured against distillate supplied and averaged over four, thirteen and fifty-two weeks, against about thirty-two to thirty-four days in 1996, measured the same way ⁵. The same barrels buy about four to five fewer days than they did in 1996, taken basis by basis and not by subtracting the printed ranges ⁵. What makes this chain legible is not that its buffer is better understood but that somebody publishes both halves of the fraction.

A queue, and half a duration. The largest transformers are quoted at around three years, and up to five, a figure counted from the order being placed and not from a failure, and one the Department of Energy carries from its own conversations with domestic producers and not from a measured series ⁴⁹. The other half of the pair is missing: this study found no published duration for a transformer failure, so where every other case has a stop and a return, iron has a return and a blank. In Figure 11 that blank is drawn as an open bar.

In place of a reserve there is a club. Voluntary sharing among the utilities that own the transformers is real, funded and federally cataloged; of its five programs exactly one adds a transformer to the country that would not otherwise exist, and the two most formal had never been used as of the Department's 2024 report ⁷⁶. Whether the club is enough is unproven, and not provable from the public record: the one dedicated inventory is sized, in its operator's own consultant's model, to a 98 percent likelihood of meeting anticipated need, and neither the inventory nor the model's inputs are public. What can be said is whom it reaches: members whose regulators let prevention be paid for. The study takes no position on what should exist instead.

5.4 The case where the clock does not end

Every case above ends: a restart announced, an output restored, a shortage declared over, a transformer delivered. Mossville does not. It is an unincorporated community in Calcasieu Parish, founded in the 1790s by African Americans, by its residents' own account to the Inter-American Commission on Human Rights; by the same account, the state permitted fourteen industrial facilities around it from the 1930s; and from 2013 a voluntary property-purchase program bought most of it out. There is no return to normal for a community that, for most of its people, no longer exists as a place ¹⁰⁸. The buyout is Sasol's, and Sasol runs a cracker and a gas-to-liquids plant, not the vinyls chain followed here. It is named as the limit case of the study's own question, not as this chain's doing.

Section 3 described instruments that arrive late and arrive public: a fence-line monitor, a pressure gauge, a published quarter ¹⁰. Each assumes somebody is still there to read it. The residents who sold cannot consult a monitor, cannot re-file a complaint, and cannot hold a seat at a table. Each also assumes the instrument arrived. In this corridor one has not: the fence-line monitoring these plants were to begin in July 2026 was deferred for two more years, by a proclamation filed on the morning it took effect, whose Annex I lists the plants exempted ¹⁵.

5.5 The asymmetry

Figure 11 sets stop against return for six cases on one scale. Three chains supply the six pairs, and most of them are one week: four of the pairs are the same saline plant, asked four different ways when it was back, and the sixth is not an event but a queue. The shape is the same each time. A thing stops in a day or two. It comes back in a season, or in years.

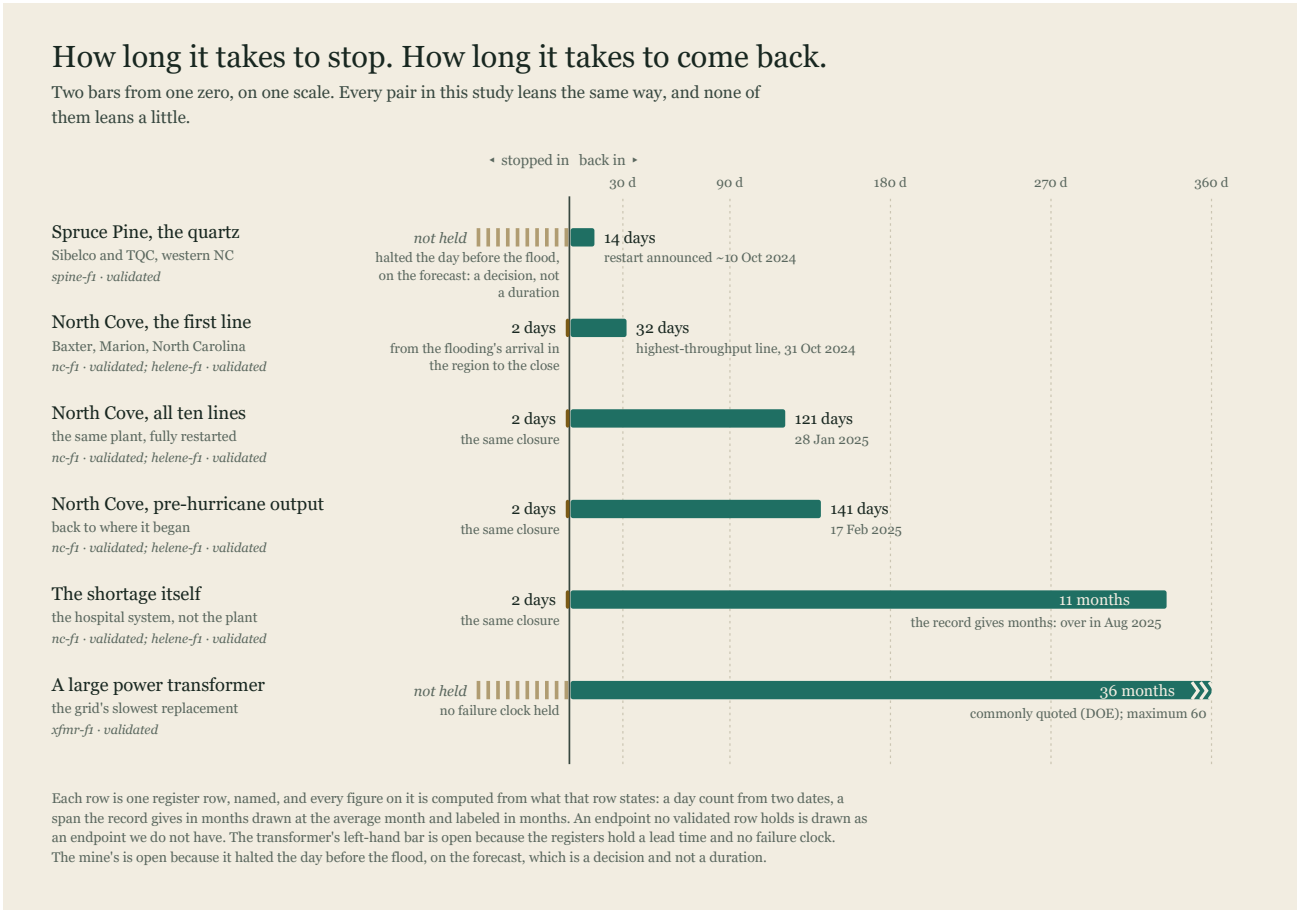


Figure 11. Stop against return for six cases, on one scale. The transformer's return bar leaves the figure, and its stop bar is open.

The transformer's bar leaves the figure and the figure is not rescaled to hold it ⁴⁹; its left-hand bar is open because the register holds a lead time and no failure duration. The asymmetry is not a fact about storms. It is what a buffer is for, and what it costs to have spent one: stopping is a decision that can be made in an afternoon, and restarting is a queue ⁴. The oil chain shows the same shape from the other end, as a buffer in days of cover that now buys about four to five fewer days than in 1996 ⁵. Lithium cannot be drawn in Figure 11. Its two delays are established ¹⁴⁹, but they are structural, what the process must take, and every bar in the figure is a realized event with two dated ends. No lithium event is in the register.

6 • One object, accounted

Figure 12 sets out what the building wire in one average American house took. The price is not the total of the lines above it. It is a second ledger, money and not quantities, set apart from them: the metal in the wire, about \$840 at the 2024 average US producer cathode price, and not the wire, which carries drawing, insulation, distribution and margin that this figure does not ¹⁵⁰. The people are named as roles and never as a count: the rows hold incommensurable units, and no total may be formed from them ¹⁵¹. The seafarers on both crossings are not counted in this study. A receipt makes everything on it look settled, which is why this one prints the lines it cannot fill.

THE BUILDING WIRE IN ONE HOME	
one average US single-family home · what it took	
copper in the wire	88 kg
195 lb of building wire; the source leaves the figure undated	dfw-f1 · validated
energy to make it	0.91-1.15 household-months
counted in months of a home's electricity use; 0.91-1.15 on the ruling's mixed basis, 0.35-1.15 across bases	dfw-c1 · validated
the long leg	about 6,700 miles
(10,760 km) Shanghai to Los Angeles, port to port	col-d1 · validated
fuel for the sea legs	3.6-5.3 gallons
bunker, this home's wire, both crossings; sea legs only	rh-bunker-comp · validated
if it is ever recycled	not measured in this study
house wire comes out as insulated scrap and goes back through a smelter; only clean scrap at the mill is measured here, and this is not it	see swr-c1
at sea, both crossings	48-52 days
at each fleet's average speed, the ore's leg and the metal's; figures that lean long	route-t1 · validated; route-t3 · validated
elapsed, ore to wall	not established
time in port, at the smelter, overland and at the mill is not sourced in this study	
hands it passed through	
miners and contractors · smelter crews · seafarers · overland freight crews · the wholesale tier · licensed electricians · homebuilders	
roles, not a headcount: the rows hold six incommensurable units and no total may be formed from them	
rests on: esc-f1 · validated; gux-w1 · validated; sea-w1 · validated; move-w1 · validated; dfw-w1 · validated	
not counted in this study: the seafarers on both crossings	
borne by	
the Peine community, Atacama · farming villages in the Guixi deposition plume	
borne by, not passed through: people outside the chain who did not consent to it and cannot leave it	
esc-g1 · validated; gux-f1 · validated	
AND ON THE OTHER LEDGER	
what the metal in it was worth	about \$840
the copper in it at the 2024 average US producer cathode price; the metal, not the wire, which carries drawing, insulation, distribution and margin this figure does not. The quantity it multiplies is undated at its source	dfw-p1 · validated
Every figure names the register row it rests on. The price is not a total of the lines above it: it is the other ledger, money rather than quantities, set apart below them. The lines marked not measured, not established and not counted are what it does not know, and says so.	

Figure 12. What the building wire in one average US home took, who handled it, who bears it, what the metal in it was worth, and the lines the study cannot fill.

7 • Limitations

- The cases are in the study because they happened and were documented. A set assembled that way carries no rate, and no likelihood is assigned to any break ¹¹.
- Inventories are private where they are commercial and uncollected where they are not. Only the oil chain's buffer is published in both halves. For lithium, the delays are public and the stocks are not ⁷.
- The record does not contain how a shortage was felt by the person at the end of it. What happened to patients during the saline shortage is recorded in a survey of their pharmacists ¹⁵².
- The queue model's ceiling is conditional on its single-queue structure, which the reproductions check in implementation and not in structure; where the divergence boundary sits leans on an assumed import ceiling (Figure 4).
- The ceiling leaves a residual of at least about a year against the commonly quoted wait, which this study cannot apportion among quote formation, the domestic side of supply, whose path no primary source covers, and the starting backlog ⁶⁹.

- Energy to make is established for one consignment whole, the wire, and for one input of another, the chip's silicon. The other four consignment tables print no figure, and every energy figure here is a national or fleet average and not a named site's.
- The refining shares for lithium and copper are readings off one printed chart, good to a point or two, and are scoped to that chart ¹⁴⁰.
- The pond residence time is one operator's filed figure, with no ceiling ⁶.
- Stages drawn as a kind of works carry a class and not a community: where no plant is named, this study holds nothing on who lives beside it. A district ring carries the communities its record names, as at the Salar de Atacama.
- Transit times are sea legs at fleet average speeds. Time in port, at the works and overland is not sourced, so no chain's total elapsed time is given.
- The induced-seismicity comparison is measured early, on small deep counts, at a depth resolution of about a kilometer ¹³⁰.
- There is no price model. The cost of a severed fuel link is the first study's finding and is cited, not re-derived ¹²¹.

8 · Established and not established

Established

- The copper chain's two crossings and their durations at fleet average speed ¹⁸, and the energy to make one home's wire ¹⁹.
- Escondida's share of mined copper ²³; the rise since 2010 in the electricity and fuel it takes to produce a tonne of Chilean copper ²⁵; and the terms and status of the Peine conciliation ²⁸.
- One domestic maker of grain-oriented electrical steel and the import share ⁴⁶; the quoted transformer wait ⁴⁹; the queue model's computed ceiling at the top of the ranges on record, conditional on its structure, and its distributional finding ¹⁵³; the position of published forecasts against the divergence wedge ⁷¹.
- The record of the transformer reserve: studied, declined for voluntary sharing, and the two most formal sharing programs unused as of 2024 ⁷⁶.
- The Spruce Pine halt and restart announcement ³, the operators' statements about stock ⁸⁷, and the fall in buyers' median chip inventory ⁸¹.
- The saline plant's three recovery dates ⁴; the continuity of the FDA's shortage listing across two hurricanes ¹¹⁵; the pharmacists' survey ¹⁰³; that institutions between the two hurricanes learned to respond and not to be less fragile, item by item ¹¹⁴; and the FDA's statement that it has no authority over allocation ¹⁰⁹.
- The two-state share of US crude production and its history ¹¹⁶; refining utilization and its decomposition ¹⁵⁴; distillate cover against 1996 ⁵.

- Port Arthur's three refinery owners ¹²³, and the civil rights complaint over the petroleum-coke facility, closed without a finding either way ¹²⁵.
- The depth-split result inside the Culberson-Reeves response area, and the absence of new deep disposal capacity in New Mexico ¹³⁰.
- The pond residence time as filed ⁶; the mine-stage shares ¹³⁹; the refining shares as a chart reading ¹⁴⁰; the residence time as inventory, and postponement as distinct from absorption ¹³⁵.
- Four instruments built after the harm, each public ¹⁰; and the buyout of most of Mossville ¹⁰⁸.

Not established, and not claimed

- Any ranking of the six chains, by duration or by likelihood ¹⁵⁵.
- The volume of water stored at Escondida, and so any number of days of cover ³¹.
- What the wire cost: only the metal in it is priced ¹⁵⁰. What a distributor keeps ²¹.
- What powers the Guixi smelter ³⁴; the mode and fuel of any overland leg.
- Spruce Pine's share of world high-purity quartz ¹⁵⁶, and how many weeks of cover its operators held.
- A date on which the quartz district reached full capacity, and therefore any comparison of full recoveries between the two valleys ¹¹².
- Any share of the wafer market ⁹²; the electricity a chip fab uses ⁷⁹.
- A duration for a transformer failure; a capacity figure for any transformer works ⁶¹; whether the sharing programs are sufficient ⁷⁶.
- What accounts for the gap between the computed ceiling and the quoted wait ⁶⁹.
- Whether the largest induced earthquakes responded to the suspension, and whether injected volumes moved into existing wells across the state line ¹³⁰.
- A physical link between lithium brine pumping and the fresh-water aquifer ¹³⁶; a link between the battery's cathode and the iron chain ¹⁴⁸.
- A total of people across stages, or a count at a stage whose record publishes none ¹².

9 • Reproducibility

The three engines, their results, and for the two seeded engines a sample of raw draws, are published with this paper; the σ fit's four large inputs are public files the reader fetches (Appendix B). All of it is mirrored in the public materials repository (github.com/DaiSiHing/roundtable-materials, under `just-in-time-world`). Appendix B lists each file with its fingerprint, each engine's seed, and who rebuilt it. The transformer queue model and the reserve-pool companion were each reimplemented independently by the Author on a different seed; the queue model's ceiling is deterministic at expectation and reproduces across the two implementations. The σ fit was recomputed by the Author from the same public inputs in an independent construction and reproduces; that tests the fit's arithmetic and not the parameter ranges it sweeps, which

are the Energy seat's. Every figure drawn from the register is redrawn at build time from its own generator and compared byte for byte with the shipped file.

Notes

1. Register row `dis-s1`. A synthesis across the six chains, ruled by two seats. It rests on the rows cited in Section 5.3.
2. Register row `sand-x1`. Sources: [1], [2], [3], [4], [5], [6]. The distance between the two sites is the study's own geometry, from their coordinates.
3. Register row `spine-f1`. Sources: [1], [7], [8], [2].
4. Register row `nc-f1`. Sources: [3], [9], [10], [11], [12], [4], [5], [6], [13].
5. Register row `oil-b2`. Source: [14].
6. Register row `li-e1`. Source: [15].
7. Register row `li-b1`. Derived from other rows of the register and no further source: `li-e1` and `li-w1`.
8. Register rows `e3-c1`, `sig-f1`. `e3-c1`: An engine's output; Appendix B names the engine, its inputs and who rebuilt it: the transformer queue model, E3. `sig-f1`: An engine's output; Appendix B names the engine, its inputs and who rebuilt it: the σ fit, and E3 for the half-month figure.
9. Register row `e3-i1`. An engine's output; Appendix B names the engine, its inputs and who rebuilt it: the transformer queue model, E3.
10. Register row `study-g1`. Sources: [16], [17], [18], [19], [20], [21].
11. Register row `dis-p1`. A ruling of the validating seat on the study's own boundaries, not a reading of a document.
12. A ruling of a referee seat, not a claim's support: seat ruling.
13. A ruling of a referee seat, not a claim's support: referee ruling · binding.
14. Source: [28].
15. Register row `chl-g1`. Source: [22].
16. Register row `dfw-f1`. Source: [23].
17. Register rows `col-d1`, `route-d8`, `route-d6`, `route-d7`, `esc-c1`. Sources: [24], [25]. `route-d8`, `route-d6`, `route-d7`: The study's own geometry: the great circle between two sited ends in the map's coordinate table, recomputed independently by the Energy seat.
18. Register rows `route-t1`, `route-t3`. Source: [26].
19. Register row `dfw-c1`. Source: [27]. A conversion by the Energy seat from the smelting band of [27], the wire's mass (row `dfw-f1`) and the household constant [28].
20. Register row `rh-bunker-comp`. Source: [29]. A conversion by the Energy seat from rows `col-d1` and `rh-bunker-p`; the fuel intensities are those of [29].
21. Register row `dfw-b1`. Source: [30].
22. Register row `scrap-g1`. Sources: [31], [32], [33], [34].
23. Register row `esc-c2`. Sources: [35], [36].
24. Register rows `col-f1`, `esc-c1`. Sources: [37], [38], [25].
25. Register row `esc-f5`. Source: [39].
26. Register row `esc-f1`. Source: [40].
27. Register row `rh-esc-power-p`. Sources: [41], [42].
28. Register row `esc-g1`. Sources: [43], [44], [45].
29. Register row `esc-c1`. Source: [25].
30. Register row `col-g1`. Sources: [46], [47].
31. Register row `esc-w1`. Source: [48].
32. Register row `gux-c2`. Sources: [49], [50].
33. Register row `gux-f1`. Sources: [51], [52].
34. Register row `rh-gux-power-p`. Sources: [53], [54].
35. Register row `gux-f2`. Sources: [49], [50].
36. Register row `gux-c1`. Source: [27].
37. Register row `swr-f3`. Source: [55].
38. Register row `swr-f5`. Source: [56].
39. Register row `salt-x2`. Source: [57].
40. Register row `swr-f1`. Source: [58].
41. Register row `swr-w1`. A statement of who works at plants the operator itself names; it rests on rows `swr-f3` and `swr-f5`.
42. Register row `swr-f2`. Sources: [55], [59].
43. Register row `route-t1`. Source: [26].
44. Register row `route-t3`. Source: [26].
45. Register row `swr-c1`. Source: [59]. A conversion by the Energy seat, with [59] for the furnace's charge materials.
46. Register rows `goes-f1`, `goes-f2`. Sources: [60], [61].
47. Register row `xfmr-x1`. Sources: [62], [63].
48. Register row `xfmr-c1`. Fits by the Energy seat to the Department of Energy's reports [60], [61]; parameters ship with the engine (Appendix B).
49. Register row `xfmr-f1`. Source: [60].
50. Register row `xfmr-f15`. Sources: [64], [65].
51. Register rows `grid-g1`, `xfmr-f5`. Sources: [60], [61].
52. A pointer, not a claim's support: TGS-127 · SECONDARY.
53. Register row `move-f1`. Source: [60].
54. Register row `goes-f2`. Sources: [60], [61].
55. Register row `xfmr-f5`. Sources: [60], [61].
56. Register row `goes-f1`. Source: [60].
57. Register row `but-w1`. A statement of who works at the one domestic works; it rests on row `goes-f1`.
58. Register row `xfmr-f2b`. Source: [60].
59. Register row `xfmr-f6`. Sources: [60], [64].
60. Register row `xfmr-c1`. Condition: the 2010 anchor. Fits by the Energy seat to the Department of Energy's reports [60], [61]; parameters ship with the engine (Appendix B).

61. Register row *xfmr-f9*. Sources: [66], [67], [68].
62. Register row *xfmr-f5*. Condition: Department of Energy, 2014. Sources: [60], [61].
63. Register row *xfmr-c1*. Condition: hazard estimated; bands stated. Fits by the Energy seat to the Department of Energy's reports [60], [61]; parameters ship with the engine (Appendix B).
64. Register row *xfmr-f1*. Condition: Department of Energy, 2024. Source: [60].
65. Register row *xfmr-f13*. Source: [69].
66. Register row *xfmr-f4*. Source: [60].
67. Register row *e3-c1*. An engine's output; Appendix B names the engine, its inputs and who rebuilt it: the transformer queue model, E3.
68. Register row *sig-f1*. An engine's output; Appendix B names the engine, its inputs and who rebuilt it: the σ fit, and E3 for the half-month figure.
69. A ruling of a referee seat, not a claim's support: seat ruling · unranked.
70. Register row *iq-f1*. Sources: [70], [71].
71. Register row *xfmr-f7*. Sources: [72], [73].
72. Register row *grid-f2*. Condition: attributed. Source: [74]. The one federal account, which itself describes the incident as reported in the press; the row is attributed and is not validated as a primary fact.
73. A pointer, not a claim's support: no-bridge condition.
74. Register row *grid-f2*. Condition: guardrail. Source: [74]. The one federal account, which itself describes the incident as reported in the press; the row is attributed and is not validated as a primary fact.
75. Register row *grid-f2*. Source: [74]. The one federal account, which itself describes the incident as reported in the press; the row is attributed and is not validated as a primary fact.
76. Register row *grid-g1*. Source: [60].
77. Register row *route-d2*. Sources: [24], [26].
78. A ruling of a referee seat, not a claim's support: seat ruling: the mode renders and the day count is withheld.
79. Register row *poly-f1*. Source: [75].
80. Register row *route-f1*. A computation by the Energy seat from rows *route-d2* and *poly-fi*.
81. Register row *fab-f2*. Source: [76].
82. Register rows *ew-f1*, *sand-x4*. Source: [77]. *sand-x4*: Derived from other rows of the register and no further source: *ew-fi* and *scrap-g1*.
83. Register row *spine-g1*. Sources: [78], [79], [80], [81], [82].
84. Register row *spine-c1*. Condition: validated for the form of the statement, never for a number. Sources: [83], [84], [85], [77].
85. Register row *spine-f2*. Source: [85].
86. Register row *spine-c1*. Condition: for the FORM, never a number. Sources: [83], [84], [85], [77].
87. Register row *spine-w1*. Source: [86].
88. Register rows *spine-w1*, *spine-g1*. Sources: [86], [78], [79], [80], [81], [82].
89. Register row *cru-f1*. Sources: [87], [88], [89].
90. Register row *poly-g1*. A class of producers; it rests on row *poly-f1*.
91. Register row *cru-g1*. One maker named at company level; it rests on row *cru-fi*.
92. Register row *waf-f1*. Source: [90].
93. Register row *waf-g1*. One maker's own description, and no agency's shares; it rests on row *waf-fi*.
94. Register rows *fab-f1*, *fab-g1*. Sources: [91], [92], [93], [94].
95. Register row *fab-f1*. Sources: [91], [92], [93].
96. Register rows *ew-f1*, *ew-g1*. Source: [77]. *ew-g1*: A recovery workforce, named as a class; it rests on row *ew-fi*.
97. Register row *sand-x4*. Derived from other rows of the register and no further source: *ew-fi* and *scrap-g1*.
98. Register row *nc-c1*. Sources: [95], [96].
99. Register rows *route-d5*, *route-s1*. *route-d5*: The study's own geometry: the great circle between two sited ends in the map's coordinate table, recomputed independently by the Energy seat. *route-s1*: A ruling of the validating seat on the study's own boundaries, not a reading of a document.
100. Register rows *route-s1*, *route-d5*. *route-s1*: A ruling of the validating seat on the study's own boundaries, not a reading of a document. *route-d5*: The study's own geometry: the great circle between two sited ends in the map's coordinate table, recomputed independently by the Energy seat.
101. Register row *conv-s1*. A ruling of the validating seat on the study's own boundaries, not a reading of a document.
102. Register row *brine-f1*. Source: [77].
103. Register row *nc-w1*. Sources: [97], [98], [99], [100].
104. Register row *nc-c1*. Condition: attributed. Sources: [95], [96].
105. Register row *brine-g1*. Classes, because the federal survey names no operation; it rests on row *brine-fi*.
106. Register row *chl-f1*. Source: [101].
107. Register row *chl-c1*. Sources: [101], [102].
108. Register row *pvc-g1*. Sources: [22], [103], [104], [105], [106], [107].
109. Register row *nc-g1*. Sources: [12], [108], [109], [110], [111].
110. Register row *helene-f1*. Source: [112].
111. Register rows *nc-f1*, *nc-f2*. Sources: [3], [9], [10], [11], [12], [4], [5], [6], [13], [113], [114], [115].
112. Register rows *sand-x1*, *salt-x1*. Sources: [1], [2], [3], [4], [5], [6]. *sand-x1*, *salt-x1*: The distance between the two sites is the study's own geometry, from their coordinates.
113. Register row *nc-w1*. Condition: attributed to the pharmacists' society, ASHP. Sources: [97], [98], [99], [100].
114. Register row *nc-g2*. Sources: [12], [13], [108], [116], [111], [117], [118].
115. Register row *nc-f2*. Sources: [113], [114], [115].

116. Register row oil-c1. Source: [119].
117. Register rows route-d4, route-o1. route-d4: The study's own geometry: the great circle between two sited ends in the map's coordinate table, recomputed independently by the Energy seat. route-o1: A ruling of the validating seat on the study's own boundaries, not a reading of a document.
118. Register rows route-o1, route-d4, oil-b2. Source: [14]. route-o1: A ruling of the validating seat on the study's own boundaries, not a reading of a document. route-d4: The study's own geometry: the great circle between two sited ends in the map's coordinate table, recomputed independently by the Energy seat.
119. Register row conv-o1. A ruling of the validating seat on the study's own boundaries, not a reading of a document.
120. Register row oil-n1. Source: [120].
121. Register row oil-s1. Source: [121].
122. Register row who-up2. Sources: [122], [123], [124].
123. Register row who-rf1. Sources: [120], [125].
124. Register row who-rf2. Source: [126].
125. Register row who-rf5. Source: [127].
126. Register row oil-c2. Source: [128].
127. Register row oil-c3. Source: [128].
128. Register row who-up3. Sources: [129], [130], [16], [131], [132].
129. Register rows who-up3, seis-x1. Sources: [129], [130], [16], [131], [132], [133], [134]. seis-x1: The well count and the absence of any volume field are derived by the Energy seat from the disposal-well records.
130. Register row seis-x1. Sources: [133], [134]. The well count and the absence of any volume field are derived by the Energy seat from the disposal-well records.
131. Register row route-d3. Sources: [24], [26].
132. Register rows route-t6, route-t5. Sources: [24], [26], [135], [136].
133. Register rows conv-l1, li-e1. Source: [15]. conv-l1: A ruling of the validating seat on the study's own boundaries, not a reading of a document.
134. Register row li-p1. Source: [137].
135. Register rows li-w1, li-b1. li-w1: Derived from other rows of the register and no further source: li-e1, by the identity that a residence time is an inventory. li-b1: Derived from other rows of the register and no further source: li-e1 and li-w1.
136. Register row li-x1. Sources: [43], [44].
137. A ruling of a referee seat, not a claim's support: seat rule.
138. Register row lng-g1. The operator's filing [15] and the tribunal record [43].
139. Register row li-c1. Source: [77].
140. Register row li-c2. Sources: [138], [139], [140].
141. Register row mine-g1. An aggregate of five producing countries; it rests on row li-c1.
142. Register row cam-g1. A statement of what this study does not hold. It cites no document because it asserts none.
143. Register row cell-g1. A statement of what this study does not hold. It cites no document because it asserts none.
144. Register row pack-g1. The maker's annual filing [137], for the one product sourced.
145. Register row li-w1. Derived from other rows of the register and no further source: li-e1, by the identity that a residence time is an inventory.
146. Register row li-b2. Derived from other rows of the register and no further source: li-e1.
147. Register row store-g1. Four kinds of party, named as classes: a statement of class, resting on no document.
148. A ruling of a referee seat, not a claim's support: li-p2 · ruled: not built.
149. Register rows li-b1, li-b2. li-b1: Derived from other rows of the register and no further source: li-e1 and li-w1. li-b2: Derived from other rows of the register and no further source: li-e1.
150. Register row dfw-p1. Source: [141].
151. Register rows esc-f1, gux-w1, sea-w1, move-w1, dfw-w1. Source: [40]. gux-w1: A statement of who works at a named operator's plant; it rests on rows gux-c2 and gux-f2. sea-w1: Seafarers on two sea legs; it rests on rows col-d1, route-t1 and route-t3. move-w1: Overland crews between two inland ends: entailed by the chain's own sited ends, and resting on no document. dfw-w1: Three trades, named as classes: a statement of class, resting on no document.
152. Register row bed-g1. Patients appear only through their pharmacists' survey; it rests on rows nc-w1 and nc-g1.
153. Register rows e3-c1, e3-i1. e3-c1, e3-i1: An engine's output; Appendix B names the engine, its inputs and who rebuilt it: the transformer queue model, E3.
154. Register rows oil-c2, oil-c3. Source: [128].
155. Register rows dis-s1, dis-p1. dis-s1: A synthesis across the six chains, ruled by two seats. It rests on the rows cited in Section 5.3. dis-p1: A ruling of the validating seat on the study's own boundaries, not a reading of a document.
156. Register row spine-c1. Sources: [83], [84], [85], [77].

References

Each address is the one this study recorded when it fetched the source: on the register row, in the fetch record, or in the saved page itself. Where the study kept a copy, its checksum is given, so a reader holding the document can tell it is the same one. Where the study had not written an address down, the document was found again at its publisher and the address accepted

only if the file fetched there matched the kept copy; an entry that still has no address is one for which that could not be done. A page or data series that changes is marked as one: its address is where the study read it, and the kept copy is what the study read.

1. Sibelco. "Impact of Hurricane Helene on Spruce Pine Operations." News release. <https://www.sibelco.com/en/news/impact-of-hurricane-helene-on-spruce-pine-operations> Pinned copy, md5 4388161512d1c3618fac2a86da45e23a.
2. The Quartz Corp. "Impact of Hurricane Helene on The Quartz Corp in Spruce Pine." <https://www.thequartzcorp.com/articles/impact-of-hurricane-helene-on-the-quartz-corp-in-spruce-pine> Pinned copy, md5 99cc10b84a16dee64cb81ca8561576d7.
3. Baxter International. "Baxter Provides Update on North Cove, N.C., Facility and Hurricane Helene Relief Plans for Community." <https://www.baxter.com/news-insights/baxter-provides-update-north-cove-nc-facility-and-hurricane-helene> Pinned copy, md5 e97c35395fdc21ddoe857962788c5e1a.
4. Baxter International. Hurricane Helene update, October 31, 2024. https://www.baxter.com/sites/default/files/2026-05/update_10-31_0.pdf Pinned copy, md5 2d4f15120598e0ec8bc07fb5519212d7.
5. Baxter International. Hurricane Helene update, January 28, 2025. https://www.baxter.com/sites/default/files/2026-05/update_1-28.pdf Pinned copy, md5 a4a227ea0b22b2ef5fcc954016106a8c.
6. Baxter International. Hurricane Helene update, February 17, 2025. <https://www.baxter.com/sites/default/files/2026-05/2-17-25-website-update.pdf> Pinned copy, md5 53c2560b7bdf3d2a0bfb5a56d94ffc5.
7. Sibelco. "Progress with Spruce Pine Recovery Activities." News release. <https://www.sibelco.com/en/news/progress-with-spruce-pine-recovery-activities> Pinned copy, md5 276b87e8e514520bcee2307bbab93f32.
8. Sibelco. "Sibelco Restarts Production and Customer Shipments at Spruce Pine Following Hurricane Helene." News release. <https://www.sibelco.com/en/news/sibelco-restarts-production-and-customer-shipments-at-spruce-pine-following-hurricane-helene> Pinned copy, md5 aab7c9840219b734b62e6a42bae515c9.
9. Baxter International. Hurricane Helene summary. <https://www.baxter.com/news-insights/hurricane-helene-summary> Pinned copy, md5 702c6539982c9d843c52d57507e3fc7e.
10. US Food and Drug Administration. Letter, Baxter, September 30, 2024. <https://www.fda.gov/media/182565/download> Pinned copy, md5 ca2877301fod6a7ff8d62046b6ddee7f.
11. US Food and Drug Administration. Letter, Baxter, October 1, 2024. <https://www.fda.gov/media/182566/download> Pinned copy, md5 26f26acc1ea88549ec9fa3f2c92e38f7.
12. US Food and Drug Administration. "Hurricane Helene: Baxter's manufacturing recovery in North Carolina." Archived copy of November 19, 2024; the live page is retired. <https://web.archive.org/web/20241119040655/https://www.fda.gov/drugs/updates-2024-hurricane-season/hurricane-helene-baxters-manufacturing-recovery-north-carolina> Pinned copy, md5 e14df2197f3222ce4ebf3374410865c2.
13. American Hospital Association. Statement of record for the Senate Special Committee on Aging hearing on pharmaceutical and medical supply chains, October 8, 2025. <https://www.aha.org/system/files/media/file/2025/10/aha-statement-of-record-for-senat-special-committee-on-aging-hearing-on-pharmaceutical-and-medical-supply-chains-testimony-10-8-2025.pdf> Pinned copy, md5 aa4d7ec69de4ed909c1f540f22311aa2.
14. US Energy Information Administration. Weekly US distillate fuel oil stocks (WDISTUS1) and product supplied (WDIUPUS2). https://www.eia.gov/dnav/pet/hist_xls/WDISTUS1w.xls
15. Albemarle Corporation. Form 10-K for fiscal year 2025. Filed February 11, 2026. US Securities and Exchange Commission, EDGAR accession 0000915913-26-000018. <https://www.sec.gov/Archives/edgar/data/915913/000091591326000018/alb-20251231.htm>
16. Railroad Commission of Texas. Notice to Operators: Disposal Well Monitoring and Reporting Requirements in the Permian Basin. February 6, 2023. <https://www.rrc.texas.gov/media/11on5x0d/2023-nto-disposal-well-monitoring-reporting-requirements-in-the-permian-basin-2-6-2023.pdf> Address found after publication; the document fetched there is the pinned copy. Pinned copy, md5 ed2b2d16854c6cdde6a14dcd857d9599.
17. US Environmental Protection Agency. "Enforcement Alert: Benzene Fenceline Monitoring at Petroleum Refineries." <https://www.epa.gov/enforcement/enforcement-alert-benzene-fenceline-monitoring-petroleum-refineries> Pinned copy, md5 010b83fb326492c5a2b11780c48268e7.
18. US Environmental Protection Agency. Final rule overview fact sheet for the HON and P&R rules. April 9, 2024. https://www.epa.gov/system/files/documents/2024-04/chem-sector-final-rule.-overview-fact-sheet_0.pdf Address found after publication; the document fetched there is the pinned copy. Pinned copy, md5 2000598093b6f3bdf72b153fdo43142.

19. Federal Register. 89 FR 42932 (2024). <https://www.federalregister.gov/documents/2024/05/16/2024-07002/new-source-performance-standards-for-the-synthetic-organic-chemical-manufacturing-industry-and> Pinned copy, md5 823db20da3e4108b3599c39e632ab76d.
20. US Environmental Protection Agency. Benzene Fenceline Monitoring Dashboard. Retrieved August 24, 2026. https://awseda.p.epa.gov/public/extensions/Fenceline_Monitoring/Fenceline_Monitoring.html?sheet=background A page that changes; the study's reading is of the copy kept on the date shown. Pinned copy, md5 08ed69b20247d8c867510698aa03e8ec.
21. Texas Historical Commission. Historical marker: Texas Company Refinery at Port Arthur (Port Arthur Works), Atlas Number 5245012334. <https://atlas.thc.texas.gov/Details/5245012334> Address found after publication; the document fetched there is the pinned copy. Pinned copy, md5 b2469accfec21dd09b6a5079848e968a.
22. Federal Register. Presidential proclamation, FR Doc. 2026-14452, with its Annex I, and the Environmental Protection Agency's Clean Air Act section 112 exemption page. No address was recorded for this document.
23. Copper Development Association. Education page on copper in the home. Archived copy of November 30, 2023; the live page is no longer served. <https://web.archive.org/web/20231130164812/https://www.copper.org/education/c-facts/home/>
24. National Geospatial-Intelligence Agency. Publication 151: Distances Between Ports. <https://msi.nga.mil/api/publications/download?key=16694076%2FSFH00000%2FPub151bk.pdf&type=view> Address found after publication; the document fetched there is the pinned copy. Pinned copy, md5 80c2d8e258a589a1c3b7cc7b886f3006.
25. Campero, C., and Harris, L. M. 2019. Water 11(5): 886.
26. International Maritime Organization. Fourth IMO Greenhouse Gas Study 2020, full report and annexes. Table 35. <https://www.cdn.imo.org/localresources/en/OurWork/Environment/Documents/Fourth%20IMO%20GHG%20Study%202020%20-%20Full%20report%20and%20annexes.pdf>
27. Liu et al. 2022. Processes 10(12): 2715. Read at an open mirror; the publisher refused the fetch.
28. US Energy Information Administration. Frequently asked questions, number 97: average annual electricity use of a US residential customer, 10,791 kilowatt-hours. No address was recorded for this document.
29. CE Delft. STREAM Freight Transport 2020. Tables 29 and 31. No address was recorded for this document.
30. US Census Bureau. Monthly Wholesale Trade Survey, inventories-to-sales ratio. Retrieved September 23, 2026. <https://api.census.gov/data/timeseries/eits/mwts> Pinned copy, md5 b5559c082eae4db17275f9db09a116f0.
31. Ministry of International Trade and Industry, Malaysia. Guidelines for Importation and Inspection of Metal Scrap. January 2022 edition. No address was recorded for this document.
32. US Census Bureau. International trade, exports by Harmonized System code: HS 7404, copper waste and scrap, by country and year. <https://api.census.gov/data/timeseries/intltrade/exports/hs> Pinned copy, md5 813c6730663c3cc527086f892b8e5cba.
33. Article, PubMed Central PMC8213787. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8213787/>
34. Fastmarkets. "US copper scrap exports fall in 2023." <https://www.fastmarkets.com/insights/us-copper-scrap-exports-fall-in-2023/>
35. BHP Group. Form 6-K, 2025. US Securities and Exchange Commission, EDGAR. <https://www.sec.gov/Archives/edgar/data/811809/000119312525160800/d938594d6k.htm>
36. US Geological Survey. Mineral Commodity Summaries 2025: Copper. <https://pubs.usgs.gov/periodicals/mcs2025/mcs2025-copper.pdf>
37. BHP Billiton. Escondida site visit, presentation to investors, April 24, 2004. <https://www.bhp.com/~media/bhp/documents/investors/reports/2004/escondidasitevisit240404.pdf> Pinned copy, md5 f121a7ad245b2be1eeb63761dd1292fd.
38. Mining Technology. Escondida copper mine, project profile. <https://www.mining-technology.com/projects/escondida/>
39. Comisión Chilena del Cobre (Cochilco). Consumo de energía en la minería del cobre, actualización año 2024. <https://www.cochilco.cl/web/download/991/2025/15151/consumo-de-energia-en-la-mineria-del-cobre-actualizacion-ano-2024.pdf> Pinned copy, md5 3804148925965feb66603c6ff11765ca.
40. Minera Escondida Limitada. Memoria Anual 2025. Filed with Chile's Comisión para el Mercado Financiero, May 27, 2026. The download is issued per session; the address is the entity page, from which the year is selected. <https://www.cmfchile.cl/institucional/mercados/entidad.php?mercado=0&rut=79587210&tipoentidad=SARYT&row=AAAwY2ACTAAAwCAAW&vig=VI&control=svs&pesania=49> Pinned copy, md5 03cdb7393e378fe08ddc13bfab9c9dbaa.
41. BHP. "BHP targets 100 per cent renewable energy at Escondida and Spence operations and elimination of water usage from aquifers in Chile." News release, October 2019. <https://www.bhp.com/news/media-centre/releases/2019/10/bhp-targets-100-per-cent-renewable-energy-at-escondida-and-spence-operations-and-elimination-of-water-usage-from-aquifers-in-chile>

42. BHP. "BHP operations in Chile start to operate with renewable energies." News release, August 2021. <https://www.bhp.com/new/s/media-centre/releases/2021/08/bhp-operations-in-chile-start-to-operate-with-renewable-energies>
43. Primer Tribunal Ambiental (Chile). Causa Rol D-12-2022, case record. <https://www.portaljudicial1ta.cl/sgc-web/ver-causa.html?rol=D-12-2022>
44. Primer Tribunal Ambiental (Chile). Sentencia, Rol D-12-2022. Archived copy. <https://web.archive.org/web/20250323000000/https://www.1ta.cl/primer-tribunal-ambiental-sentencia-d-12-2022/>
45. Minería Chilena. "Tribunal constata avances en implementación de medidas ambientales de conciliación entre el CDE y comunidad indígena con mineras." Reproduces the tribunal's own release of February 24, 2026, whose page refused the fetch. <https://www.mch.cl/tribunal-constata-avances-en-implementacion-de-medidas-ambientales-de-conciliacion-entre-el-cde-y-comunidad-indigena-con-mineras/>
46. Servicio Nacional de Pesca y Acuicultura (Sernapesca). "Pescadores y pescadoras pasarán a administrar la Caleta Coloso gracias a la Ley de Caletas." November 18, 2025. <https://www.sernapesca.cl/noticias/pescadores-y-pescadoras-pasaran-a-administrar-la-caleta-coloso-gracias-a-la-ley-de-caletas/> Pinned copy, md5 f69d581a8a446eed58c60fa979c917f3.
47. Antofagasta TV. "Escondida | BHP inicia obras de proyecto para modernizar caleta de pescadores de Coloso." June 28, 2024. Bylined to the company. <https://www.antofagasta.tv/categoria/antofagasta/escondida-bhp-inicia-obras-proyecto-modernizar-caleta-pescadores-coloso> Pinned copy, md5 e182f49b02bc8a189ea665b6d42d78e3.
48. BHP Billiton. Form 6-K, July 25, 2013: "BHP Billiton Approves Escondida Water Supply Project." US Securities and Exchange Commission, EDGAR accession 0001181431-13-041413. <https://www.sec.gov/Archives/edgar/data/811809/000118143113041413/rrd386633.htm> Pinned copy, md5 8e9e99019cd2d40ea2567fd381c94cd5.
49. International Copper Study Group. The World Copper Factbook 2025. <https://icsg.org/download/2025-10-the-world-copper-factbook/>
50. Jiangxi Copper Company Limited. 2024 Annual Report. <https://en.jxcc.com/uploadFiles/file/20250430/b393e724d4b840bb98e11cccd75d5771.pdf> Pinned copy, md5 98c5bdf76c9a661049488d7bda858485.
51. "Health risks to local residents from the exposure of heavy metals around the largest copper smelter in China." Ecotoxicology and Environmental Safety. <https://www.sciencedirect.com/science/article/abs/pii/S014765131831371X>
52. Pure Earth. Guixi smelter, project record. <https://www.pureearth.org/project/guixi-smelter/>
53. Global Energy Monitor. Guixi power station, wiki entry. https://www.gem.wiki/Guixi_power_station
54. Power Technology. Jiangxi Guixi power plant, plant profile. <https://www.power-technology.com/data-insights/power-plant-profile-jiangxi-guixi-power-plant-china/>
55. Southwire Company. SCR copper rod systems, product page. <https://www.southwire.com/scr-technologies/scr-copper-rod-systems> Pinned copy, md5 501e4de1fcfe48249e5b28d27cc1cfe0.
56. Southwire Company. "Four Manufacturing Sites Earn The Copper Mark." December 8, 2025. <https://www.southwire.com/blog/s/manufacturing-sites-earn-copper-mark> Pinned copy, md5 f12ccdf4a8aaaeec7a2b220e5d5ad8f2.
57. Southwire Company. Specification 10028: Romex SIMpull copper Type NM-B cable. https://cabletechsupport.southwire.com/en/cablespec/download_spec/?spec=10028&country=US Pinned copy, md5 bb001197e5f90cdb935f2941df678892.
58. Southwire Company. About us. <https://www.southwire.com/about-us> Pinned copy, md5 d872edf7a378685ad9405ad34bafac8f.
59. Southwire Company. Shaft furnace, data sheet. The bare address is refused; the link on the company's SCR page carries a session token. <https://www.southwire.com/medias/Southwire-Shaft-Furnace.pdf> Pinned copy, md5 c155a033c76cc56e8b2627d981e2002d.
60. US Department of Energy. Large Power Transformer Resilience: Report to Congress. Signed July 10, 2024. <https://www.energy.gov/sites/default/files/2024-10/EXEC-2022-001242%20-%20Large%20Power%20Transformer%20Resilience%20Report%20signed%20by%20Secretary%20Granholm%20on%207-10-24.pdf> Address found after publication; the document fetched there is the pinned copy. Pinned copy, md5 89c09245f41fac3e62582a6ff1487393.
61. US Department of Energy. Large Power Transformers and the U.S. Electric Grid. April 2014 update. <https://www.energy.gov/sites/prod/files/2014/04/f15/LPTStudyUpdate-040914.pdf> Address found after publication; the document fetched there is the pinned copy. Pinned copy, md5 788cac21c48cfb9a8a04584eaa3bc6c4.
62. ABB. Environmental product declaration, TrafoStar 500 MVA power transformer. <https://library.e.abb.com/public/566748ad75116903c1256d630042f1af/ProductdeclarationStarTrafo500.pdf> Address found after publication; the document fetched there is the pinned copy. Pinned copy, md5 7010442cd22b1f413d8fa4e8cdac9bde.
63. Hitachi Energy. Environmental product declaration, 40 MVA power transformer. https://www.epditaly.it/wp-content/uploads/2016/12/EPD_40MVA_Hitachi_rev3-11.03.2022-29.04.2022bis.pdf Address found after publication; the document fetched there is the pinned copy. Pinned copy, md5 057440e92eabd4d9586ba6c28f94d6cc.

64. National Infrastructure Advisory Council. Addressing the Critical Shortage of Power Transformers. June 2024 draft. https://www.cisa.gov/sites/default/files/2024-06/DRAFT_NIAC_Addressing%20the%20Critical%20Shortage%20of%20Power%20Transformers%20to%20Ensure%20Reliability%20of%20the%20U.S.%20Grid_Report_06052024_508c.pdf Address found after publication; the document fetched there is the pinned copy. Pinned copy, md5 54f0cc87a8aae1e8b34a527a05952e94.
65. US Bureau of Labor Statistics. Producer Price Index, power and distribution transformers (PCU335311335311), via the Federal Reserve Bank of St. Louis. <https://fred.stlouisfed.org/graph/fredgraph.csv?id=PCU335311335311> Pinned copy, md5 3d3e7a8ece367a6d2b4636bcf8a3f4bf.
66. Hitachi Energy. "Hitachi Energy invests US\$37 million to expand transformer manufacturing facility in South Boston, Virginia." October 2022. <https://www.hitachienergy.com/us/en/news-and-events/press-releases/2022/10/hitachi-energy-invests-us-37-million-to-expand-transformer-manufacturing-facility-in-south-boston-virginia>
67. Virginia Economic Development Partnership. Hitachi Energy, South Boston. Press release, September 2025. <https://www.vedp.org/press-release/2025-09/hitachi-energy-south-boston>
68. Hitachi Energy. "Hitachi Energy breaks ground on the nation's largest facility for the production of large power transformers in South Boston, Virginia." June 2026. <https://www.hitachienergy.com/us/en/news-and-events/press-releases/2026/06/hitachi-energy-breaks-ground-on-the-nation-s-largest-facility-for-the-production-of-large-power-transformers-in-south-boston-virginia>
69. Bloomberg News. "GE Vernova CEO Sees Order Backlog Stretching Into 2028." March 12, 2025. Syndicated copy at Energy Connects. <https://www.energyconnects.com/news/utilities/2025/march/ge-vernova-ceo-sees-order-backlog-stretching-into-2028/> Pinned copy, md5 18ce4dac2c7d7ab687204969ff5e62dd.
70. Lawrence Berkeley National Laboratory. Queued Up, 2025 edition: interconnection queue data file through 2024. https://eta-publications.lbl.gov/sites/default/files/2025-08/lbnl_ix_queue_data_file_thru2024_v2.xlsx
71. US Energy Information Administration. Form EIA-860, 2024. <https://www.eia.gov/electricity/data/eia860/xls/eia8602024.zip>
72. North American Electric Reliability Corporation. 2025 Long-Term Reliability Assessment. https://www.nerc.com/globalassets/our-work/assessments/nerc_ltra_2025.pdf Pinned copy, md5 3c9ed67fc5ccdb1e1689dd713135e4aa.
73. North American Electric Reliability Corporation. 2024 Long-Term Reliability Assessment, corrected July 2025. https://www.nerc.com/globalassets/our-work/assessments/2024-ltra_corrected_july_2025.pdf Pinned copy, md5 a1b6c1b19f3d919e2f8807f3a1c03194.
74. US Department of Energy. Assessment of Large Power Transformer Risk Mitigation Strategies. October 2016. <https://www.energy.gov/sites/prod/files/2017/01/f34/Assessment%20of%20Large%20Power%20Transformer%20Risk%20Mitigation%20Strategies.pdf> Address found after publication; the document fetched there is the pinned copy. Pinned copy, md5 56c335c3b83dae844964e26f8cb5a76b.
75. International Energy Agency Photovoltaic Power Systems Programme, Task 12. Life Cycle Inventories and Life Cycle Assessments of Photovoltaic Systems. Report IEA-PVPS T12-19:2020. <https://iea-pvps.org/wp-content/uploads/2020/12/IEA-PVPS-LCI-report-2020.pdf> Address found after publication; the document fetched there is the pinned copy. Pinned copy, md5 f44857a6d2c4f4fe555846fb87ca5ccf.
76. US Department of Commerce. "Results from Semiconductor Supply Chain Request for Information." January 2022. <https://www.commerce.gov/news/blog/2022/01/results-semiconductor-supply-chain-request-information> Pinned copy, md5 2c5fd516d704c9ef3c92db08a59bc9d8.
77. US Geological Survey. Mineral Commodity Summaries 2025. Cited for its chapters on salt, lithium and silicon, and for carrying no line for high-purity quartz. <https://pubs.usgs.gov/periodicals/mcs2025/mcs2025.pdf> Pinned copy, md5 b695b2badc1b444a358c2971c04354b6.
78. The Quartz Corp. Company history. <https://www.thequartzcorp.com/articles/history>
79. Sibelco. Spruce Pine, location page. <https://www.sibelco.com/en/locations/spruce-pine>
80. National Labor Relations Board. Case 10-RC-369950, docket. <https://www.nlrb.gov/case/10-RC-369950>
81. US Department of the Interior. "Interior Department releases final 2025 list of critical minerals." Press release. <https://www.doi.gov/pressreleases/interior-department-releases-final-2025-list-critical-minerals>
82. Congressional Research Service. In Focus IF13145. <https://www.congress.gov/crs-product/IF13145>
83. Cassella, Megan. "How a tiny town hit by Helene could upend the global semiconductor chip industry." CNBC, October 3, 2024. <https://www.cnbc.com/2024/10/03/helene-quartz-mine-semiconductor-north-carolina.html> Pinned copy, md5 76ea8240adedc85365f9aab54b6a9a11.
84. The same article, as carried by NBC News. <https://www.nbcnews.com/business/business-news/tiny-town-hit-helene-upend-global-semiconductor-chip-industry-rcna173933> Pinned copy, md5 f6cb57fb6a833b4d1381a2e2e0a6603.

85. Potter, Brian. "Does All Semiconductor Manufacturing Depend on Spruce Pine Quartz?" Construction Physics, November 8, 2024. <https://www.construction-physics.com/p/does-all-semiconductor-manufacturing> Pinned copy, md5 cdc17ae6707452a9ceb121a82f678a47.
86. The Quartz Corp. "Update on Hurricane Helene impact on The Quartz Corp." <https://www.thequartzcorp.com/articles/update-on-hurricane-helene-impact-over-the-quartz-corp> Pinned copy, md5 4e83cc3d4dd4a649b99753312fc80b9e.
87. Momentive Technologies. "Momentive Technologies expands in Northeast Ohio, creating 97 new jobs." <https://www.momentive.tech.com/news/momentive-technologies-expands-in-northeast-ohio-creating-97-new-jobs> Pinned copy, md5 c32763752b3ffbbaf664a1a95481727.
88. Business Wire. "Momentive Technologies Completes Acquisition of CoorsTek's Crucibles Business." November 1, 2022. <https://www.businesswire.com/news/home/20221101005097/en/Momentive-Technologies-Completes-Acquisition-of-CoorsTeks-Crucibles-Business>
89. JobsOhio. "Momentive Technologies expands in Northeast Ohio, creating 97 new jobs." <https://www.jobsohio.com/newsroom/news-press/momentive-technologies-expands-in-northeast-ohio-creating-97-new-jobs> Pinned copy, md5 42c833267ec65de489e0e330c1ebdbd2.
90. SUMCO Corporation. Growth strategy. <https://www.sumcosi.com/english/aboutus/growth.html> Pinned copy, md5 4f29bcef48de5a466e18eae71729c038.
91. Texas Instruments. "Texas Instruments breaks ground on new 300-mm semiconductor wafer fabrication plants in Sherman, Texas." May 18, 2022. <https://www.ti.com/about-ti/newsroom/news-releases/2022/2022-05-18-texas-instruments-breaks-ground-on-new-300-mm-semiconductor-wafer-fabrication-plants-in-sherman--texas.html> Pinned copy, md5 ab1c64570ace9aefda447e89227f6a34.
92. SEMI. "Global 200mm Fab Capacity on Pace to Record Growth to Meet Surging Demand and Address Chip Shortage, SEMI Reports." 2021. <https://www.semi.org/en/news-media-press/semi-press-releases/global-200mm-fab-capacity-record-growth> Pinned copy, md5 8614d54a4c4ea54c44ef997d95f18eao.
93. SEMI. "Global 200mm Fabs to Reach Record High Capacity by 2026, SEMI Reports." <https://www.semi.org/en/news-media-press-releases/semi-press-releases/global-200-mm-fabs-to-reach-record-high-capacity-by-2026-semi-reports> Pinned copy, md5 a0c60coba628a0918c0c419b00622165.
94. Texas Instruments. "Texas Instruments announces award agreement for CHIPS and Science Act funding." December 20, 2024. <https://www.ti.com/about-ti/newsroom/news-releases/2024/2024-12-20-texas-instruments-announces-award-agreement-for-chips-and-science-act-funding.html> Pinned copy, md5 ce3a79e9e522f3b9e23777fb5740fb3b.
95. American Hospital Association. "UPDATE: Baxter to Increase Allocations for IV Solution Supplies to Hospitals Effective Today, Oct. 9." Special bulletin, October 9, 2024. Cited for the share of supply, in the association's words. <https://www.aha.org/special-bulletin/2024-10-09-update-baxter-increase-allocations-iv-solution-supplies-hospitals-effective-today-oct-9> Pinned copy, md5 3ad723fc84277997515835831481d5a8.
96. Baxter International. Form 8-K, October 8, 2024. US Securities and Exchange Commission. Cited for the plant's closure; it carries no share of supply. <https://www.sec.gov/Archives/edgar/data/10456/000162828024042579/bax-20241008.htm> Address found after publication; the document fetched there is the pinned copy, apart from a script the server appends. Pinned copy, md5 dbd45dcfc37da22155c914cb7e210e1f.
97. American Society of Health-System Pharmacists (ASHP). "New ASHP Survey Shows How Hospitals Are Handling IV Fluid Shortages." November 8, 2024. <https://news.ashp.org/news/ashp-news/2024/11/08/new-ashp-survey-shows-how-hospitals-are-handling-iv-fluid-shortages> Pinned copy, md5 6640f4e2b3744953dc55c6477e6f18a6.
98. American Society of Health-System Pharmacists (ASHP). Sterile fluid shortage severity survey, 2024. <https://news.ashp.org/-/media/assets/drug-shortages/docs/2024/ASHP-Sterile-Fluid-Shortage.pdf> Address found after publication; the document fetched there is the pinned copy. Pinned copy, md5 15315a482926d4ff94f773aaef068007.
99. Administration for Strategic Preparedness and Response, TRACIE. IV fluid shortage strategies, October 9, 2024. <https://files.asprtracie.hhs.gov/documents/iv-fluid-shortage-strategies.pdf> Address found after publication; the document fetched there is the pinned copy. Pinned copy, md5 b65d73b86334143d2e10b0b49971af76.
100. US Senators. Letter to the Department of Health and Human Services on IV fluids, 2024. https://www.kaine.senate.gov/imo/media/doc/letter_to_hhs_on_iv_fluids.pdf Address found after publication; the document fetched there is the pinned copy. Pinned copy, md5 8b774a40b37e08off175dcff35ad9a25.
101. Euro Chlor (Cefic). "The European chlor-alkali industry: an electricity intensive sector." https://www.eurochlor.org/wp-content/uploads/2019/04/3-2-the_european_chlor-alkali_industry_-_an_electricity_intensive_sector_exposed_to_carbon_leakage.pdf Address found after publication; the document fetched there is the pinned copy. Pinned copy, md5 1a50ab8f06ef3f582accf4e007c6bd84.

102. Euro Chlor. Production statistics. Fetched at eurochlor.org/production/; the saved page gives the address below as its own. <https://eurochlor.org/industry/> Pinned copy, md5 cc49ebb7421ba6f64fd642158bde3278.
103. Earthjustice. "Louisiana court ruling reverses lower court decision and upholds air permits for Formosa Plastics' massive petrochemical complex in Cancer Alley." Press release, 2024. A release by counsel to the plaintiffs; no court document is held. <https://earthjustice.org/press/2024/louisiana-court-ruling-reverses-lower-court-decision-and-upholds-air-permits-for-formosa-plastics-massive-petrochemical-complex-in-cancer-alley>
104. Inter-American Commission on Human Rights. Report No. 43/10, Petition 242-05, Admissibility: Mossville Environmental Action Now v. United States. March 17, 2010. Cited for the residents' own account of the community, recorded by the Commission under the position of the petitioners; not a finding of the Commission. <https://cidh.oas.org/annualrep/2010eng/USAD242-05EN.doc> Pinned copy, md5 odaodc4bcb7de883c7bee4444406d00a.
105. Sasol. "Sasol Announces Voluntary Property Purchase Program." July 18, 2013. It does not name Mossville. <https://www.sasol.com/sasol-announces-voluntary-property-purchase-program-residents-designated-areas-affected-companys> Pinned copy, md5 13e3bdddeb939d67f4e1e9d2b5943a99.
106. Sasol. Written response to Concerned Citizens of Mossville and two university clinics, hosted by the Business & Human Rights Resource Centre. Dated 2022 by its file name only. The company's own account. https://media.bhrrc.org/media/documents/BHRCResponse_100422.pdf Pinned copy, md5 bbo60bc487435fc548cd3e6723fdd2b1.
107. Louisiana Illuminator. "Appeals court rules in favor of Formosa Plastics facility in St. James." January 26, 2024. <https://lailuminator.com/2024/01/26/appeals-court-rules-in-favor-of-formosa-plastics-facility-in-st-james/>
108. Baxter International. North Cove fact sheet, May 2025. No address was recorded for this document. Pinned copy, md5 2058e8826b7692f04c9c886330b1e273.
109. US Department of Health and Human Services. Fact sheet: HHS continues action to increase access and supply of IV fluids following Hurricane Helene. October 18, 2024. The live page redirects to an archive; an archived copy was read. <https://www.hhs.gov/about/news/2024/10/18/fact-sheet-hhs-continues-action-increase-access-supply-iv-fluids-hurricane-helene.html>
110. United States Code, Title 42, section 1320a-7b. <https://uscode.house.gov/view.xhtml?req=granuleid:USC-prelim-title42-section1320a-7b&num=0&edition=prelim>
111. US Department of Health and Human Services. White paper on preventing drug shortages and supply-chain vulnerabilities. April 2, 2024. <https://aspe.hhs.gov/reports/preventing-shortages-supply-chain-vulnerabilities> Pinned copy, md5 3489904bb68ff3be98761afc41bfccfb.
112. National Hurricane Center. Tropical Cyclone Report: Hurricane Helene (AL092024). https://www.nhc.noaa.gov/data/tcr/AL092024_Helene.pdf Pinned copy, md5 64cf9a311c1f1478b12bc6b1047551ca.
113. US Food and Drug Administration. "FDA Commissioner Scott Gottlieb, M.D., updates on some ongoing shortages related to IV fluids." Statement, January 16, 2018. Archived copy of January 25, 2018; the address is the page's original one. <https://www.fda.gov/NewsEvents/Newsroom/PressAnnouncements/ucm592617.htm> Pinned copy, md5 b60c98b8db8312574d00f3ca00b5e7ec.
114. US Food and Drug Administration, via PR Newswire. "Statement by FDA Commissioner Scott Gottlieb, M.D., updating on some ongoing shortages related to IV fluids." <https://www.prnewswire.com/news-releases/statement-by-fda-commissioner-scott-gottlieb-md-updating-on-some-ongoing-shortages-related-to-iv-fluids-300583356.html> Pinned copy, md5 7fa658cb77409e5236cfa2fde85dd6a8.
115. US Food and Drug Administration. Drug Shortages database, index. Archived copy of September 6, 2024. <https://web.archive.org/web/20240906163654/https://www.accessdata.fda.gov/scripts/drugshortages/default.cfm> An archived capture, served inside the archive's own wrapper; the study's reading is of the copy kept. Pinned copy, md5 2ca983dd520284355361a4dca1d5b9a5.
116. United States Code, Title 21, section 356c. <https://uscode.house.gov/view.xhtml?req=granuleid:USC-prelim-title21-section356c&num=0&edition=prelim>
117. US Food and Drug Administration, Drug Shortages Task Force. Report on drug shortages, 2019. <https://www.fda.gov/media/131130/download> Pinned copy, md5 007fa59cd31621bc644e6f26a1aae646.
118. Baxter International. "Baxter's Response and Recovery Efforts Following Hurricane Helene." <https://www.baxter.com/news-insights/baxter-response-recovery-efforts-following-hurricane-helene> Pinned copy, md5 294eb817e572boa065aa7342aed14abb.
119. US Energy Information Administration. Crude oil production, United States, Texas and New Mexico: series MCRFPUS2, MCRFPTX2, MCRFPNM2. Retrieved August 24, 2026. https://www.eia.gov/dnav/pet/hist_xls/MCRFPUS2m.xls A series that changes; the study's reading is of the copy kept on the date shown. Pinned copy, md5 21fb8e86a9cc174af734593c66cea09c.

120. US Energy Information Administration. Refinery Capacity Report, capacity as of January 1, 2026. <https://www.eia.gov/petroleum/refinerycapacity/refcap26.xlsx> Address found after publication; the document fetched there is the pinned copy. Pinned copy, md5 f271ae8cce3e0f6555bea4f4481d4252.
121. The Roundtable. The Uneven Month. Study 1, research paper and results. <https://thethirdattractor.org/studies/the-uneven-month/>
122. Railroad Commission of Texas. About the Railroad Commission of Texas; Commissioners. Retrieved August 24, 2026. <https://www.rrc.texas.gov/about-us/> A page that changes; the study's reading is of the copy kept on the date shown. Pinned copy, md5 4f00a2942587a7f805bd81c5f1254995.
123. New Mexico State Investment Council. Land Grant Permanent Fund. Retrieved August 24, 2026. <https://www.sic.state.nm.us/investments/permanent-funds/land-grant-permanent-fund/> Pinned copy, md5 ef988734a12d2462c38d1b0a395c66e5.
124. US Energy Information Administration. "Onshore crude oil production on federal lands has increased in recent years." Today in Energy. Retrieved August 24, 2026. <https://www.eia.gov/todayinenergy/detail.php?id=65804> Pinned copy, md5 b709f13cbe5a03d64d56e6ac794beb54.
125. Motiva Enterprises. About us. Retrieved August 24, 2026. <https://www.motiva.com/about-us> A page that changes; the study's reading is of the copy kept on the date shown. Pinned copy, md5 3508a37be9b5fb7ad021878e3d043686.
126. US Census Bureau. 2020 Decennial Census and American Community Survey: Port Arthur city and Jefferson County, Texas; State of Texas. <https://api.census.gov/data/2020/dec/p1> Pinned copy, md5 8c09b494baf52c4ff5764ab65f63c62d.
127. US Environmental Protection Agency, Office of External Civil Rights Compliance. Administrative closure letter, Complaint 02R-21-R6. December 26, 2023. <https://www.epa.gov/system/files/documents/2023-12/2023-12-26-final-02r-21-r6-tceq-ox-bow-admin-closure-letter-to-recipient.pdf> Pinned copy, md5 c3ffb5f024c68d24c4bbb47c2578a94e.
128. US Energy Information Administration. Refinery utilization, gross inputs and operable capacity: series MOPUEUS2, MGIRIUS2, MOCLEUS2. Retrieved August 24, 2026. https://www.eia.gov/dnav/pet/hist_xls/MOPUEUS2m.xls A series that changes; the study's reading is of the copy kept on the date shown. Pinned copy, md5 64858cb62dae878855fc6a96cefaae40.
129. Railroad Commission of Texas. Seismicity Response; Seismicity Review. Retrieved August 24, 2026. <https://www.rrc.texas.gov/oil-and-gas/applications-and-permits/injection-storage-permits/oil-and-gas-waste-disposal/injection-disposal-permit-procedures/seismicity-review/seismicity-response/> A page that changes; the study's reading is of the copy kept on the date shown. Pinned copy, md5 3a1045d76ec181eeaabc855a124ffdba.
130. Railroad Commission of Texas. Seismic Response Statement of Guidance. January 31, 2022. https://www.rrc.texas.gov/media/buhgzt00/2022-01-31_seismic_response_sog_final.pdf Address found after publication; the document fetched there is the pinned copy. Pinned copy, md5 1220405432924cb96a1740c56a0703ca.
131. New Mexico Oil Conservation Division. Seismicity Response Protocol, table, revised November 23, 2021. No address was recorded for this document. Pinned copy, md5 f25c232e6bed53b8b90189dfcc0a00d1.
132. New Mexico Energy, Minerals and Natural Resources Department, Oil Conservation Division. Notice: Electronic Seismic Information Forms and GIS Layers for Seismic Areas. December 20, 2021. <https://www.emnrd.nm.gov/ocd/wp-content/uploads/sites/6/12-20-21-NOTICE-Seismic-forms-and-GIS-Layers.pdf> Address found after publication; the document fetched there is the pinned copy. Pinned copy, md5 863fc293bb9e19499e699421a2fe61f0.
133. US Geological Survey, Earthquake Hazards Program. FDSN event web service: earthquakes of magnitude 2.0 and above, West Texas and southeastern New Mexico, January 1, 2014 to August 24, 2026. Retrieved August 24, 2026. The live catalog is revised over time; the pinned file is the copy of record. <https://earthquake.usgs.gov/fdsnws/event/1/query?format=csv&starttime=2014-01-01&endtime=2026-08-24&minlatitude=30.5&maxlatitude=33.2&minlongitude=-105.2&maxlongitude=-100.3&minmagnitude=2.0&orderby=time-asc> Pinned copy, md5 3766a0ed393b68f2ee6c511d8832f3b9.
134. New Mexico Energy, Minerals and Natural Resources Department, Oil Conservation Division. Well_Injection feature service (OCDView), Eddy and Lea counties. Retrieved August 24, 2026. A live service; the pinned file is the copy of record. https://gis.emnrd.nm.gov/arcgis/rest/services/OCDView/Well_Injection/FeatureServer/0 Pinned copy, md5 1279faf25c249585bafb82d53cb65f10.
135. Sociedad Química y Minera de Chile (SQM). Form 20-F for fiscal year 2025. US Securities and Exchange Commission, EDGAR. <https://www.sec.gov/Archives/edgar/data/909037/000090903726000023/sqm-20251231.htm>
136. US Census Bureau. International trade, imports by Harmonized System code and mode of transport, calendar year 2024. Retrieved September 23, 2026. <https://api.census.gov/data/timeseries/intltrade/imports/hs> Pinned copy, md5 654cfo14347b08f70af4021e514fb1e5.
137. Tesla, Inc. Form 10-K for fiscal year 2025. US Securities and Exchange Commission, EDGAR accession 0001628280-26-003952. <https://www.sec.gov/Archives/edgar/data/1318605/000162828026003952/tsla-20251231.htm> Address found after publication; the document fetched there is the pinned copy, apart from a script the server appends.

138. International Energy Agency. Global Critical Minerals Outlook 2025: Executive summary. Retrieved August 24, 2026. <https://www.iea.org/reports/global-critical-minerals-outlook-2025/executive-summary> Pinned copy, md5 250740a13aa1f8f75e4e83248f33478d.

139. International Energy Agency. Global Critical Minerals Outlook 2025: Overview of outlook for key minerals. Retrieved August 24, 2026. <https://www.iea.org/reports/global-critical-minerals-outlook-2025/overview-of-outlook-for-key-minerals> Pinned copy, md5 17e8f7d781b24d33f037cdd653f014ed.

140. International Energy Agency. Global Critical Minerals Outlook 2025, full report. Page 93: "Geographical distribution of refined material production for key minerals in the base case." The file the publisher serves today is not the copy the study read; the copy as read is kept, and the reading was made on it. No address was recorded for this document. Pinned copy, md5 6332f45aaf20c5f2ee8b6012243304d7.

141. US Geological Survey. Mineral Commodity Summaries 2026: Copper. <https://pubs.usgs.gov/periodicals/mcs2026/mcs2026-copper.pdf> Pinned copy, md5 12ca7112f4c8ae5a17c4821642cebffe.

Appendix A · the register ledger

Every row the study keeps. Standing runs open, sourced, validated, integrated. For a validated row the last column says what was validated, in words its validating seat has read. For a row below validated it gives the question the row was opened to ask, labeled as a question; such a row carries no claim in the published chapters. Generated from the registers at build time.

copper · 83 rows

ROW	STANDING	VALIDATOR	WHAT IT HOLDS
den-world-mine	validated	-	
den-world-refined	validated	-	
den-lifecycle-energy	validated	-	
esc-f1	validated	institutions-commons	Minera Escondida's own statutory annual report: 4,000 employees of its own and 6,532 operational contractors at December 31, 2025. Two figures, never summed. The filing does not say whom 'operational' leaves out.
esc-f2	open	energy-biophysical-economics	<i>question asked:</i> truck-fleet diesel consumption (or fleet size) from BHP/primary
esc-f3	validated	author	
esc-c1	validated	energy-biophysical-economics	Desalting the mine's water takes about 3.4 kWh per cubic meter and pumping it 3,100 meters up and 170 km (about 105 miles) inland about 14 (band 11 to 15), so the climb costs about four times the desalting; at the plant's stated flow that is about 1.4 TWh of electricity a year, about what 130,000 US homes use.
esc-c2	validated	econ-stats-referee	Escondida produced about 5.7% of the world's mined copper in its 2025 fiscal year, one tonne in eighteen (company filing over the USGS world estimate); the share is of mined copper, not refined, and the year before it was nearer one in twenty.
esc-g1	validated	institutions-commons	The Peine conciliation, from the tribunal's own record: the community filed before the state did; the agreement was approved in December 2024 on a divided bench, the dissent holding that no measure repairs the drawdown itself and that recovery around the year 2200 assumes a cessation nobody signed; about US\$47 million and fourteen measures, among them a standing table where the community sits; implementation uneven as of early 2026. Characterized by venue and status; never as resolved.
esc-f4	validated	econ-stats-referee	

esc-f5	validated	energy-biophysical-economics	Chile's national electricity use per tonne of copper in concentrate, 2024, with its 2010 baseline, and open-pit fuel use per tonne (Cochilco). National averages by process; not any one mine's.
esc-f6	dropped	econ-stats-referee	<i>dropped; carries no claim</i>
esc-f7	sourced	econ-stats-referee	<i>question asked:</i> desal investment (audit A8; distinct from col-f2 capacity)
esc-f8	open	energy-biophysical-economics	<i>question asked:</i> Escondida/Spence PPA split, if BHP filings disclose (energy memo rec 5 — collapses apportionment estimate)
esc-w1	validated	complex-systems-polycrisis	A water reservoir exists at the Escondida mine, between the pipeline from the coast and the concentrator; the operator's 2013 filing names it and gives no capacity. Its volume, and so any number of days of cover, is not established and never renders as a quantity.
esc-g2	open	institutions-commons	<i>question asked:</i> NEW (instcommons rec 7): pull the approval resolution PDF (fojas 32157) — full Mo-M13 obligations, timelines, breach machinery; fills 'graduated sanctions: not ...
esc-ghost1	sourced	econ-stats-referee	<i>question asked:</i> Same-scale ghost: Manhattan Island outline over the Escondida district (concept F, ghost #1 of the two-ghost budget)
esc-f9	open	econ-stats-referee	<i>question asked:</i> Copper ore-grade decline series: ~40% (19th-c Montana) -> ~4% (early-20th-c US avg) -> ~0.4% (now) — spoken by Hagens, unchallenged by Conway (TGS ep.127) ...
esc-f10	open	econ-stats-referee	<i>question asked:</i> Conway (TGS ep.127): net zero requires 'another three [Chuquicamata-scale] mines every year' to 2050 [SECONDARY-INTERVIEW -> verify vs IEA critical-minerals reports ...
esc-i1	validated	-	
esc-i2	validated	-	
col-f1	validated	energy-biophysical-economics	The mine's concentrate leaves by slurry pipeline to its own port (operator presentation, 2004); this seat validated one thing only, that the concentrate runs downhill from about 3,100 meters while the desalinated water is pumped up the same corridor, and nothing about whether the line is pumped.
col-f2	sourced	energy-biophysical-economics	<i>question asked:</i> EWS desal plant capacity from BHP/Bechtel primary
col-i1	integrated	author	
col-d1	validated	energy-biophysical-economics	Port-to-port distances for the three ocean legs (NGA Pub. 151); the longest, Shanghai to Los Angeles, is 5,810 nautical miles: 10,760 km, or about 6,700 miles.
route-d8	validated	energy-biophysical-economics	The concentrate's leg from the port to the smelter, and the cathode's back, are each at least about 280 miles (450 km) in a straight line; the routes themselves are longer.
route-d7	validated	energy-biophysical-economics	The finished wire's last leg is at least about 650 miles (1,050 km) in a straight line, from the mill at Carrollton to the Dallas area; the route itself is longer.
route-d6	validated	energy-biophysical-economics	The cathode's US leg is at least about 1,850 miles (3,000 km) in a straight line, from Long Beach to the mill at Carrollton; the route itself is longer.
col-g1	validated	institutions-commons	Caleta Coloso's fishers' organizations administer the cove under Chile's Caletas law from November 2025 (the fisheries agency's own account), and the port's owner funds the cove's new facilities through its foundation (the company's account, with the fishers' president quoted on the new infrastructure on the agency's page). Who will own the facilities, the agreement's term, and any exchange behind the funding are not established.
route-t1	validated	energy-biophysical-economics	At the container fleet's average speed at sea, the 5,810 nautical miles (about 6,700 miles) from Shanghai to Los Angeles is 14 to 15 days; a fleet-wide

			annual average, not a voyage record, and it leans long because a scheduled ocean crossing tends to beat its fleet's mean.
route-t3	validated	energy-biophysical-economics	At the bulk-carrier fleet's average speed at sea, the ore's 9,150 to 9,700 nautical miles (about 10,500 to 11,200 miles) from northern Chile to Shanghai is 33 to 37 days, more than twice the metal's crossing, because the ore travels farther on a slower kind of ship; the same fleet-average caution applies.
route-t2	validated	energy-biophysical-economics	
gux-f1	validated	institutions-commons	A peer-reviewed study documents arsenic, cadmium, copper, lead and zinc in the soils and vegetables of named villages around the Guixi smelter. The scale, about fifteen villages and ten thousand people, is a remediation organization's figure. This study states no health outcome.
gux-f2	validated	econ-stats-referee	Jiangxi Copper's own 2024 annual report lists its provincial state parent as the largest shareholder at 44.06% at year end; that is one holder in a top-ten table, and no sum across holders is used for the owner-operator point.
gux-c1	validated	energy-biophysical-economics	Smelting copper concentrate to cathode took about 9 GJ per tonne as a 2017 China average (modern band 9 to 14), against 250 to 400 kWh per tonne for the electrolytic refining step alone; national averages, not the Guixi plant's own figures.
gux-c2	validated	econ-stats-referee	Guixi is the only single plant that refines more than a million tonnes of copper a year, by international capacity tables and the operator's own annual report; the claim is about refining, and the plant is not the world's largest smelter.
gux-f3	dropped	UNCONVENED-EARTH-SYSTEMS	<i>dropped; carries no claim</i>
gux-f4	validated	econ-stats-referee	
gux-i1	validated	-	
cath-i1	integrated	author	
swr-f1	validated	econ-stats-referee	Southwire describes itself, on its own website, as one of the leading manufacturers of wire and cable; that is the company's word and its own account, and no rank, share, headcount or revenue is established.
swr-f5	validated	author	That Southwire operates a Building Wire Plant in Carrollton, Georgia, beside its Copper Rod Plant (the operator's own newsroom, December 2025). A plant and its purpose; no process step, no tonnage.
swr-w1	validated	institutions-commons	Rod-mill and building-wire-plant crews work at Southwire's Carrollton works: a role at plants the operator itself names, and no more. No count, and nothing about where they live.
swr-f2	validated	energy-biophysical-economics	What the maker specifies for the shaft furnaces it sells: fuel 'natural gas and/or LPG', and 'up to 20% high quality scrap' in the charge (Southwire product sheet and SCR systems page). A product specification; not a statement about the Carrollton furnace.
swr-c1	validated	energy-biophysical-economics	Clean copper scrap charged to a shaft furnace reaches molten copper for roughly three to six percent of the ore route's primary energy. Excludes collecting, sorting and shipping the scrap; applies to clean scrap only; one term in it is assumed.
swr-f4	open	energy-biophysical-economics	<i>question asked:</i> Energy intensity per tonne of the Carrollton rod line (melt + roll + draw). Split from swr-f2.
swr-i1	integrated	author	
swr-f3	validated	econ-stats-referee	Southwire states that its own Carrollton rod mill is rated at 54 tonnes an hour; that is a rated figure in the company's words, the page does not say which of its Carrollton mills it describes, and no annual output is established.

dfw-f1	validated	econ-stats-referee	
dfw-b1	validated	econ-stats-referee	What a wire distributor keeps on hand is not published in the federal monthly wholesale survey, which reports only the parent category, electrical and electronic goods, and not the wiring-supplies line (the annual survey was not checked); and even that line would be dollars of stock over dollars of sales, which moves with the price of copper and is not a count of wire, so no stock of wire and no days of cover are established here.
dfw-p1	validated	econ-stats-referee	The copper in one average American home's building wire, at the year's average US producer cathode price, was worth about \$840 in 2024; that is the metal, not the wire, which carries drawing, insulation, distribution and margin this figure does not.
dfw-u1	sourced	econ-stats-referee	<i>question asked:</i> where American copper actually goes: the end-use split, and the reason copper's break is felt as a building that waits rather than a light that goes out
gux-w1	validated	institutions-commons	Smelter and refinery crews work at the Guixi works: a role at a named operator's named plant, and no more. No count.
sea-w1	validated	institutions-commons	Seafarers crewed both ocean crossings: entailed by the two sea legs. Role only. This study holds no count of them, and the receipt says so on its own line.
move-w1	validated	institutions-commons	Overland freight crews carried the wire's copper from port to mill and from mill to home: entailed by two inland endpoints. The mode of the port-to-mill leg is not sourced. Role only, no count.
dfw-w1	validated	institutions-commons	Three kinds of hands exist at the install stage, as classes and no more: the wholesale tier, licensed electricians and homebuilders. The household is not a hand. No count and no place.
dfw-c1	validated	energy-biophysical-economics	Making the 88 kg of copper wire in one house took energy equal to about one month of that home's electricity use (0.35 to 1.15 months depending on whether primary or delivered energy is compared), and over forty years the wire carries several hundred times that; the making is energy of all kinds, not electricity alone.
scrap-f1	open	author	<i>question asked:</i> name at least one facility to earn a site pin, else stays a ring
scrap-f2	validated	complex-systems-polycrisis	
scrap-c1	dropped	-	<i>dropped; carries no claim</i>
scrap-f4	open	econ-stats-referee	<i>question asked:</i> NEW: volume flat but VALUE doubled — \$2.76B (2017) -> \$5.38B (2024), unit value \$2,744 -> \$5,625/t (+105%). Decompose price vs composition (spec-gates pushing sorting ...)
scrap-f3	validated	econ-stats-referee	
scrap-g1	validated	complex-systems-polycrisis	US copper-scrap exports to all destinations stayed flat, between about 0.78 and 1.01 million tonnes every year from 2017 to 2024, while their destinations reorganized: China's share fell from about 68% to 10% and came back to 41%, and Malaysia's rose from under 1% to 25% and fell to 7%. Two rules from opposite sides did it, China's 2020 reclassification of high-grade scrap and Malaysia's 2022 purity standard; the 2021 dip is not attributed to Malaysia. This is US exports, not world scrap trade.
rh-esc-power	open	-	
rh-esc-power-p	validated	energy-biophysical-economics	Escondida and the Spence mine together contracted 6 TWh a year of renewable electricity from Enel Generación Chile (from August 2021) and Colbún (from January 2022); the volumes are contractual and annual, shared between two mines, and say nothing about hour-by-hour supply.
rh-esc-power-env	open	energy-biophysical-economics	<i>question asked:</i> ENV/ECO: footprint of those named plants (land, siting, lifecycle) — 'renewable' still stands somewhere
rh-esc-power-econ	open	econ-stats-referee	<i>question asked:</i> ECON: PPA scale, seller, what the contract means for Chile's grid buildout

rh-esc-power-comp	validated	-	
rh-gux-power	open	-	
rh-gux-power-p	validated	energy-biophysical-economics	What powers the Guixi smelter is not on public record: a 1,880 MW four-unit coal station stands beside it (Global Energy Monitor), and whether any of its output reaches the smelter is stated nowhere a reader can check.
rh-gux-power-env	open	energy-biophysical-economics	<i>question asked:</i> ENV/ECO: the power plant's own emissions/water on top of the smelter's documented plume
rh-gux-power-econ	open	econ-stats-referee	<i>question asked:</i> ECON: plant ownership, employment, smelter-town dependency (company town economics)
rh-gux-power-comp	sourced	energy-biophysical-economics	<i>question asked:</i> COMPARISON: one feel-it line for the smelter's draw
rh-swr-power	open	-	
rh-swr-power-p	open	energy-biophysical-economics	<i>question asked:</i> POWERS: Georgia grid mix at the rod mill (Southern Co.) + SCR furnace fuel — what actually melts the cathode?
rh-swr-power-env	open	energy-biophysical-economics	<i>question asked:</i> ENV/ECO: that grid mix's footprint (incl. Vogtle share if material) at the mill's meter
rh-swr-power-econ	open	econ-stats-referee	<i>question asked:</i> ECON: Southwire as Carroll County employer; energy cost share of rod production
rh-swr-power-comp	sourced	energy-biophysical-economics	<i>question asked:</i> COMPARISON: one feel-it line for the rod mill's draw
rh-bunker	open	-	
rh-bunker-p	validated	energy-biophysical-economics	
rh-bunker-env	validated	energy-biophysical-economics	
rh-bunker-econ	open	econ-stats-referee	<i>question asked:</i> ECON: freight cost share of delivered wire price (how cheap moving a tonne across the Pacific actually is)
rh-bunker-comp	validated	-	Carrying one home's worth of copper across the Pacific twice, as ore and as metal, takes four or five gallons of ship fuel (3.6 to 5.3; 5.3 to 7.6 GJ per tonne of copper); sea legs only, measured at the ship, with the overland legs excluded because their mode is not sourced here and their distances are straight-line floors only, from which no fuel follows.

disruption · 6 rows

ROW	STANDING	VALIDATOR	WHAT IT HOLDS
dis-p1	validated	econ-stats-referee	The study does not rank disruptions by likelihood, by ruling and with the Editor's acceptance: its cases are here because they happened and were documented, which gives no rate, so it describes recurrence, exposure and concentration and orders nothing.
dis-p2	validated	complex-systems-polycrisis	
dis-p3	validated	institutions-commons	

dis-p4	validated	energy-biophysical-economics	
dis-s1	validated	complex-systems-polycrisis	The six chains answer the disruption question in different kinds (a curve, a transit time, days of cover, a queue), fixed by where the stock sits and whether the process can be hurried. More data fills in a kind and never turns one kind into another, so an ordering across the six would order definitions. A finding about these six chains, which were chosen and are not a sample.
dis-s2	validated	complex-systems-polycrisis	

iron · 35 rows

ROW	STANDING	VALIDATOR	WHAT IT HOLDS
goes-f1	validated	author	Who makes grain-oriented electrical steel in the United States: Cleveland-Cliffs is the sole active domestic supplier and can meet 12 to 20 percent of domestic demand (DOE 2024), closed at company level; the named works is scoped, not claimed.
but-w1	validated	institutions-commons	That mill crews work at the Butler works, as a role. No count, and no finer role than the operator's source supports.
goes-f2	validated	energy-biophysical-economics	Transformer core steel is rolled to nine thousandths of an inch, which is a catalog grade (0.23 mm, the maker's M-3), and it is that thin because the eddy-current part of a core's loss rises with the square of the sheet's thickness; the cost shares and import shares on this row were not this seat's to validate.
goes-i1	integrated	author	
xfmr-f1	validated	econ-stats-referee	The Department of Energy reports that large power transformers, once delivered inside a year, are now commonly quoted at 36 months with a recorded maximum of 60; these are figures the department carries from conversations with producers, not a measured series, and they give no lower bound.
xfmr-f2b	validated	econ-stats-referee	The Department of Energy describes large power transformers as highly customized and usually custom-built to their use; that language is what is validated, and no count of designs or unit cost is established.
xfmr-f2a	open	econ-stats-referee	question asked: LPT unit cost (the \$14M figure) - split from xfmr-f2, and econ would not spend a round on it
xfmr-f3	validated	econ-stats-referee	In 2019, on the Commerce Department's survey as the Department of Energy reports it, 137 of the 754 large power transformers bought for American use were made domestically, about 18%; the class is units of 100 MVA and above, and four in five were imported complete.
xfmr-f4	validated	econ-stats-referee	American demand for large transformers rose: the Department of Energy puts it at about 750 units in 2019, on a basis of units above 60 MVA, and projects about 900 a year by 2027, while reporting that manufacturers described far larger jumps in their own orders; the report does not reconcile the two, and no growth multiple is established.
xfmr-f5	validated	econ-stats-referee	The average installed large power transformer was about 38 to 40 years old, and about 70% were 25 years or older, on industry estimates the Department of Energy published in 2014; the department's 2022 estimate of the installed fleet is 4,900 to 6,799 units; forty years is a typical design life, not a failure age, and the survival figures in the study are the study's own fitted estimates.
xfmr-f6	validated	complex-systems-polycrisis	A 2020 federal survey of 87 domestic transformer-component makers, reported by the Department of Energy, found 89% had difficulty finding qualified or experienced workers and named welding and coil winding among the training needs; a 2024 federal advisory report puts the time from hire to fully trained and productive at six to eight months. No source gives a duration for winding a coil, and none renders.
xfmr-c1	validated	complex-systems-polycrisis	The queue model's input bands (fleet age, failure hazard, demand growth, capacity and build time) were delivered by the energy seat, are loaded and checked when the engine starts, and were reproduced by the Author and the referee. This seat validates that the engine uses them as given. It does not validate the hazard and demand physics behind them, which remain the energy seat's.

xfmr-x1	validated	complex-systems-polycrisis	One manufacturer's environmental declaration puts about 40 tonnes of copper in a 500 MVA transformer, and a second maker's 40 MVA unit carries about six tonnes, so smaller units hold more copper per unit of rating. Two declared design points and a direction, not a band for the class and not a coefficient for the fleet; the copper-to-transformer edge stays parked.
xfmr-f7	validated	energy-biophysical-economics	The grid regulator's published forecasts run from about 2 to 2.5 percent a year nationally to 5.6 percent in the fastest-growing region, and all of them clear the computed boundary; the queue becomes permanent only at sustained growth between roughly 7¾ and 10½ percent a year, depending on what counts as cleared, on an optimistic bound and with the forecasts themselves rising from one edition to the next.
xfmr-f8	parked	energy-biophysical-economics	<i>question asked:</i> EIA-860 vintage pull — replaces the Gamma age fit wholesale (1960s-70s installation waves a smooth gamma can't represent); upgrade, not gate
xfmr-f9	validated	energy-biophysical-economics	Hitachi Energy's transformer works at South Boston, Virginia, operating since 1968, announced expansions in October 2022 and September 2025 and broke ground in June 2026; the record gives dates and dollars and no capacity figure, so none is drawn from it.
xfmr-f10	open	energy-biophysical-economics	<i>question asked:</i> Distribution-class intrinsic build time B — own primary needed before the queued second-pass run (both DOE primaries are LPT-only; 1-3 mo placeholder ASSUMED)
xfmr-f11	validated	energy-biophysical-economics	
xfmr-f12	validated	complex-systems-polycrisis	
e3-c1	validated	econ-stats-referee	The queue model's ceiling: even at the fastest order growth any maker has reported, the supply paths on record yield a standard quote of about 24 to 25 months (21 to 28 across the model's uncertainty), reproduced on two independent implementations; it is a ceiling at the most generous corner, never an estimate of any wait, and reaching 36 months would take intake of roughly three and a half to four times the 2019 level.
e3-i1	validated	complex-systems-polycrisis	In a priority-first queue, the priority class waits about the build time at every priority share run, and the standard class holds the queue. A mechanism, never a level.
sig-f1	validated	econ-stats-referee	Orders from buyers who could be jumping the queue are between about one in five and one in twenty (a fitted estimate with a wide envelope), and even at the top of that range the effect on the median quoted wait is at most half a month, so it does not explain the gap between the model's ceiling and the quoted 36.
xfmr-f13	validated	econ-stats-referee	In March 2025 GE Vernova's chief executive said in an interview that the company's order backlog for gas turbines, power transformers and switchgear together stretched into 2028; that one man said this is all that is established, it is not a transformer-line horizon, and no unit order book has been found for any maker.
iq-f1	validated	econ-stats-referee	Power plants and batteries holding signed or draft interconnection agreements came to about seven and a half times what the country connected in 2024 (Lawrence Berkeley's queue data over EIA's additions); these are agreements, not transformer orders, about a third of past agreements were later withdrawn, and the figure is context for the queue, not a forecast.
xfmr-f14	open	energy-biophysical-economics	<i>question asked:</i> US domestic LPT production path 2020-24 — the un-primaried half of supply after the import side went realized-history
xfmr-f15	validated	econ-stats-referee	Transformer prices rose about 80% from the start of the pandemic on a quoted-price basis (a federal advisory council, single source) and 63 to 68% on the federal producer price index for the industry; the two bases measure different things and both are given.
xfmr-f5b	open	econ-stats-referee	<i>question asked:</i> DISTRIBUTION-class ageing: ~75% of distribution transformers past expected service life - split out of xfmr-f5, which asserted it on LPT evidence
age-i1	integrated	author	
move-f1	validated	author	What moving a large power transformer involves: units of roughly 100 to 400 tons, and only about thirty Schnabel railcars in North America (DOE 2014 and 2024, verbatim).

grid-g1	validated	institutions-commons	A federal report in 2017 weighed a strategic transformer reserve and recommended industry-led sharing instead, and DOE's 2024 report did not change that. Of five cataloged sharing programs one adds a transformer to the country, and the two most formal had never been used as of 2024. Utilities reported holding spares equal to 116% of the most-crucial units in 2016. DOE names rate recovery as a barrier to joining the one program that adds stock. That program's operator holds its inventory to a 98 percent likelihood of meeting anticipated need in its own consultant's model (DOE 2024, p. 31); the inventory is not disclosed and DOE's report does not give the model's inputs, so the figure cannot be checked from the public record. The pooling curves are a model's and are good for direction and order only; that a club has its weakest incentive to admit those who would gain most is an inference, not a computed result. The study takes no position on a federal reserve.
grid-g2	validated	complex-systems-polycrisis	
grid-f1	validated	author	
grid-f2	validated	complex-systems-polycrisis	A federal document, citing media reports, says the 2013 incident at a California transmission substation was met by a re-dispatch that avoided a blackout and that the damage took nearly a month to repair. That attributed restatement is the ceiling; no primary source exists, and neither a count of transformers nor a figure in days is established.
sst-f1	open	energy-biophysical-economics	<i>question asked:</i> SST entrants (Amperesand, Heron Power, DG Matrix), supply-chain contrast (SiC/ferrite vs GOES), Nvidia 800-VDC architecture pull [SECONDARY-NYT -> company/Nvidia primary]
sst-f2	open	energy-biophysical-economics	<i>question asked:</i> data-center load whipsaw (training-mode 500 MW swings) as grid-stability driver [SECONDARY-NYT -> grid-operator primary]

lithium · 20 rows

ROW	STANDING	VALIDATOR	WHAT IT HOLDS
li-x1	validated	complex-systems-polycrisis	One community and one tribunal treated a copper operation and a lithium operation as jointly responsible for one named aquifer, and the December 2024 agreement binds them to shared measures and a standing table. The edge between the two chains is institutional. Whether pumping brine and pumping fresh water act on the same water is disputed and is not settled here; a disruption to copper does nothing to lithium.
li-e1	validated	energy-biophysical-economics	Albemarle says of its own ponds on the Salar de Atacama that evaporation takes about eighteen to twenty-four months and may be prolonged by weather; one operator's figure for one operator's ponds, and the other operator on the same salt flat publishes none.
li-w1	validated	econ-stats-referee	Because brine spends eighteen to twenty-four months in Albemarle's evaporation ponds, by the company's own filing, those ponds hold about that many months of their own output at any moment; this is a derivation from that residence time, stated as a ratio and never a tonnage, it assumes steady operation, and the company sets no upper limit on the time.
route-d3	validated	energy-biophysical-economics	The lithium chain's two sea distances, reused from the copper chain's and carrying its conditions: 9,150 to 9,700 nautical miles (16,946 to 17,964 km, or about 10,500 to 11,200 miles) from northern Chile to China for the carbonate, and 5,810 (10,760 km, or about 6,700 miles) from Shanghai to Los Angeles for the packs (NGA Pub. 151); nothing new is measured.
route-t5	validated	energy-biophysical-economics	At the container fleet's average speed at sea, the packs' 5,810 nautical miles from Shanghai to Los Angeles is 14 to 15 days, a figure that leans long; that this trade crosses by sea in containers is on the federal record (US Census, 2024: 99 percent of HS 8507 from China by weight went by vessel, and 99 percent of that containerized), which establishes how the trade moves, not how one product does.
route-t6	validated	energy-biophysical-economics	At the container fleet's average speed at sea, the carbonate's 9,150 to 9,700 nautical miles from northern Chile to China is about three to three and a half weeks (22 to 25 days), a figure that leans long; the containerized hull is what SQM, the salt flat's other operator, describes for its own product in its annual report, which also names a slower break-bulk alternative that is not quantified here.
li-c1	validated	energy-biophysical-	Mine production of lithium in 2024: Australia 36.7 percent, Chile 20.4, China 17.1, the top three together 74.2 (USGS, lithium content); these are shares of mining, not of refining, and of a

		economics	world total from which the United States' output is withheld.
li-c2	validated	econ-stats-referee	Read off one published International Energy Agency chart for 2024, to within a point or two: China refines about 70% of the world's lithium and the top three countries about 96%, against about 44% and just under 60% for copper; these are readings of a chart, not measurements, battery-grade graphite and magnet rare earths sit higher still, and no comparison with the study's other four materials is established.
mine-g1	validated	institutions-commons	Who: nobody, and the reason is worth a sentence. This stage is a world total, not a place, and adding up is the operation that removes people. Five producing countries stand behind the figure, with five workforces, five sets of neighbors and five regulators; the number sums their tonnes and nothing else. Where the chain reaches a place again, the people return.
li-n1	sourced	econ-stats-referee	<i>question asked:</i> The carbonate plant this chain's brine goes to is named in the record, and it is Albemarle's, at La Negra
lng-g1	validated	institutions-commons	Who: Albemarle runs the carbonate plant, and it is also one of the companies in the Peine settlement at the salt flat. The company links the two places; the community does not. Peine's case is about groundwater at the salt flat, not about this plant. Around the plant is an industrial district of the Antofagasta region with people living in it, and this study can say no more about them than that: it holds no Chilean regulatory record for the plant itself.
li-p2	dropped	complex-systems-polycrisis	<i>dropped; carries no claim</i>
cam-g1	validated	institutions-commons	Who: the makers of cathode powder are specialty chemical producers. This study names none of them and has not sourced where their plants are. The work is preparing the precursor, firing it in kilns, and quality control. No headcount is given, and this study holds nothing on the communities beside these plants. A class, standing in until a plant is named.
cell-g1	validated	institutions-commons	Who: the makers of lithium-ion cells. This study names none of them and has not sourced where their plants are. The work is coating electrodes, assembling cells, and the slow first charging and testing that follows. No headcount is given, and this study holds nothing on the communities beside these plants. A class, standing in until a plant is named.
li-p1	validated	author	Where one named product's modules and packs are assembled, from the maker's own annual filing (Lathrop, California and Shanghai). One product; not a statement about the industry.
pack-g1	validated	institutions-commons	Who: the firms that build cells into modules, packs and containers. Where that is done is not established in general. In the one case this study has sourced, the maker's own annual filing names dedicated factories for the product in Lathrop, California and in Shanghai: neither where the cells are made nor where the battery is installed. Other makers may do it differently. The work is module assembly, enclosures, cooling and commissioning. No headcount is given. This study has not looked for a community record at either factory, so it states none.
li-b1	validated	energy-biophysical-economics	How much warning a break gives can be worked out from how long each process step takes, wherever that time is published; how much stock stands ready to absorb a break is published nowhere in this chain, in tonnes or in days.
conv-11	validated	energy-biophysical-economics	
li-b2	validated	complex-systems-polycrisis	Restarting a chain behind an evaporation pond cannot be faster than the pond's residence time, whatever anyone does: the pipeline that delays a break by its residence time charges the same time again on the way back. A floor given by physics, on the residence time the lithium chapter sources; no interruption is on record to observe it.
store-g1	validated	institutions-commons	Who: four kinds of party, because at this stage the unit changes. The owner, a utility or an independent operator. The public utility commission that approved the installation and decided how its cost is recovered, which is where the cost decision is actually made, and a body whose hearings the public can attend. The ratepayers of the territory it serves, who pay for it. And whoever lives beside the site, named separately, because living next to something is not the same as paying for it. No figure is put on anyone's bill: nothing in this study supports one. Every other 'who' in this study is somebody else: a mine's workforce, a smelter's neighbors, a town along a fence line. At the end of this chain the reader stops being a spectator and becomes a party, because the installation is paid for through the rate base they are in.

oil · 25 rows

ROW	STANDING	VALIDATOR	WHAT IT HOLDS
oil-b1	validated	energy-biophysical-economics	
oil-b2	validated	energy-biophysical-economics	US distillate stocks divided by distillate product supplied gave about 27 to 29 days of cover in mid-August 2026 against about 32 to 34 in the same week of 1996, measured the same way over four-, thirteen- and fifty-two-week averages; a ceiling on comfort, not a countdown, since tank bottoms and pipeline fill are counted in the stock.
conv-o1	validated	energy-biophysical-economics	
route-o1	validated	energy-biophysical-economics	
route-d4	validated	energy-biophysical-economics	The diesel's last leg is at least about 250 miles (400 km) in a straight line, from Port Arthur, where the map places the refining, to the Dallas area; the route itself is longer.
oil-s1	validated	econ-stats-referee	The Roundtable's first study found that a severed diesel link adds a median of about \$23 billion a month to American household costs, falling at 8.9% of after-tax income at the bottom of the income distribution against 1.6% at the top; it priced a shock and modeled no delay to any delivery.
oil-g1	validated	econ-stats-referee	
who-up1	open	institutions-commons	<i>question asked:</i> Name the operator CLASS and the post-2020 consolidation - the ownership-side view of oil-c1's U
who-up2	validated	institutions-commons	The Railroad Commission of Texas, by its own account, was created in 1891 to regulate railroad charges and passed its last rail duties to the state transport department in 2005. New Mexico's Land Grant Permanent Fund dates from statehood in 1912 and distributed \$1.34 billion in 2024. How much Permian oil pays into it is not sourced, so the fund is context and not attribution.
who-up3	validated	institutions-commons	What two regulators did about earthquakes from wastewater disposal, from their own documents: Texas drew Seismic Response Areas, suspended 23 deep disposal permits from January 12, 2024 and cut injection around Stanton, and changed what wells must report in 2022 and 2023; New Mexico adopted its own distance-based protocol in November 2021. Whether any of it worked is not this row's question.
who-up4	validated	institutions-commons	
oil-c1	validated	energy-biophysical-economics	Texas and New Mexico's share of US crude production fell from 32 percent in 1981 to 23 percent in 2002 and then rose to 59 percent in 2025 (EIA state series); these are two states, not the Permian Basin.
seis-x1	validated	complex-systems-polycrisis	Earthquake counts fell inside Texas's seismic response areas, and fell by more in a nearby part of the same basin with no response area, so the record cannot separate the interventions from a basin-wide decline by place. Split by depth inside the treated area it partly can: deep events nearly stopped while shallow ones fell about as the basin did, held loosely on small counts and an early measurement. Whether the largest events responded cannot be tested on this record for about a century. New deep disposal capacity did not open across the state line; volumes into existing wells there are unpublished and untested.
perm-i1	integrated	author	
who-rf1	validated	institutions-commons	The federal refinery capacity report names the parent of each of Port Arthur's three refineries: Saudi Aramco (Motiva), Valero (Premcor) and TotalEnergies. Ownership only; the capacity share is another row's.
who-rf2	validated	institutions-commons	Port Arthur had 56,039 people and Jefferson County 256,526 at the 2020 Census; median household income was \$45,752 against \$76,292 for Texas, and the poverty rate 29.1% against

13.8% (ACS 2023). These describe the town that hosts the complex. They are not a count of anyone's exposure.			
who-rf3	validated	institutions-commons	
who-rf4	validated	institutions-commons	
oil-c2	validated	energy-biophysical-economics	US refinery utilization climbed from 78 percent in 1985 to a peak near 96 percent in 1998 and has stayed in a narrow high band since, with idle capacity above 15 percent only in 2009 and 2020; a description of the series since 1985, not a claim about why.
oil-c3	validated	energy-biophysical-economics	Between 2019 and 2025 refinery throughput fell 1.7 percent while capacity fell 3.4 percent, so utilization is higher on less work done; the recovery after 2020 itself was demand returning, and this seat's prediction that closures drove it was tested and failed.
oil-n1	validated	energy-biophysical-economics	The seven largest US refineries are all on the Gulf Coast, which holds 54.4 percent of operable capacity, and the three in Port Arthur, Texas, together hold 7.0 percent (EIA Refinery Capacity Report 2026, calendar-day basis); why refining sits there, whether geology, ports or permits, a capacity table cannot say.
know-i1	integrated	author	
who-rf5	validated	institutions-commons	In 2021 a Port Arthur community group and two legal organizations filed a civil-rights complaint against the state environmental regulator over its oversight of a petroleum-coke facility, not against a refinery. EPA investigated for two years, the state withdrew from negotiations, and the file closed without a finding either way and can be re-filed. West Port Arthur, the neighborhood named, carries no figure because nobody tabulates its boundary.
who-rf6	open	institutions-commons	<i>question asked:</i> Test whether the town's largest SO2 source is the facility with the OLDEST PERMIT rather than the largest plant
study-g1	validated	institutions-commons	At four links in this study, the instrument that lets the people living there see a harm was built after the harm was established, and was built public. Four, not five: New Mexico's case is the Texas one seen from the other side of a line. 'Cannot be unpublished' holds for records that are filed and not for services that are merely kept running. At Guixi no such instrument exists at all.

salt · 32 rows

ROW	STANDING	VALIDATOR	WHAT IT HOLDS
brine-f1	validated	econ-stats-referee	The United States produces about 40 million tonnes of salt a year, brine the cheapest form at about ten dollars a tonne against fifty-six for rock salt (US Geological Survey); the roughly 14 million tonnes going to chlorine makers is derived from the survey's shares, and the survey names no individual operations.
brine-f2	validated	institutions-commons	
brine-g1	validated	institutions-commons	The Gulf belt's brine and dome operators, and the parishes and counties over the domes, named as classes because the federal survey names no operation. Wellfield and plant roles; no count. A class standing in until an operation is named.
chl-c1	validated	energy-biophysical-economics	Electricity to make a tonne of chlorine by membrane electrolysis, and the European industry's annual draw (Euro Chlor). No world share is established.
chl-f1	validated	complex-systems-polycrisis	Electrolysis makes chlorine and caustic soda in a fixed ratio, about 1.1 tonnes of caustic for each tonne of chlorine on the industry association's figures. What makes that lock bite is a storage asymmetry: chlorine cannot wait and caustic can, so production follows chlorine demand. This seat validates the mechanism; the ratio is the energy seat's reading of the primary.
chl-f2	open	complex-systems-polycrisis	<i>question asked:</i> chlorine does not ship well (liquefaction, rail tank cars, storage limits) -> chlor-alkali co-locates with its consumers; the geography row that explains the Gulf belt
chl-g1	validated	institutions-commons	Chlor-alkali producers named from a federal primary: OxyChem at Geismar; Westlake at Plaquemine, Geismar and Calvert City. And that the property-line monitoring these

			plants were to begin in July 2026 was deferred for two years by a proclamation filed the morning it took effect, whose Annex I lists the plants exempted.
pvc-f1	sourced	author	<i>question asked:</i> the PVC path to medical film: resin -> plasticized film (DEHP/alternatives row) -> bag; name the film/bag material grade; the copper cross-edge (wire insulation) ...
pvc-g1	validated	institutions-commons	The communities beside the vinyls stage, from the public record: an operating plant named ahead of an unbuilt one; the Formosa St. James permits characterized by venue and status (vacated 2022, reinstated January 2024, further review denied September 2024) and never as resolved. Mossville, in Calcasieu Parish, founded in the 1790s by African Americans by its residents' own account to the Inter-American Commission on Human Rights, was largely bought out under the voluntary property purchase program Sasol announced in July 2013 for the areas beside its Westlake expansion; Sasol says 584 property owners took part and that those who declined still live there. The buyout was Sasol's and not this chain's. The program's terms and valuations are not established.
nc-c1	validated	econ-stats-referee	The American Hospital Association stated that the North Cove plant made approximately 60%, or 1.5 million bags a day, of the intravenous solutions used in American hospitals; the figure is the association's alone, its origin is not given, and the company's own filing confirms the closure but not the share.
nc-f1	validated	complex-systems-polycrisis	Baxter's North Cove plant closed on September 29, 2024 on the company's own record, and the company's dated updates give its return: the first line at 32 days, all ten lines at 121, pre-storm output at 141. The quartz mine closed by the same storm never published a full-capacity date, so the two recoveries compare restarts to restarts only.
helene-f1	validated	complex-systems-polycrisis	On the National Hurricane Center's report, the core of Hurricane Helene brought catastrophic flash flooding to the western North Carolina region that holds both of this study's sites on the morning of September 27, 2024, after two days of rain that had already saturated the ground. That morning is the date of record for the flooding's arrival, at the precision of the region: the hour it reached either site is not on record.
helene-t1	validated	complex-systems-polycrisis	The National Hurricane Center's best track for Helene gives 25 positions of the storm's center, at the times the report lists, from September 23, 2024, to the evening of the 28th; a 26th dated row marks dissipation and carries no position. The line the map draws is where the center was, not where the rain fell: the center passed well to the west of both valleys, and the flooding that closed them came from the storm's rain over ground the days before had already saturated.
helene-r1	validated	complex-systems-polycrisis	NOAA's daily rainfall analysis, summed over the four days from noon on September 24 to noon on the 28th, puts 250 millimeters or more over both of this study's valleys and 500 or more over a core to their south and west, peaking at about 650 millimeters in the analysis. That is where a gauge-corrected radar analysis puts the rain, cell by cell, and not a measurement at any place: at the storm's record gauge the analysis reads about two-thirds of what the gauge caught.
nc-f2	validated	complex-systems-polycrisis	The same firm lost a single-point saline plant to a hurricane twice in seven years, Puerto Rico in 2017 and North Carolina in 2024, and both times the storm hit a supply already listed as short: the regulator's own database, archived three weeks before Helene's landfall, shows saline in shortage.
nc-g1	validated	institutions-commons	Baxter International operates North Cove, with more than 2,500 employees by two separate accounts. FDA says in its own words that it has no authority over allocation or distribution. HHS and FDA assert, with no magnitude, that group-purchasing contracting contributes to thin hospital buffers; the statutory safe harbor is what makes that contracting lawful. No primary gives hospitals' days of inventory, and none is printed.
nc-w1	validated	complex-systems-polycrisis	How long hospital stocks lasted renders as an observed interval and not as a days-of-inventory figure: it was days from the plant's closure to the first documented allocation of supplies. What patients experienced is recorded only through a survey of their pharmacists, qualitatively. The patient-impact wording is the Institutions & Commons seat's.
conv-s1	validated	energy-biophysical-economics	
route-s1	validated	energy-biophysical-	

		economics	
route-d5	validated	energy-biophysical-economics	The finished bag's last leg is at least about 850 miles (1,350 km) in a straight line, from the plant at Marion to the Dallas area; the route itself is longer.
nc-f3	open	econ-stats-referee	<i>question asked:</i> HHS Secretary letter referenced by the AHA Oct-9 bulletin - the next primary for the 60% share
nc-g2	validated	institutions-commons	Between Hurricane Maria and Hurricane Helene, from the primary record, the response got faster: in 2024 the same FDA toolkit brought in imports under temporary authorization from nine of Baxter's own plants abroad, the first three weeks after the storm; by the end of the year, in the company's own words, nearly 200 airplanes had delivered nearly 18,000 tons of product; and one Defense Production Act order was used. What did not change: the firm's redundancy plan stayed firm-held and inspected only as appropriate, a proposal to pay hospitals for buffer stock was solicited and not adopted, the strategic reserve excludes commercial products, a six-month reserve bill was unenacted in October 2025, and the 2024 federal paper on shortages does not mention saline. Institutions learned to respond; they did not learn to be less fragile.
resp-i1	integrated	author	
bed-f1	validated	econ-stats-referee	
bed-f2	sourced	author	<i>question asked:</i> stage 6: single-use medical plastics -> regulated medical waste -> incineration (dioxin history = the PVC disposal rabbit hole, one card); the honest non-cycle: ...
bed-g1	validated	institutions-commons	The people at the bedside, split by relation: pharmacists, nurses and supply staff appear through their professional bodies' public record; patients appear only as a share in a survey of their pharmacists; the purchasing structure decides and is cited, not restated. The record holds no patient's own voice, and this study adds none.
salt-x1	validated	complex-systems-polycrisis	One storm closed a saline plant and a quartz district roughly fifteen to twenty kilometers apart in the same week. A shared hazard, not one chain reaching the other: different owners, different regulators, different customers. One observation held in two registers, not two.
salt-x2	validated	complex-systems-polycrisis	The house wire the copper chapter follows is insulated and jacketed in PVC, on the maker's own specification and the listing standard. The salt chain joins the copper chain there at a kind of material, not at a facility: nothing says any particular PVC works supplies the wire mill. Whether a chlorine disruption would stop wire is not established.
salt-x3	validated	complex-systems-polycrisis	
rh-chl-rail	open	institutions-commons	<i>question asked:</i> chlorine on rails: the regulation-history card (public record only, per guardrail)
rh-wfi-water	open	energy-biophysical-economics	<i>question asked:</i> the OTHER ingredient: water-for-injection — the bag is salt + water + power; one card on WFI stills (energy comparison per standard)
rh-maria-2017	validated	institutions-commons	

sand · 34 rows

ROW	STANDING	VALIDATOR	WHAT IT HOLDS
spine-c1	validated	econ-stats-referee	No agency publishes Spruce Pine's share of the world's high-purity quartz, and the 70 to 90 percent figures in circulation trace to industry analysts; what is validated is that absence and the way the study renders it, with the counter-view in its own voice, and no share number is established.
spine-f1	validated	complex-systems-polycrisis	Sibelco, one of Spruce Pine's two quartz operators, halted on September 26, 2024 and announced a restart of production and shipments about fourteen days later, on its own statements, with ramp-up continuing. Neither operator published a date for full capacity, so comparisons with this clock render restarts and first lines only.

spine-f2	validated	energy-biophysical-economics	High-purity quartz is made into the crucible that holds molten silicon while a crystal is pulled; it is not the silicon, and the crucible is short-lived: by one published analysis it degrades in use and cracks after it cools.
spine-g1	validated	institutions-commons	Spruce Pine's two operators: Sibelco, and The Quartz Corp, a 50/50 Imerys and Norsk Mineral venture with its purification plant in Norway. A 2025 union representation petition at Sibelco's Crystal Plant ended with no union certified, the tally unverified. Quartz carries no federal critical-mineral designation.
spine-w1	validated	complex-systems-polycrisis	Both Spruce Pine operators pointed to stocks held in the chain. One said in its own words that feedstock and finished stocks held elsewhere in the chain would carry its customers through the closure, and called that slack a deliberate choice made after the pandemic; the other said its final product stock had not been impacted. No quantity of cover is stated. The buffers bought the weeks; whether substitution would have bought months was never tested.
spine-c2	validated	econ-stats-referee	
spine-i1	integrated	author	
route-d2	validated	energy-biophysical-economics	The sea distance between ports of the kind each end of the sand chain's Pacific leg would use: 4,839 to 5,185 nautical miles (8,962 to 9,603 km, or about 5,600 to 6,000 miles), Yokohama or Kobe to Los Angeles (NGA Pub. 151); anchored at both ends, since the study names neither the wafer works nor the US port, and not a measurement of this cargo's voyage.
route-t4	sourced	energy-biophysical-economics	<i>question asked:</i> Days at sea for the sand chain's Pacific leg (wafers, a Japanese port to a US west-coast port).
route-f1	validated	energy-biophysical-economics	
cru-f1	validated	author	A named fused-quartz crucible maker, at company level: Momentive Technologies, four Ohio plants, by its own account. No plant is identified as making the crucible this chain follows.
cru-c1	validated	econ-stats-referee	What a silicon wafer consumes in crucible quartz is not established: the one account held, a secondary analysis, says a crucible is short-lived and gives no crucible weight and no count of ingots per crucible, and no maker's specification of a crucible's weight is held; so no quartz per ingot or per wafer is stated, and nothing here says how quartz demand moves with wafer output.
cru-g1	validated	institutions-commons	Who: the makers of fused-quartz crucibles are specialist firms. One is named in this study, at company level only; no plant is identified as making the crucible this chain follows. The work is furnace operation, forming and quality control. No headcount is given, and this study holds nothing on the communities beside these works. A class, standing in until a plant is named.
poly-f1	validated	energy-biophysical-economics	Solar-grade polysilicon takes about 49 kWh of electricity per kilogram (IEA PVPS), which puts the silicon in one small chip at about a watt-hour; the electronic-grade figure is higher and unpublished, and the chip fab's own electricity, which this seat expects to be the larger term, is a figure this study named and did not pull.
poly-g1	validated	institutions-commons	Who: polysilicon producers are chemical manufacturers and heavy users of electricity, as this study's own figure for them shows. This study names none of them and has not sourced where their plants are. The work is process operation, maintenance and electrical trades. No headcount is given, and this study holds nothing on the communities beside these plants. A class, standing in until a plant is named.
waf-f1	validated	econ-stats-referee	SUMCO calls itself a front-runner in silicon wafers on its own website, and its one published share, more than half, is of wafers for leading-edge logic chips only; we found no public agency that publishes wafer makers' shares, and no share of the wafer market is established.
waf-g1	validated	institutions-commons	Who: the firms that pull silicon crystals and slice them into wafers. One, Japan's SUMCO, describes itself as among the front-runners; no public agency publishes the shares, and this study has not sourced where the works are. The work is crystal pulling, slicing and measurement. No headcount is given, and this study holds

			nothing on the communities beside these works. A class, standing in until a plant is named.
sand-s1	validated	author	
fab-f1	validated	complex-systems-polycrisis	After the chip shortage a US maker announced and began a large analog-chip plant in Texas, with about three years between announcement and first output: a capacity response in the same shape the transformer queue model describes. The wider claim, that older-generation chip capacity went unbuilt before 2020, is only partly sourced, and the chapter's own aside says so.
fab-f2	validated	econ-stats-referee	The Commerce Department's 2021 inquiry, drawing more than 150 responses, found the median inventory of the chips buyers highlighted had fallen from 40 days in 2019 to under 5 in 2021, and named analog chips used in power management among them; it is a median across respondents, not a measure of any one chip.
fab-f3	open	econ-stats-referee	<i>question asked:</i> PMIC lead-time/price record 2021-22 (40-52+ wk lead times; +10% 1H22 on 2021 rises) — trade-record primaries (distributor lead-time data if obtainable); the Apple ...
fab-f4	validated	econ-stats-referee	
fab-g1	validated	institutions-commons	Texas Instruments signed an award agreement in December 2024 for up to \$1.6 billion in federal CHIPS funding across three fabs in two states; an award agreement is not a disbursement. The company's own forward statements: as many as 3,000 direct jobs at a four-fab Sherman build, and 2,000 company jobs across the three funded fabs. Promises in the company's voice, never combined. Local incentives and the community beside the site are not on this study's record.
fab-g2	open	institutions-commons	<i>question asked:</i> Grayson County / Texas state-local incentives (Chapter 313 / JETI agreements, county abatements) and the community record around the Sherman fabs. Split from fab-g1.
fab-i1	integrated	author	
dev-f1	validated	econ-stats-referee	
ew-f1	validated	complex-systems-polycrisis	The federal commodity summaries report that post-consumer scrap provided about 150,000 tonnes of US copper in 2024, against 720,000 tonnes from manufacturing scrap, and report silicon recycling in one word: insignificant. So the copper on a circuit board can return and the silicon does not, and board copper is bounded above by that post-consumer stream and sits well below it. A ceiling, not a share.
ew-g1	validated	institutions-commons	A recovery workforce, formal and informal, exists where the world's electronics are taken apart. A class only: no count, no place, and no facility named.
sand-x1	validated	complex-systems-polycrisis	One storm closed a quartz district and a saline plant roughly fifteen to twenty kilometers apart in the same week. A shared hazard, not one chain reaching the other. One observation held in two registers, not two.
sand-x2	validated	complex-systems-polycrisis	
sand-x3	open	complex-systems-polycrisis	<i>question asked:</i> EV/lithium pull on GOES (goes-f2, iron register) is the same same-machines competition pattern: chains competing for one industrial asset — motif row
sand-x4	validated	complex-systems-polycrisis	Copper recovered from circuit boards re-enters the copper chain at its scrap pool. A join at a kind of material, never at a named smelter, and small: it renders as where the sand chain's copper goes, not as a route a disruption travels.
rh-spine-helene	open	institutions-commons	<i>question asked:</i> the storm itself: Helene's inland flood signature (NOAA/USGS record) — one card, public record, feeds sand-x1
rh-cz-energy	validated	energy-biophysical-economics	

Appendix B · engines, seeds, reproductions

E3 and E3-C are seeded and have their results and a raw-draw sample beside them; the σ fit is deterministic. Files are named here by FILE NAME, never by a folder in our working repository, which no reader has; the table at the foot of this appendix gives each file’s fingerprint, so anyone holding the study’s published materials can confirm they hold the same file. Below, a **reimplementation** is a second hand’s own code with a different seed where it draws, and nothing imported from the engine. A **probe** runs the engine’s own code and shares it. The two are not the same kind of evidence and are listed apart. Verdicts are read from the files that hold them.

E3, the transformer queue

Engine: `e3_transformer_queue_mc.py`, seed 33. Three objects were reproduced and they differ in standing:

OBJECT	STANDING	REPRODUCED BY	RESULT
the divergence boundary	current	reimplementation: <code>e3_reimpl5_seed7.py</code> , own construction, a deterministic skeleton with no draws	reproduces
the mechanical ceiling (the one-class intake probe)	current	reimplementation: <code>e3_ceiling_expectation_author.py</code> , arrivals at expectation	REPRODUCES; worst difference 0.01 month across eight rows. A check on implementation, not on the single-queue structure the ceiling is conditional on.
the same ceiling, as a Monte Carlo	current	referee cross-seed run of the Author’s engine, seed 11, in the referee’s verification package; Author, <code>e3_ceiling_mc_author.py</code> , seed 7	medians agree to 0.05 month
the σ map	superseded grid	reimplementation: <code>e3_reimpl4_seed7.py</code> , seed 7	reproduces that grid; the current map’s top member has not been reproduced outside the engine
engine behavior generally		probes (share the engine’s code): the referee’s verification scripts	referee probes of the engine, not a rebuild

E3-C, the reserve/pool companion

Engine: `reserve_pool_mc.py`, seed 41. Reimplementation: `reserve_reimpl_seed7.py`: REPRODUCES (6 legs, all pass).

In the seat’s words: INDEPENDENT compatibility OVERSTATES pooling. Real compatibility is clustered by design family. Every pooling figure here is an UPPER BOUND on the voluntary substitute’s reach. That bound is on the pool’s reach: whether some other member’s spare fits. What pooling adds is reach multiplied by the hole a holder’s own spares leave, and the same clustering moves the hole, in a direction this engine does not model. Read every gain here for its direction and its order, never for its size.

The σ fit

Engine: `eb_d039_sigma_fit.py`, with its results file beside it. **It has no seed and draws nothing:** the fit is deterministic, and re-running it returns the same file. Its own numbers render — the arrival envelope a chapter gives as “between about one in five and one in twenty” is this file’s.

Recomputed by a second hand, in its arithmetic and not in its choices. The Author rebuilt the fit from the same public inputs, with an independent plant aggregation and corner sweep

(`verify_sigma_fit_author.py`), and it reproduces the seat’s results on every check. The parameter ranges the fit sweeps were taken from the seat’s fit and not chosen afresh, so the recomputation confirms that the script computes what it says and does not test the ranges themselves. The reader should weigh the result as the wide envelope it reports rather than as a settled number.

Running it

The small inputs travel with the engine: `ferc1_acct353_additions.json`, `ppi_PCU3353113353111.json` are committed and ship beside it. The three EIA-860 annual zips and the LBNL interconnection-queue workbook do not — together about 74 MB, and this study pins large raws rather than committing them. **Fetch the three EIA-860 zips and the LBNL workbook listed below into the same folder as the script, unopened, and run it; it reads the zips directly and names anything missing.**

FILE	MD5	SOURCE
<code>eia860_2022.zip</code>	<code>2fbba665a021a7b4943349d72405cddd</code>	<code>https://www.eia.gov/electricity/data/eia860/archive/xls/eia8602022.zip</code>
<code>eia860_2023.zip</code>	<code>12ad0f63fabf0025ebf53f6728f13c18</code>	<code>https://www.eia.gov/electricity/data/eia860/archive/xls/eia8602023.zip</code>
<code>eia860_2024.zip</code>	<code>684352896d6a62b5d2d98a403b30e120</code>	<code>https://www.eia.gov/electricity/data/eia860/xls/eia8602024.zip</code>
<code>lbnl_ix_queue_data_file_thru2024_v2.xlsx</code>	<code>f67f732679a0f3d223b34e24fc84ee44</code>	<code>https://eta-publications.lbl.gov/sites/default/files/2025-08/lbnl_ix_queue_data_file_thru2024_v2.xlsx</code>

Fingerprints

SHA-256 of each file named above, taken with line endings normalized to LF, so that the value is the same on every platform. Computed when this paper was built.

FILE	SHA-256
<code>e3_transformer_queue_mc.py</code>	<code>ef84b1a408fddb8d9f73d9cb030637fbadbc845d54a5fd185fdc8e1b95a823e8</code>
<code>e3_reimpl5_seed7.py</code>	<code>6099fefbd340c4a949eb9a29ad0fb44c4a5633909cfaf40ac79867f112ea5db7</code>
<code>e3_ceiling_expectation_author.py</code>	<code>5dd4641956191d0b368fb838fedab2e3363e72564c91e75423408ce446a89523</code>
<code>e3_ceiling_mc_author.py</code>	<code>4548bcee417de1967cbef9faec1a178e099236596249dd9d7d965ca71428090d</code>
<code>e3_reimpl4_seed7.py</code>	<code>4c2c10500f76443d6332d48d56041c6ce19806161d3d6d71188744059c4d5922</code>
<code>reserve_pool_mc.py</code>	<code>5be1e2690146c809bb7478c9ea5ec8c7d0296dd31ca78d6c90e0aae031c8b00e</code>
<code>reserve_reimpl_seed7.py</code>	<code>d6a72fedafcf1f74e6ba529a1253f51e9485bd002339eff48b889b8f0afe0419</code>
<code>eb_d039_sigma_fit.py</code>	<code>fa8fea807380d0d7f71f2c546c3134ca989de47eac49d6c17a13075330265a2b</code>
<code>eb_d039_sigma_fit_results.json</code>	<code>01ed4d45ff2cea420b08e26293b69bb9ae1811029b561059f6f380c8f501ee2c</code>